DTCP Firmware Updates

Dmytro Stavskyi

01/21/2026

Objective

- Be able to communicate with the AFE chip using SPI (receive any valid data from the chip, e.g., register values)

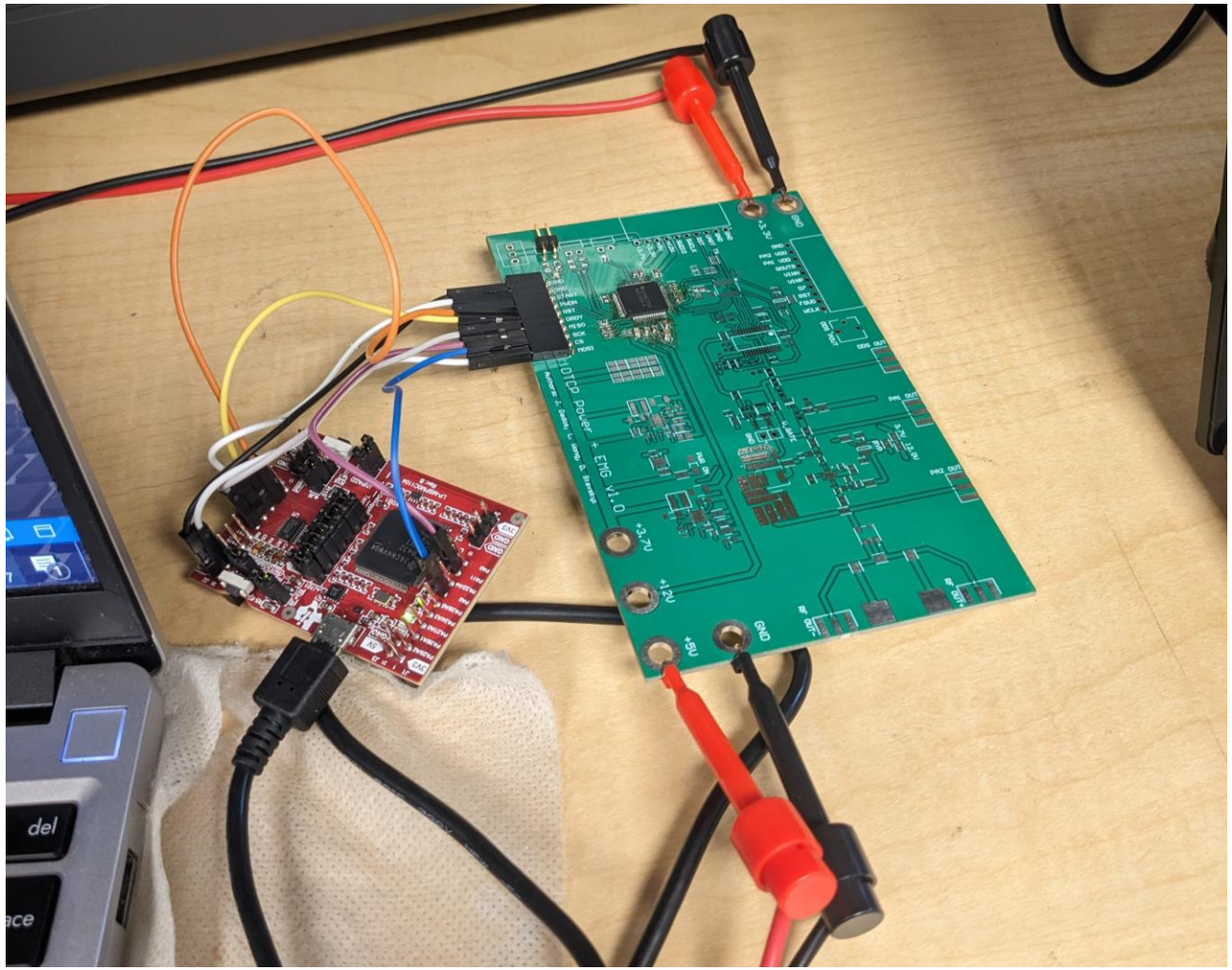
Methods

Assemble a separate board with only AFE in it for component isolation

Use the debug probs and the LaunchPad to program the chip

Measure the SCLK, CS and MOSI / MISO using oscilloscope

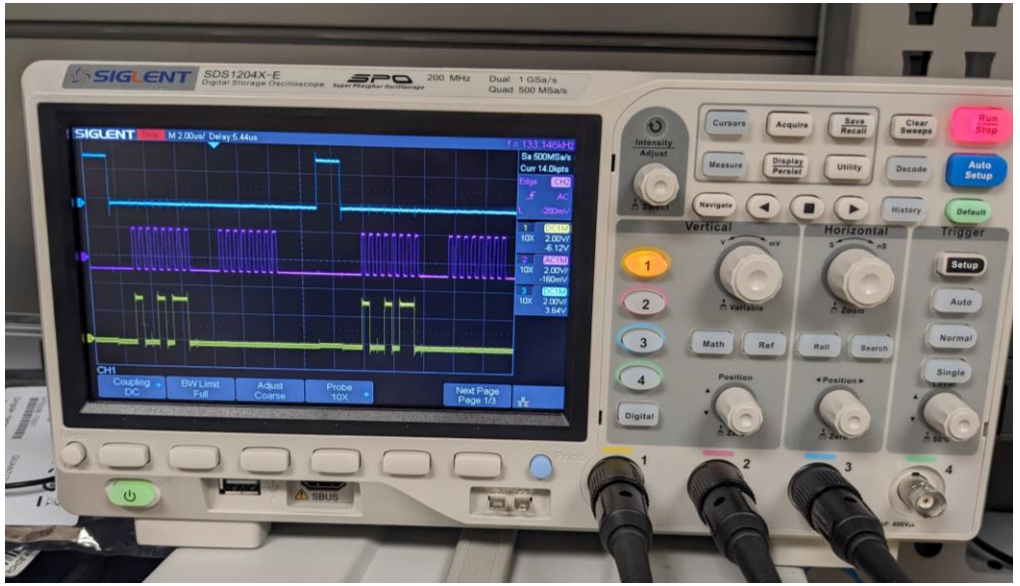
Measure the current consumption on the power supply

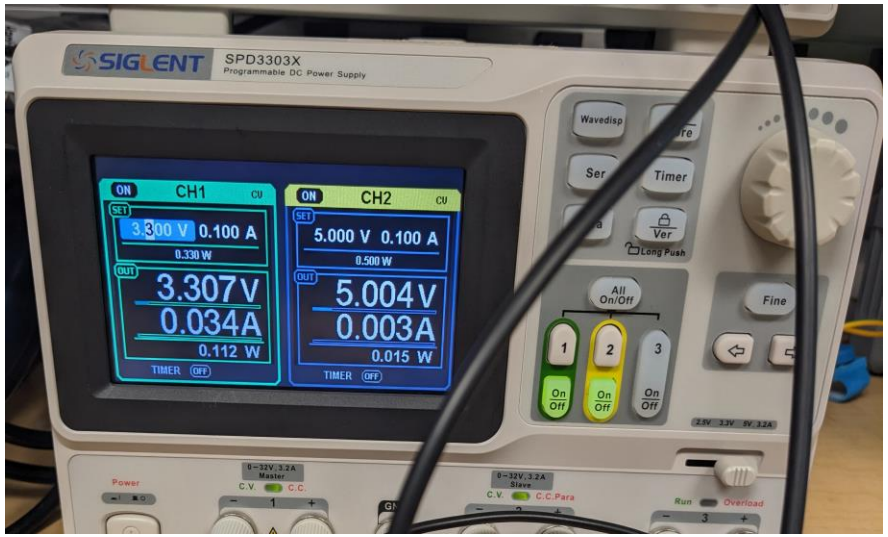


Results

At this point, we can write to registers and read from them and receive proper values

Due to incorrect configuration for one pin, the current consumption was 35 mA, but after the fix, it went back to 1~2 mA





Observation / Conclusion

- In RDATAC (Read Data Continuous) mode, you cannot read or write to registers
  - After issuing that mode using SPI, trying to read a register yields nothing, but DRDY (Data Ready) is never asserted
- Possible sources of errors: incorrect values for configuration registers

Next Steps

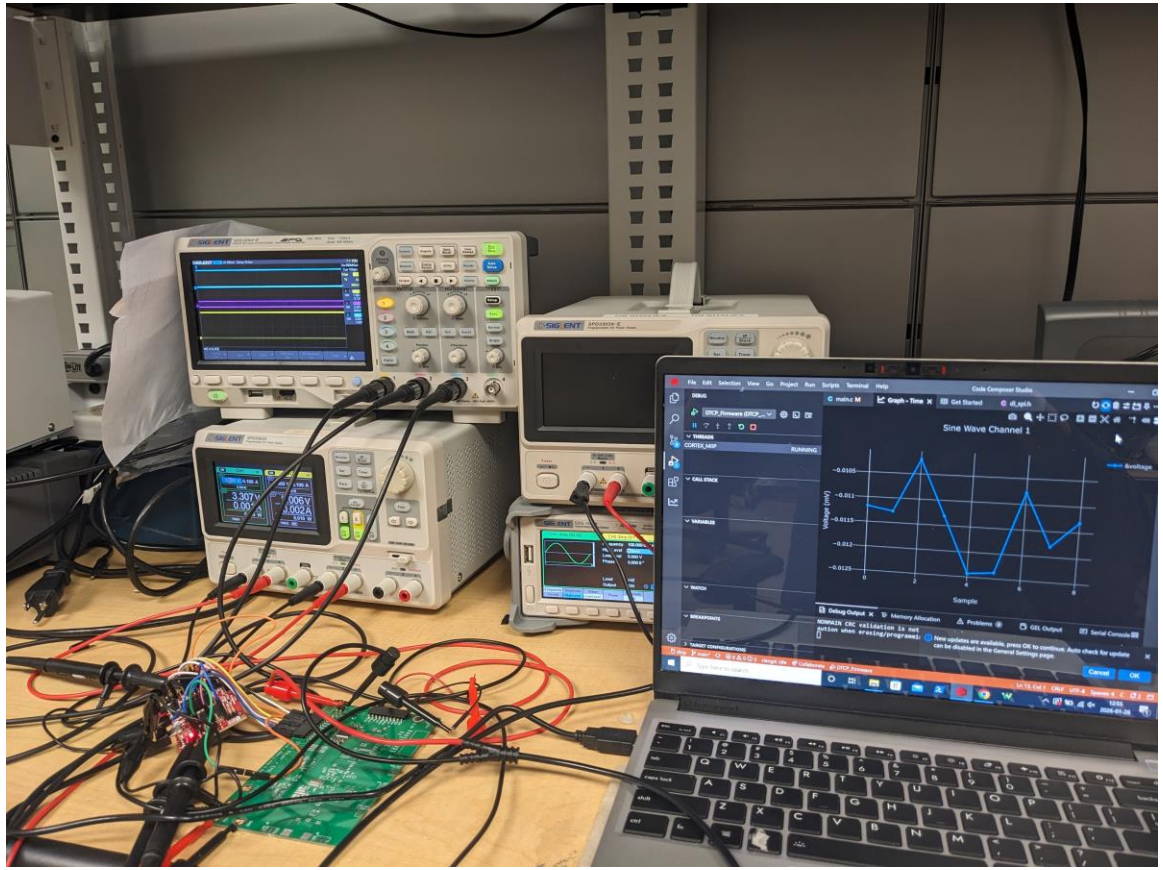
- Get conversion data from the analog channels

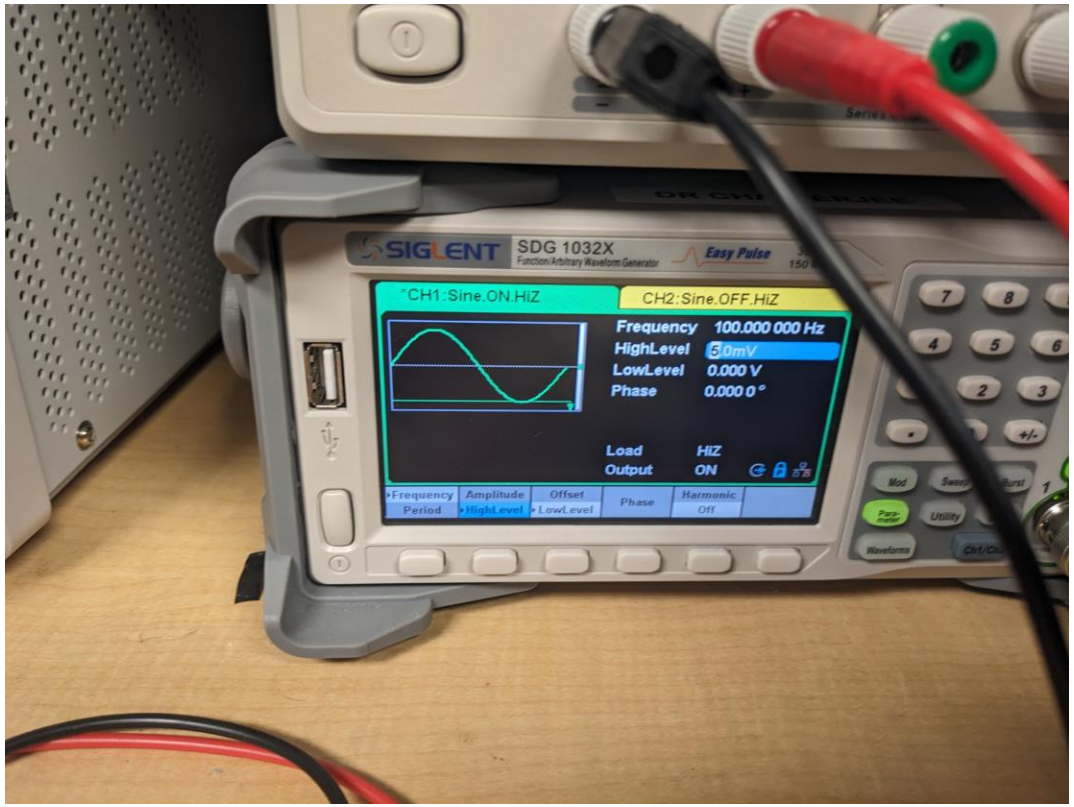
01/28/2026

Objective

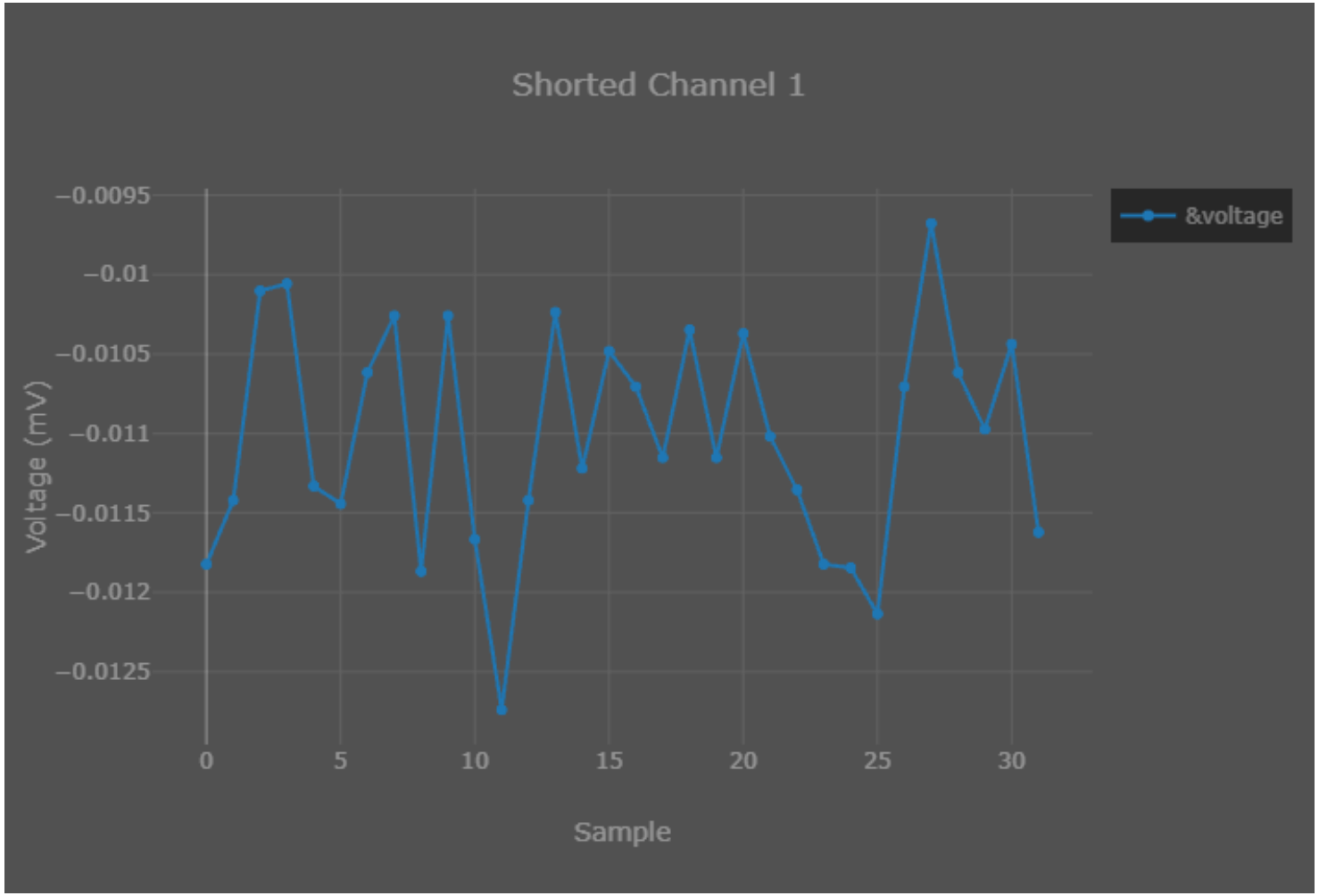
- Get conversion data from the analog channels

Methods

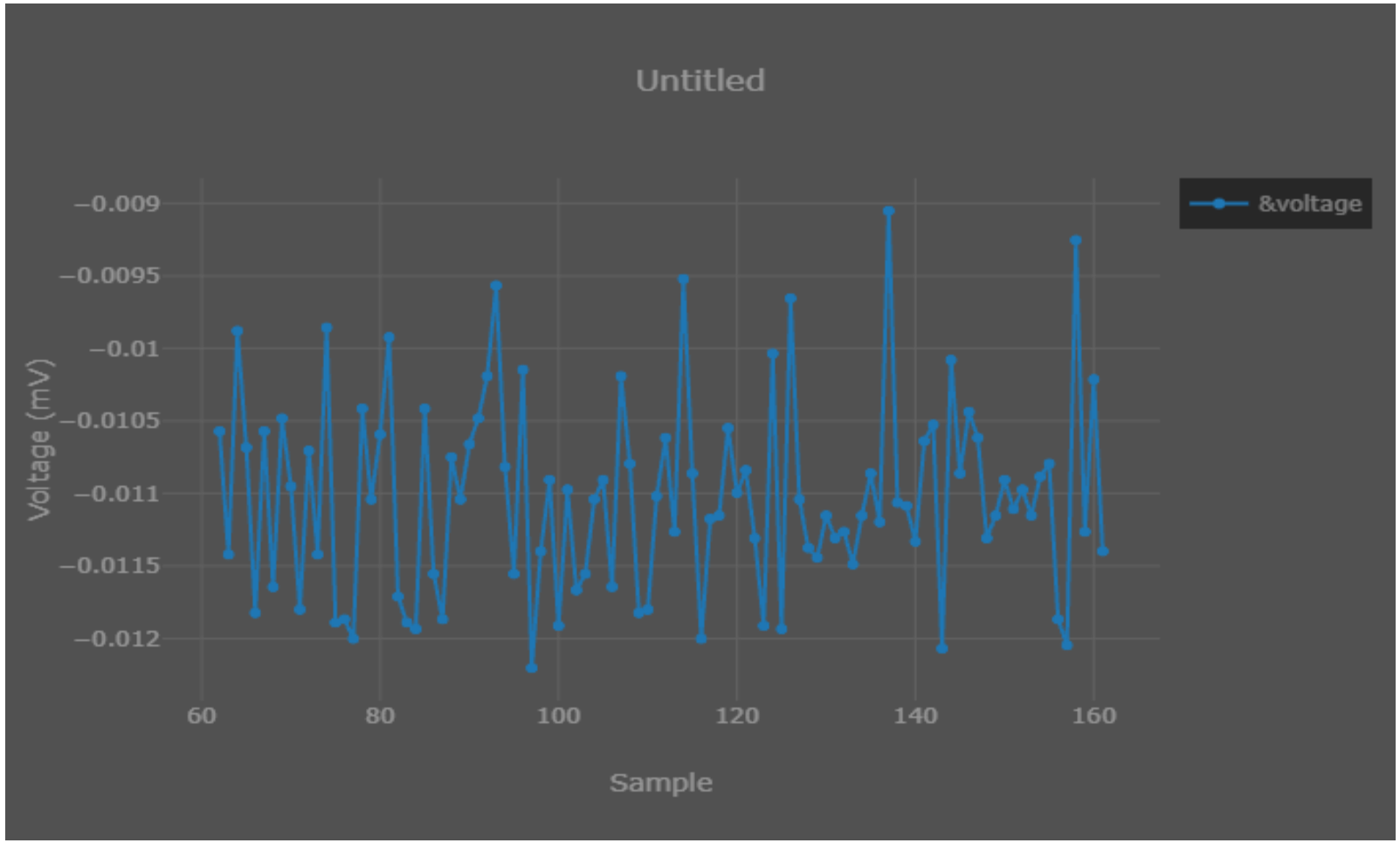

Sine Wave Channel 1


SIGLENT
SDG 1032X
Function/Arbitrary Waveform Generator
Easy Pulse
*CH1:Sine.ON.HiZ
CH2:Sine.OFF.HiZ
Frequency 100.000 000 Hz
HighLevel 5.0mV
LowLevel 0.000 V
Phase 0.000 0 °
Load HiZ
Output ON
Frequency
Amplitude
Offset
Phase
Harmonic
Period
HighLevel
LowLevel
Off

Results (shorted inputs; nominal sampling rate of 1000)



Results (100 Hz 5 mV sine wave; nominal sampling rate of 1000)



Observation / Conclusion

- There were timing issues with the SPI, but adding some delays fixed the issues
- The first 24 bits are status bits, and they are what they should be, so we can assume that we are reading the data correctly

Next steps

- Improve firmware (interrupts, etc.)
- Test the conversion with different configuration values

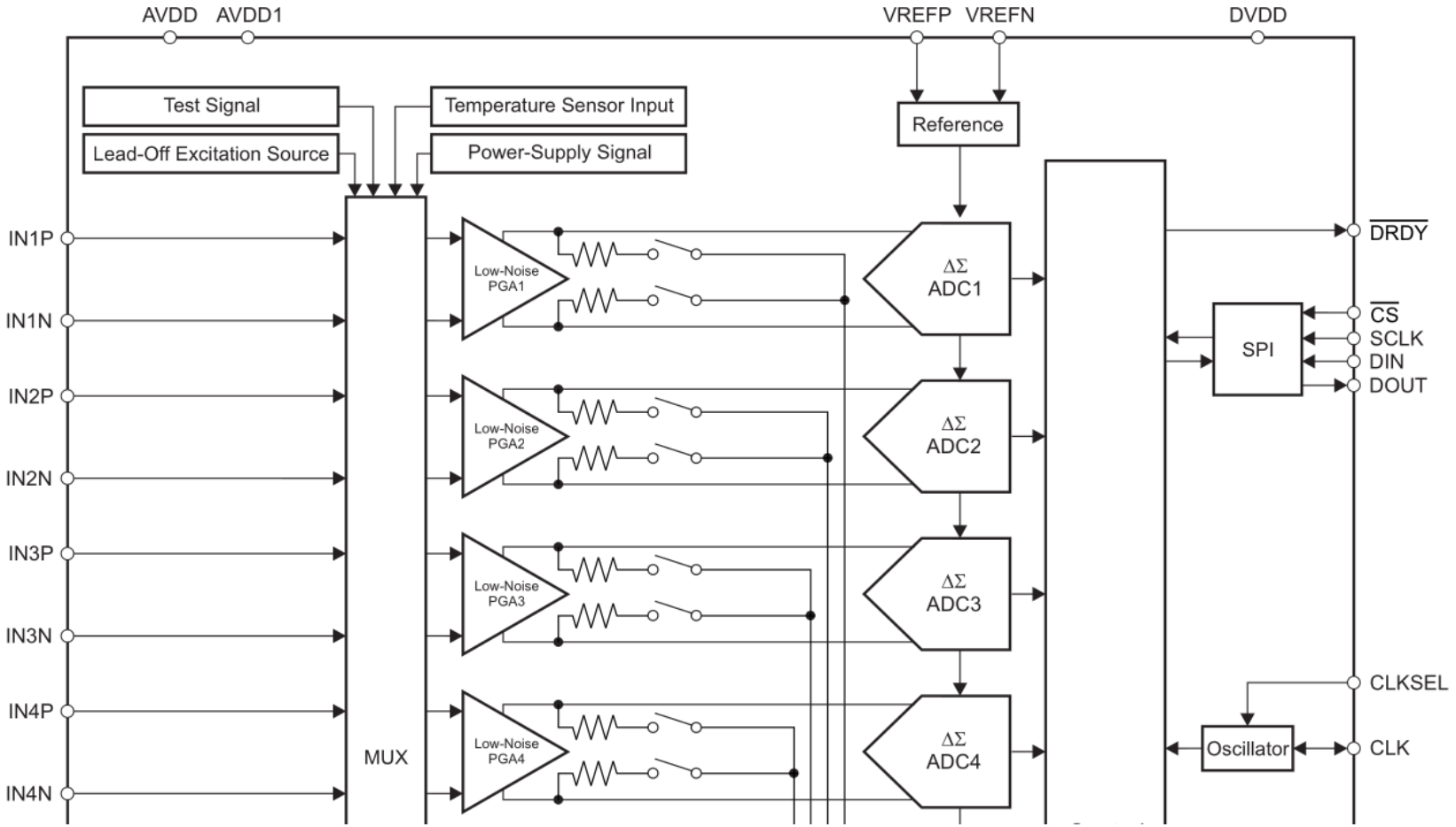
02/04/2026

Objective

- Determine if the chip can properly process any data
- Measure a wave provided by the function generator (e.g., 10 mV 100 Hz sine)

Methods

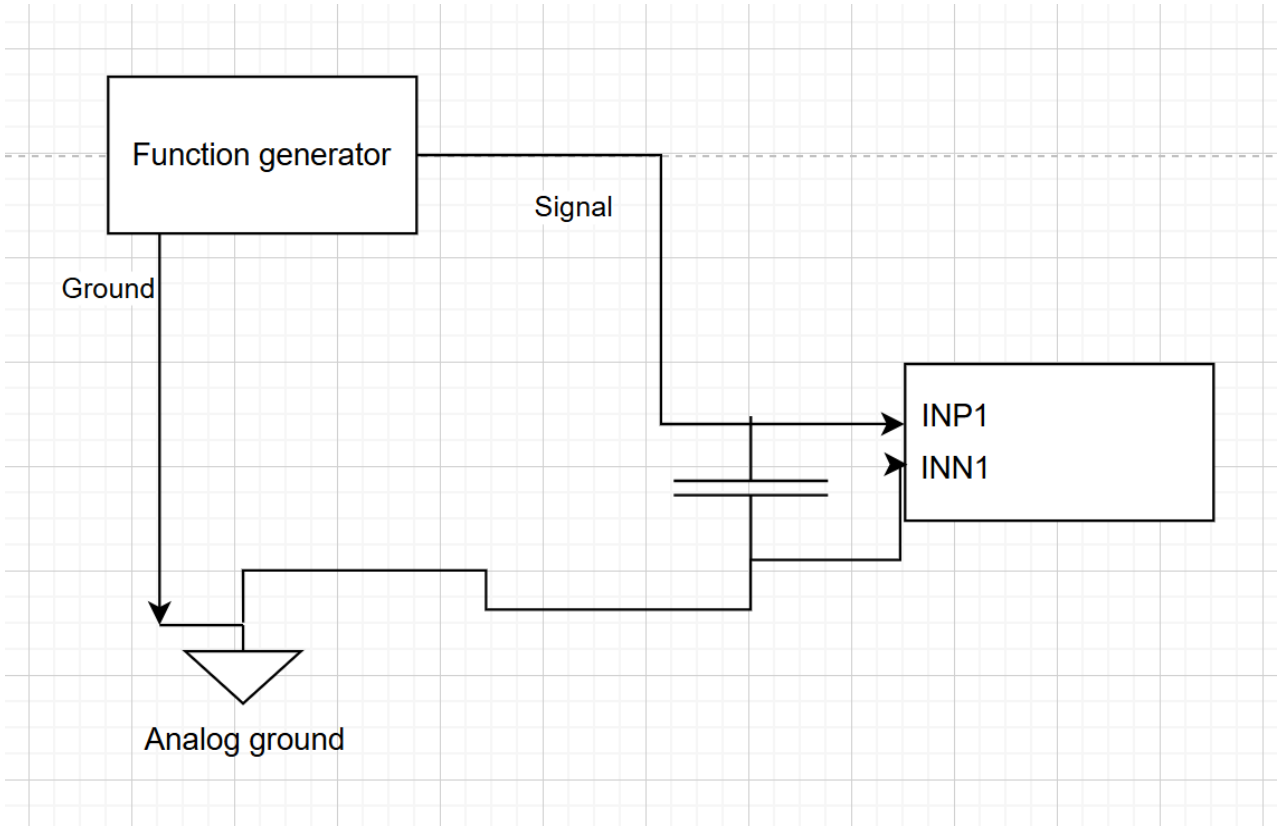
9.2 Functional Block Diagram



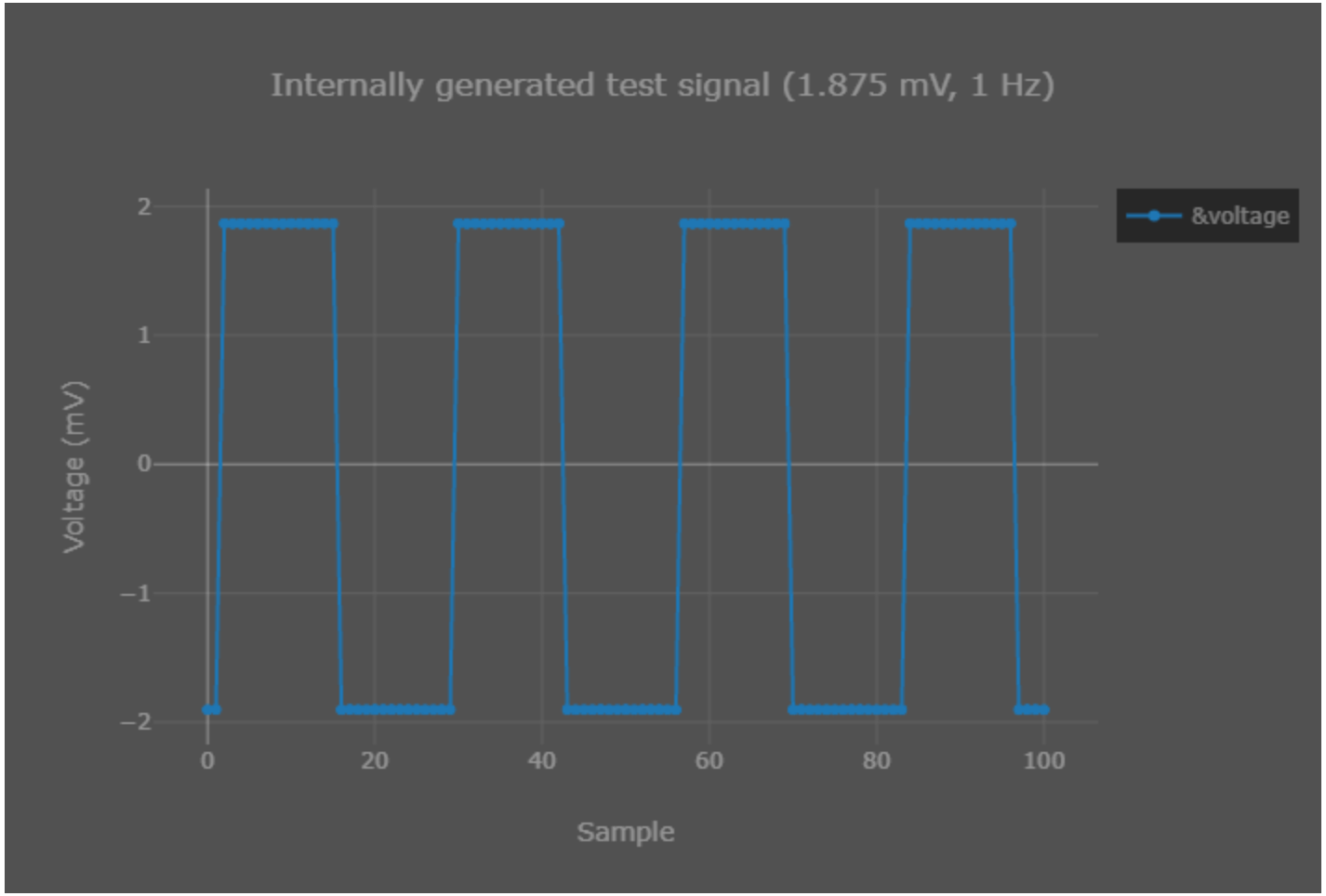
You can route the input to the channel from the normal external terminals to some other source like a test signal or a power-supply signal using a MUX

Methods


SIGLENT
SDS1204X-E
Digital Storage Oscilloscope
200 MHz
Dual 1 GSa/s
Quad 500 MSa/s
Intensity Adjust
Vertical
Math
Position
Digital
Coupling DC
BW Limit Full
Adjust Coarse
Probe 10X
Next Page Page 1/3
SBUS


Function generator
Signal
Ground
INP1
INN1
Analog ground

Results (Test signal measurement)

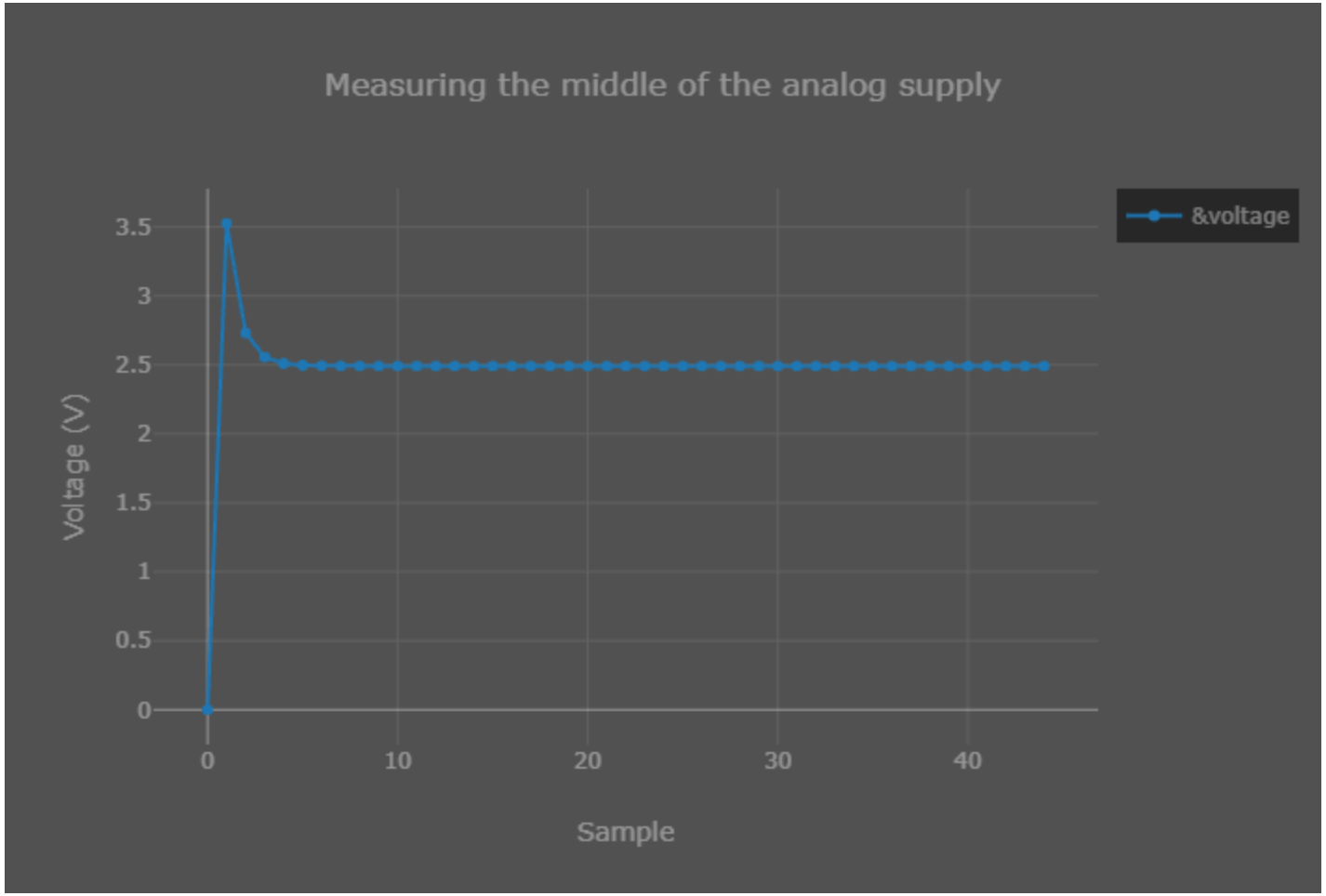


Top: 1.8487 mV

Bottom: -1.897537 mV

Measured Amplitude: 1.873 mV

Results (Supply measurement)



The input (based on the datasheet) is
(AVDD + AVSS) / 2
= (5 V + 0) / 2
= 2.5 V

Results (Sampling rate)

Frequency of the data ready signal (986.45 Hz) which is equivalent to the sampling rate (1 kHz)

Observation / Conclusion

- When using the normal input to the channel, the measurements are still essentially noise (even when I did a 1 Hz square wave)
- Since the test signal measurements are fine, the problem is probably to the left of the MUX, so the input terminals and/or function generator and/or grounding

Observation / Conclusion

9.3.1.2 Analog Input

The analog inputs to the device connect directly to an integrated low-noise, low-drift, high input impedance, programmable gain amplifier. The amplifier is located following the individual channel multiplexer.

The ADS1299-x analog inputs are fully differential. The differential input voltage ($V_{INxP} – V_{INxN}$) can span from $–V_{REF}$ / gain to V_{REF} / gain. See the *Data Format* section for an explanation of the correlation between the analog input and digital codes. There are two general methods of driving the ADS1299-x analog inputs: pseudo-differential or fully-differential, as shown in Figure 20, Figure 21, and Figure 22.

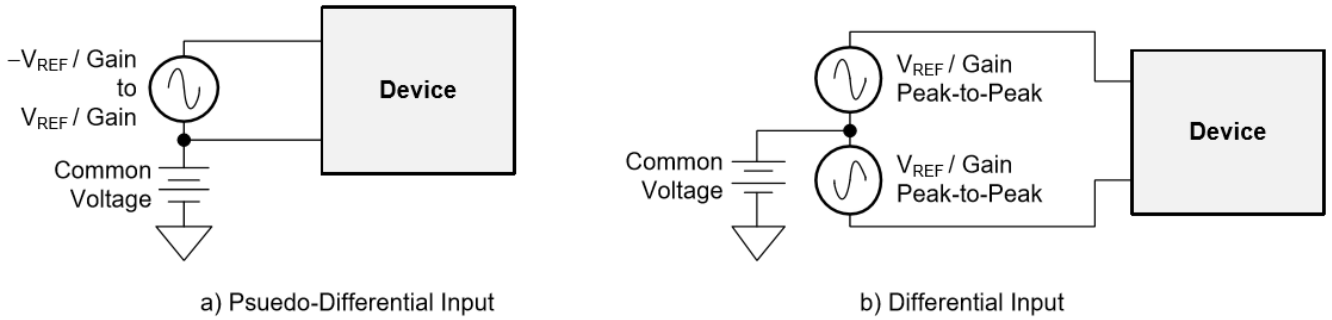


Figure 20. Methods of Driving the ADS1299-x: Pseudo-Differential or Fully Differential

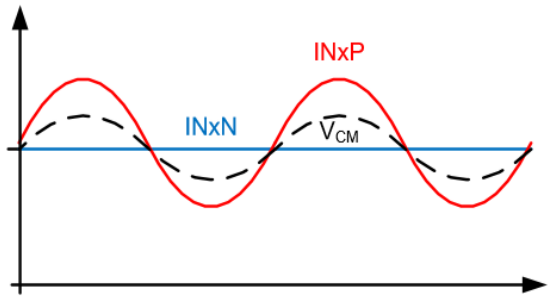


Figure 21. Pseudo-Differential Input Mode

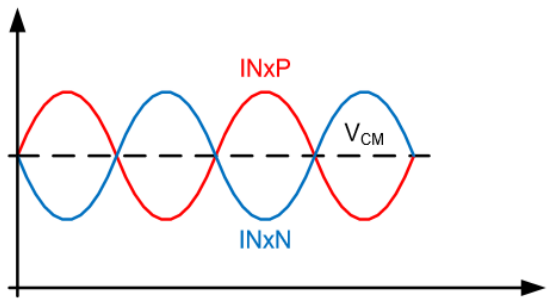


Figure 22. Fully-Differential Input Mode

The datasheet recommends the common voltage to be mid-supply (2.5 V in our case)

Vref / gain = 4.5 / 24 = 0.1875 V = 187.5 mV

Next steps

- Research the bias amplifier
- Test with a different channel

02/11/2026

Dmytro Stavskyi and Luis Wong

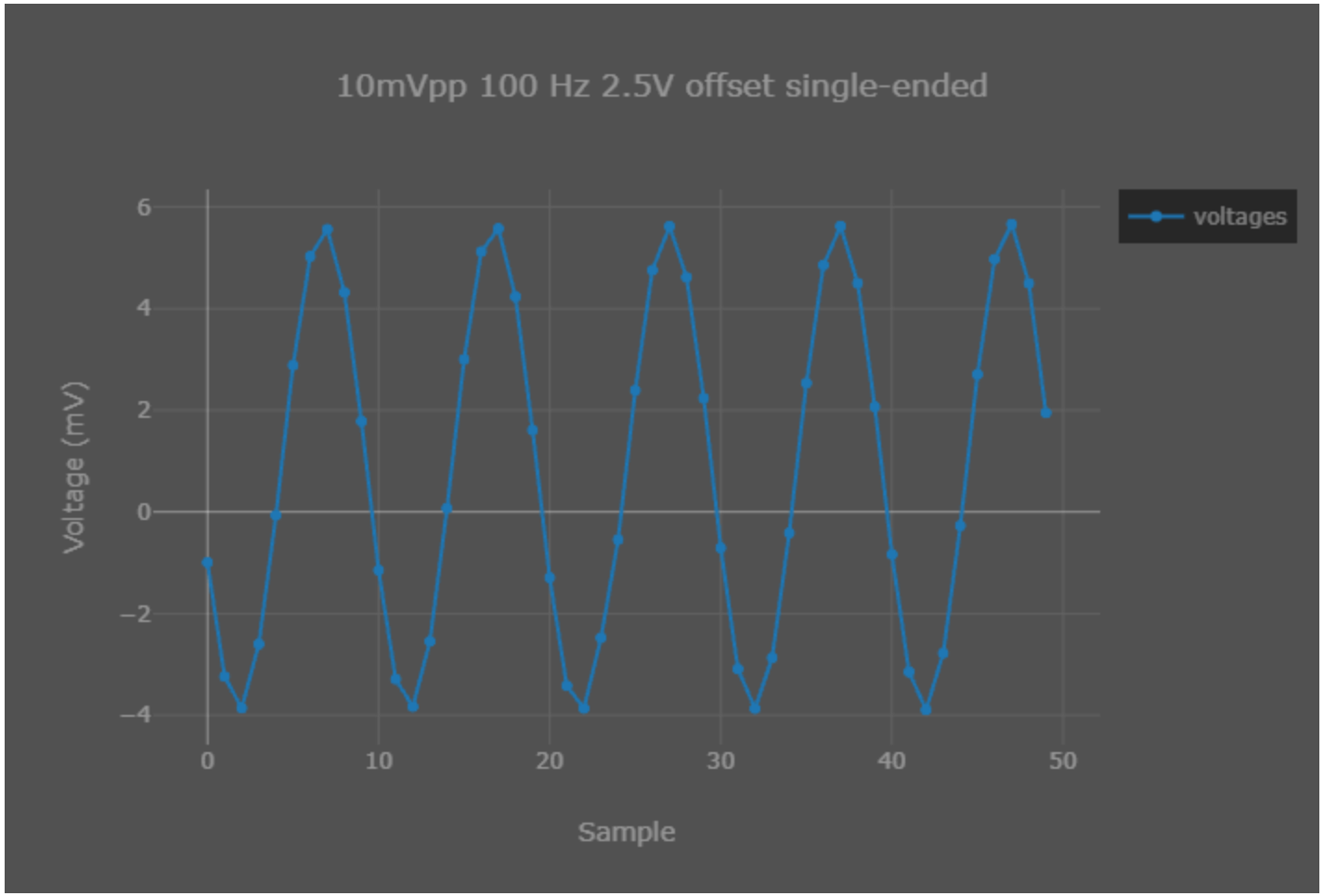
Objective

- Capture normal input data provided by the function generator

Methods

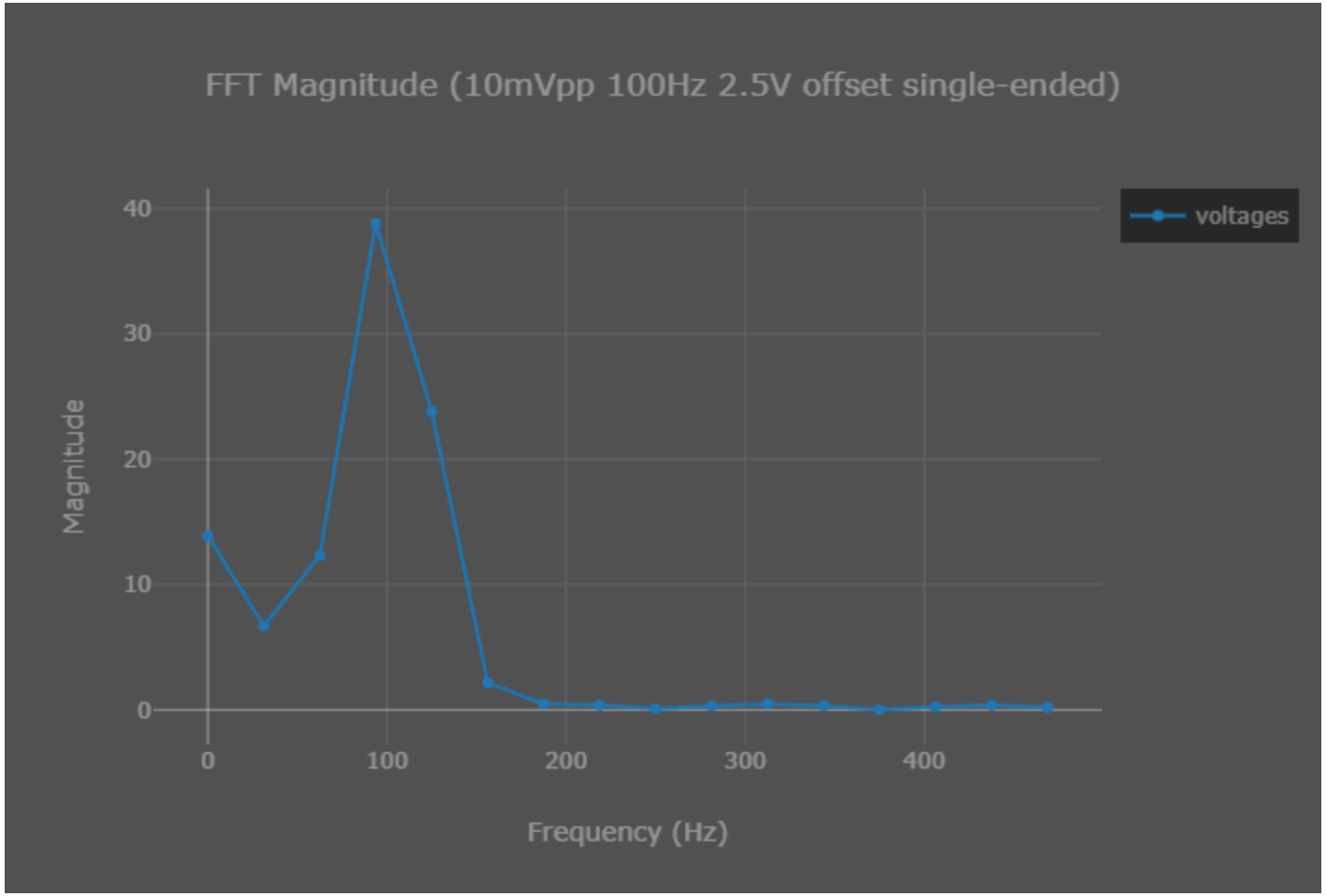
- Instead of continuously updating and plotting a variable, we save a number of points (e.g., 50) in an array and then plot those points
- Use FFT to calculate the frequencies of the captured signal
- For single-ended measurements, we applied the sine wave signal to the positive terminal (offset provided by the function generator) and either analog ground or some DC offset to the negative terminal

Results (5mV 100Hz 2.5 V offset single-ended)



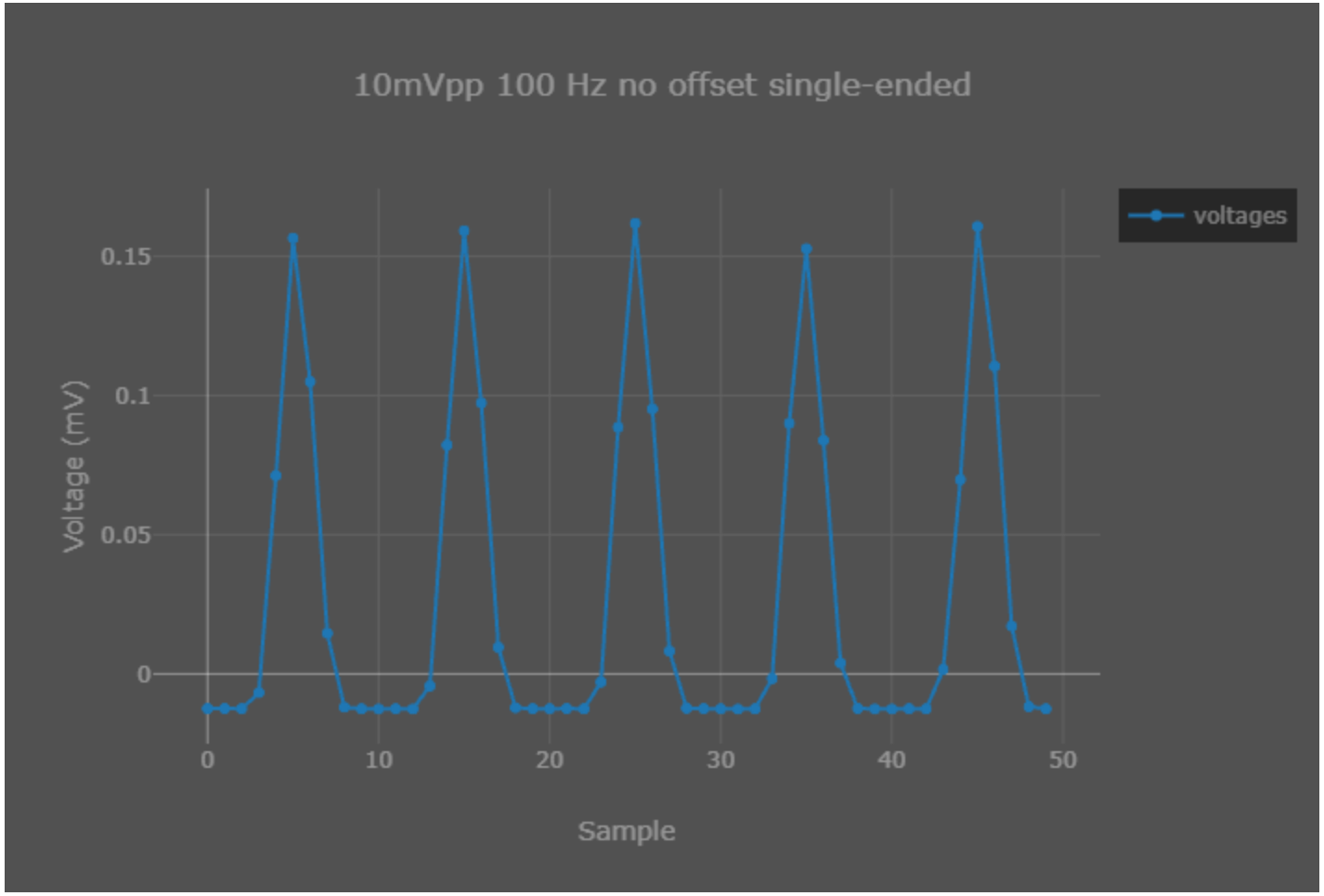
- Result was achieved by storing the data points in an array rather than a single variable.
- The graph is generated every time a breakpoint is reached, by default this is located at the location of the variable we are watching.
- According to the CCS documentation the graph runs through the debug probe and can only update every 0.1s under extremely ideal conditions.
- The [Sampling Rate] setting in CCS is arbitrary, and the true sampling rate is determined by the peripheral.

Results (5mV 100Hz 2.5 V offset single-ended FFT)



- Used an FFT to confirm that the frequency of the signal captured is 100Hz just like we generated.

Results (5mV 100 Hz no offset single-ended)



- No offset causes the device to essentially only capture noise.

The usable input common-mode range of the front-end depends on various parameters, including the maximum differential input signal, supply voltage, PGA gain, and the 200 mV for the amplifier headroom. This range is described in Equation 4:

$$AVDD - 0.2\ V - \left(\frac{Gain \times V_{MAX\_DIFF}}{2}\right) > CM > AVSS + 0.2\ V + \left(\frac{Gain \times V_{MAX\_DIFF}}{2}\right) \quad (4)$$

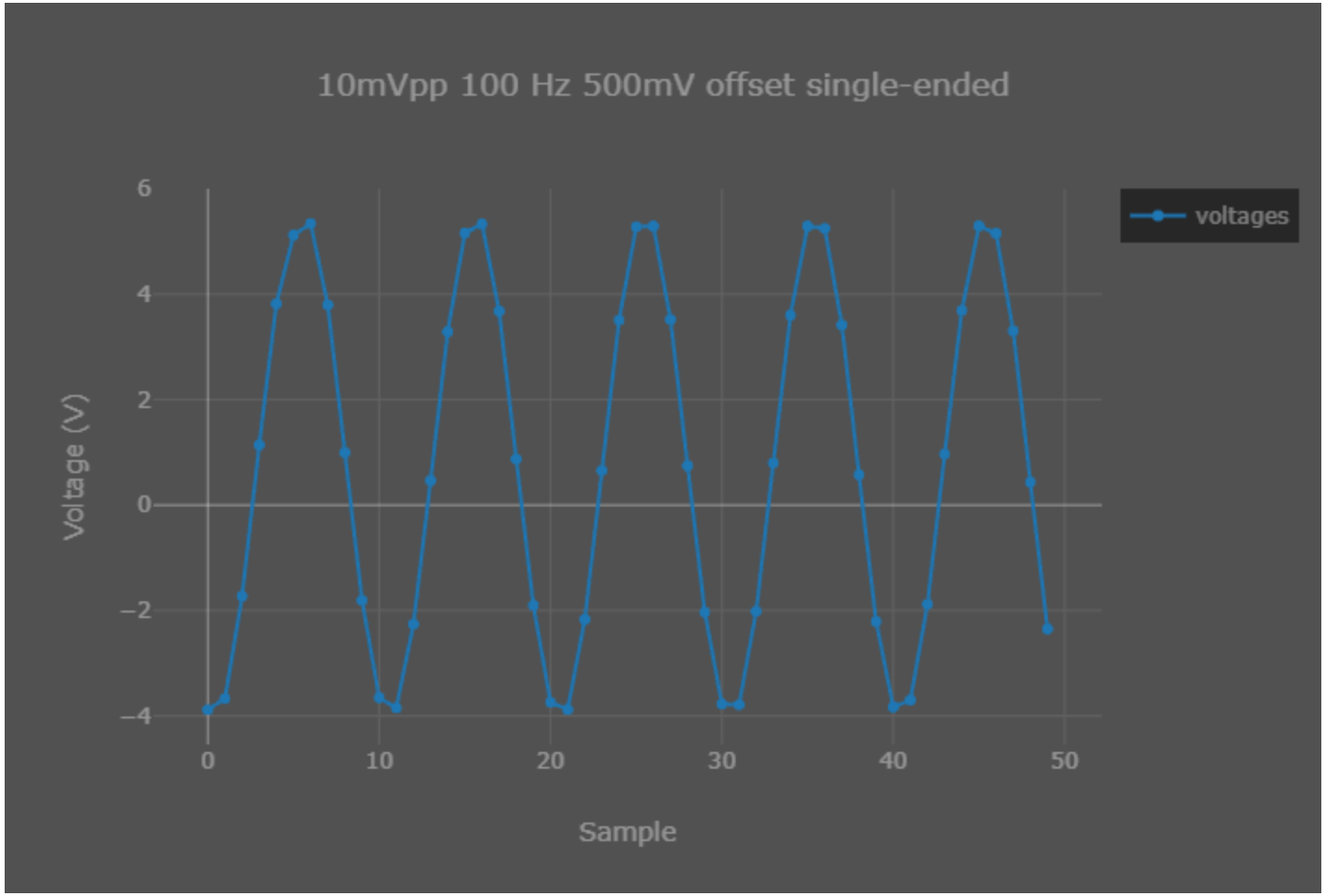
where:

V_{MAX_DIFF} = maximum differential signal at the PGA input

CM = common-mode range

- AVDD = 5V
- Gain = 24
- AVSS = 0V
- V_{max_diff} = maximum differential signal at the PGA input = 10mV
- 0.32V < CM < 4.68V

Results (5mV 100 Hz 500mV offset single-ended)



Observation / Conclusion

- We need to offset the signal internally which can be achieved through the bias circuit.
- We did not wire the bias circuit when creating this PCB, so it may not work properly

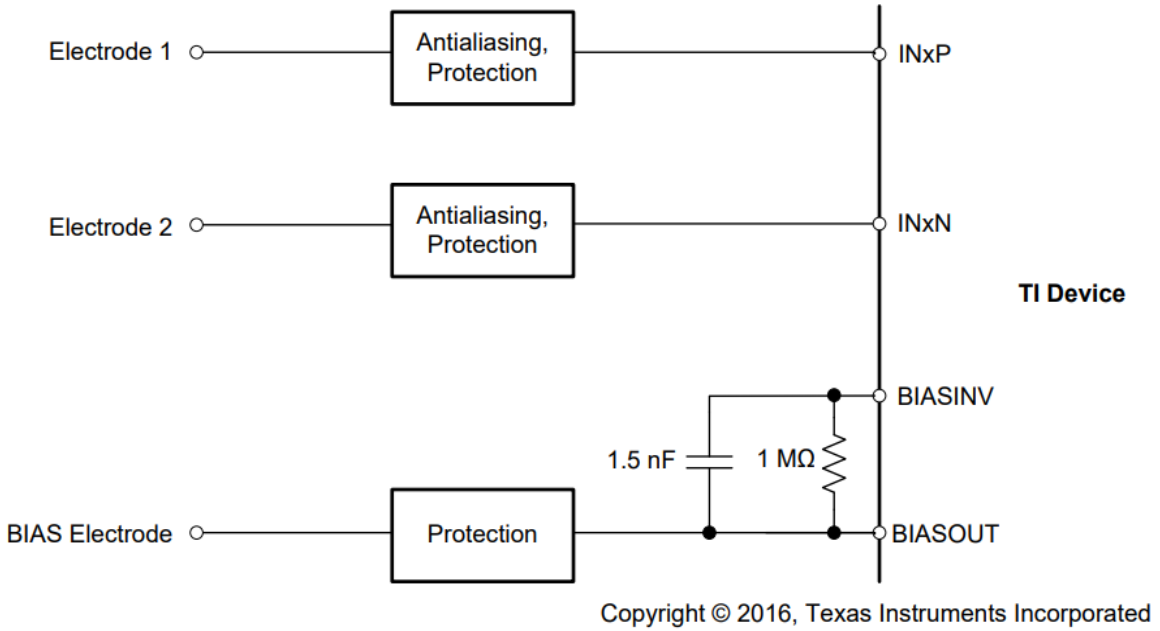
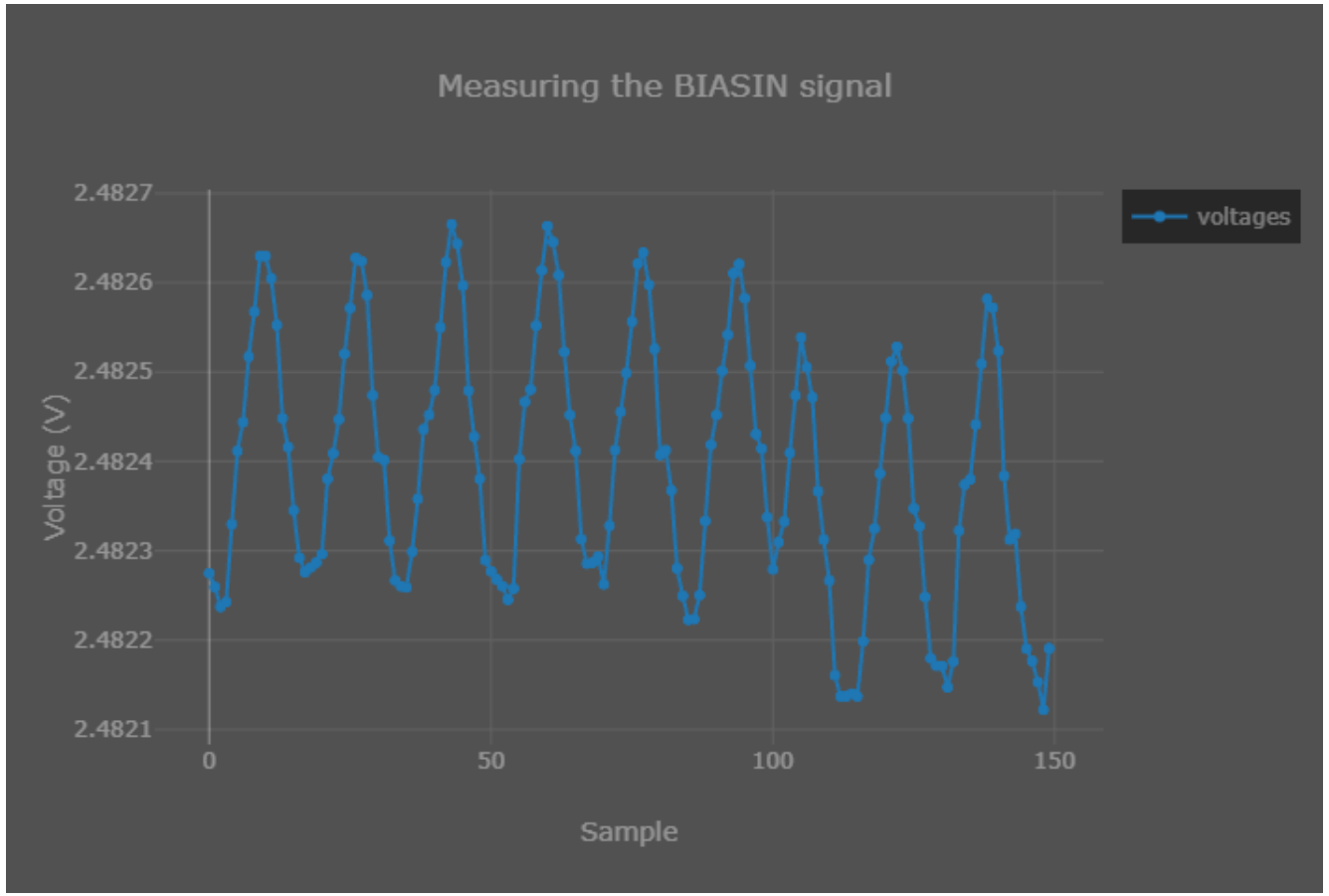
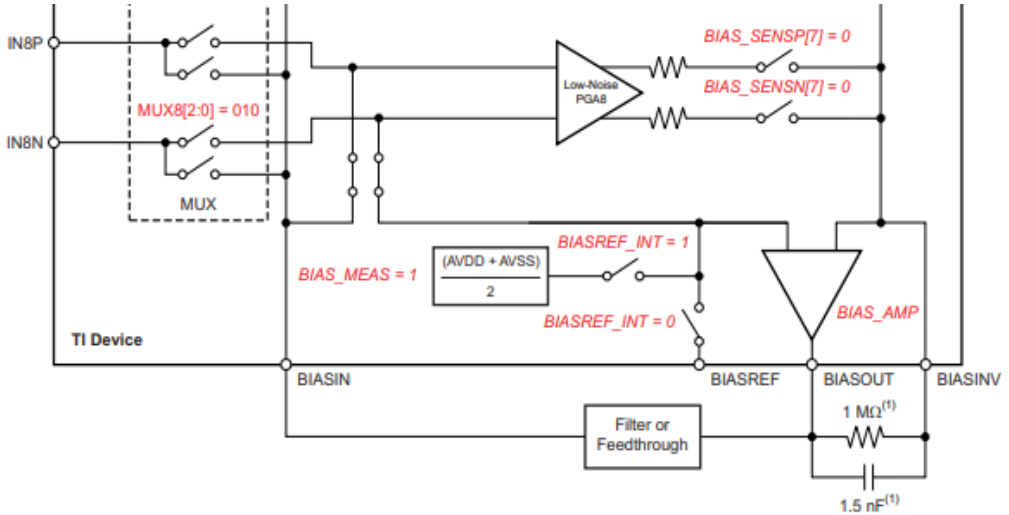


Figure 68. Setting Common-Mode Using BIAS Electrode

Observation / Conclusion



Right now, BIASIN and BIASOUT are shorted.
There is an option to route the BIASIN signal to a channel using MUX for measurement.



(1) Typical values for example only.

Figure 34. BIASOUT Signal Configured to be Read Back by Channel 8

Next steps

- Test the bias amplifier more
- Begin revising the schematic of the AFE circuit for the new design

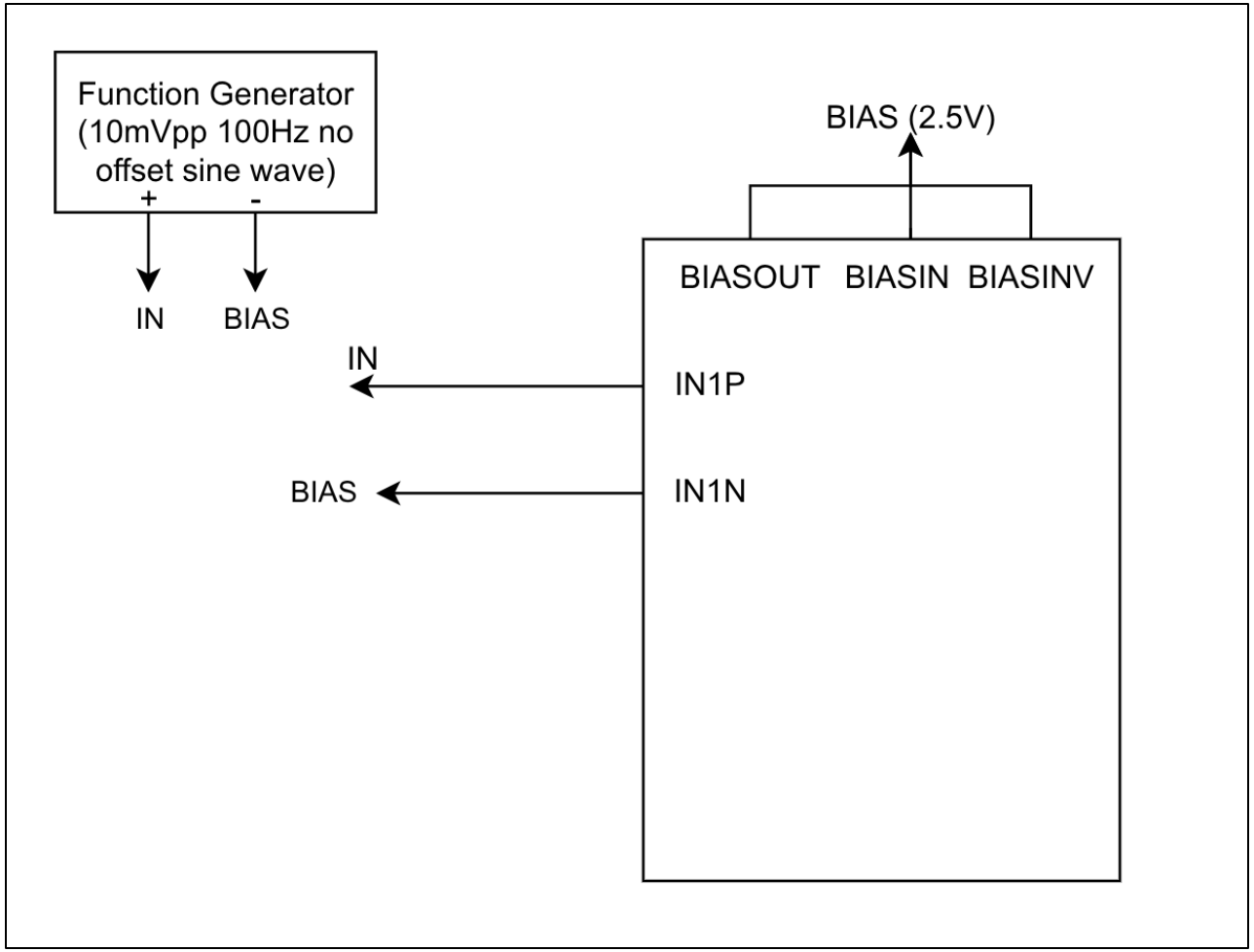
02/18/2026

Dmytro Stavskyi and Luis Wong

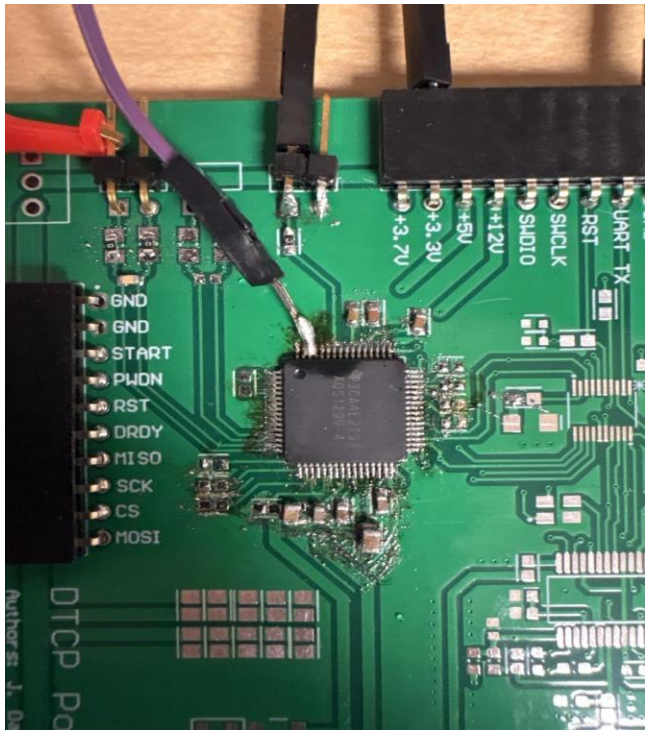
Objective

- Capture a normal input without providing offset in the function generator
  - Offset the external signal internally

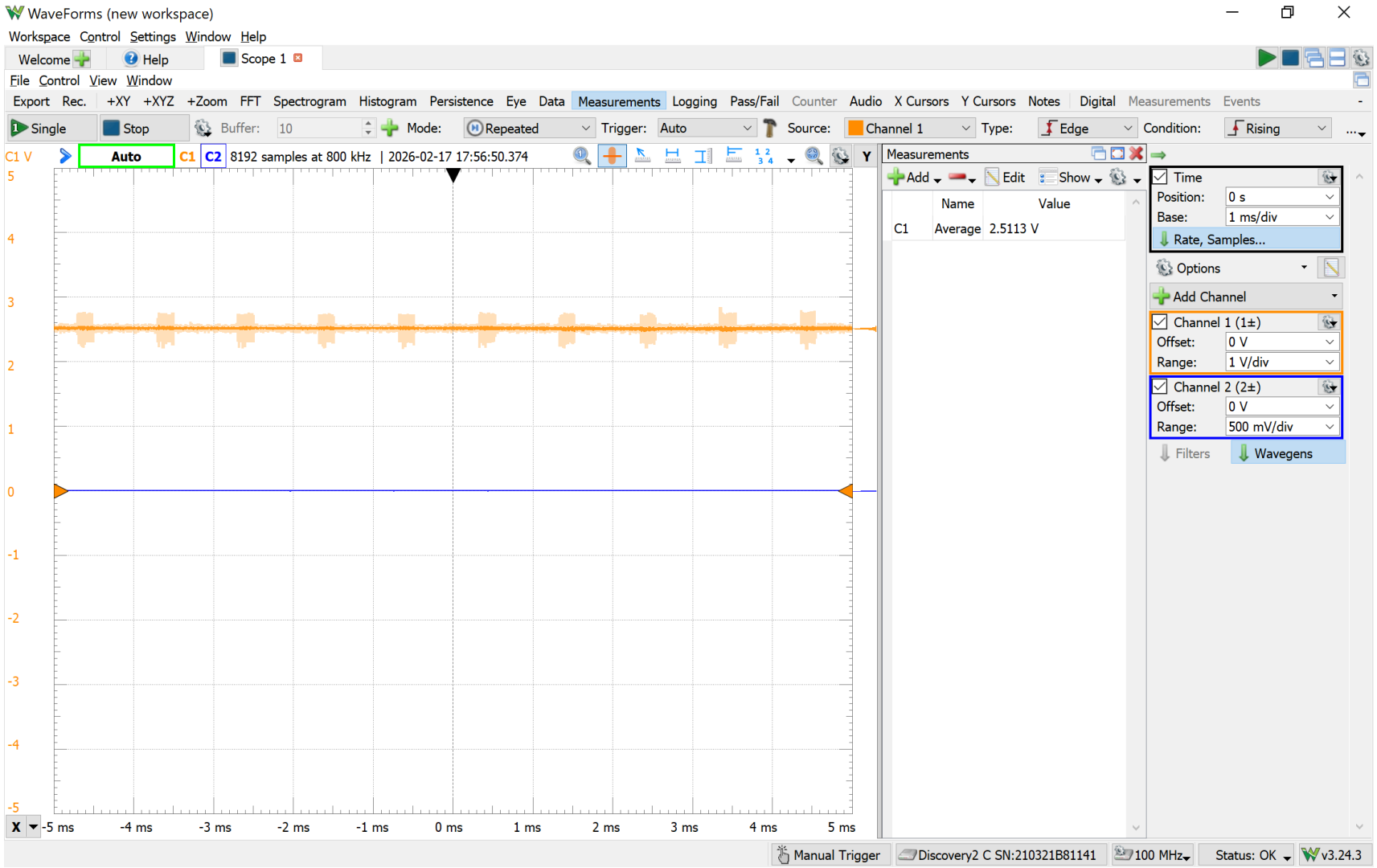
Methods



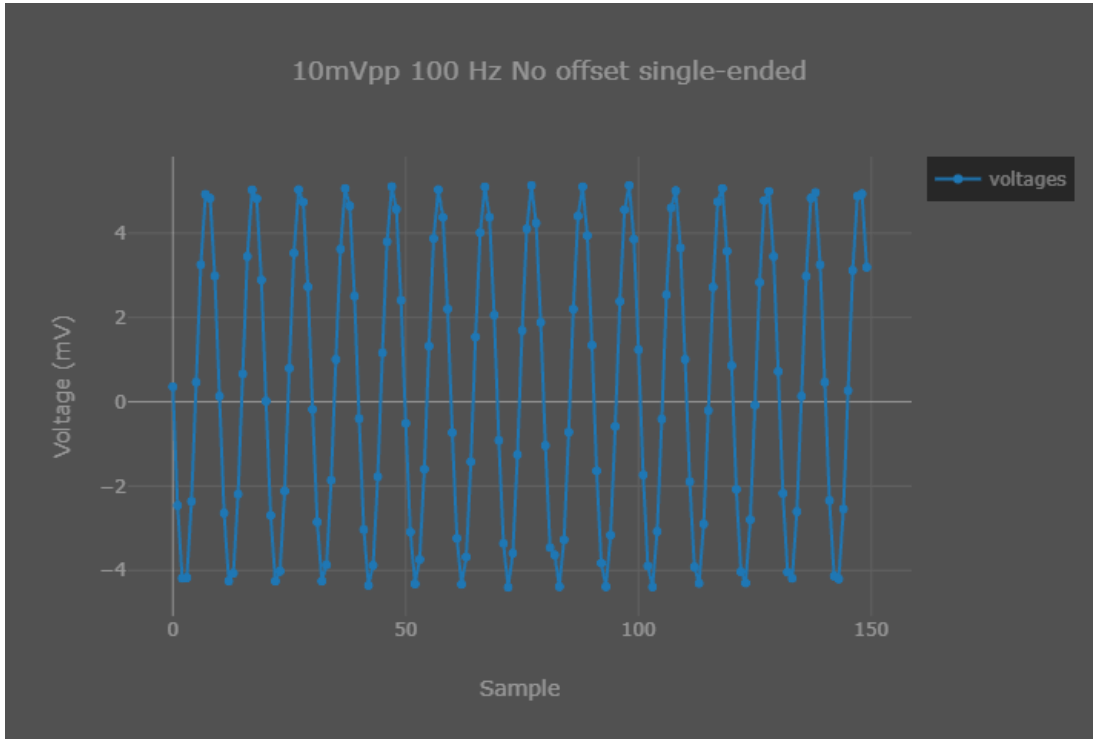
- BIASOUT, BIASIN, and BIASINV are shorted
- The bias is connected to the negative terminal of the channel and to the reference terminal of the input signal

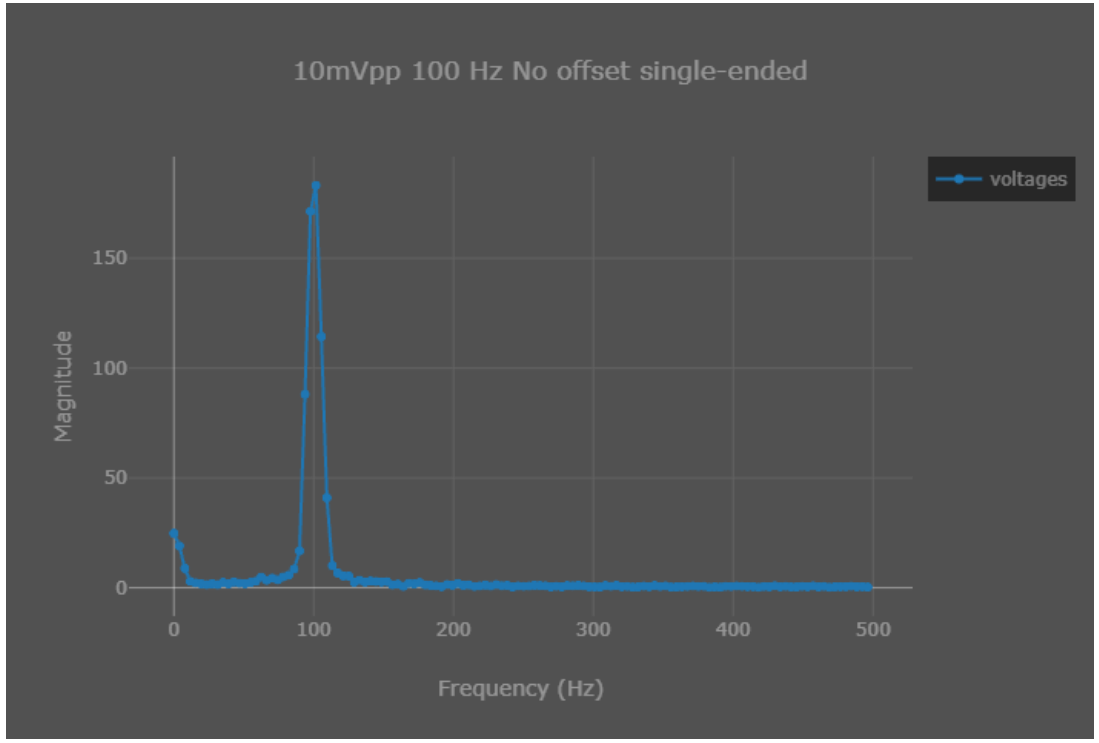


Results (Measurement of the bias)



Results (no offset in the function generator)





Observation / Conclusion

- SRB1 pin connected to bias

9.6.1.15 MISC1: Miscellaneous 1 Register (address = 15h) (reset = 00h)

This register provides the control to route the SRB1 pin to all inverting inputs of the four, six, or eight channels (ADS1299-4, ADS1299-6, or ADS1299).

Figure 64. MISC1: Miscellaneous 1 Register

7	6	5	4	3	2	1	0
0	0	SRB1	0	0	0	0	0
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 26. Miscellaneous 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7:6	Reserved	R/W	0h	**Reserved** Always write 0h
5	SRB1	R/W	0h	**Stimulus, reference, and bias 1** This bit connects the SRB1 to all 4, 6, or 8 channels inverting inputs 0 : Switches open 1 : Switches closed
4:0	Reserved	R/W	0h	**Reserved** Always write 0h

Next steps

- Improve the code
- Research the biasing more
- Continue to revise the AFE schematic for the next design

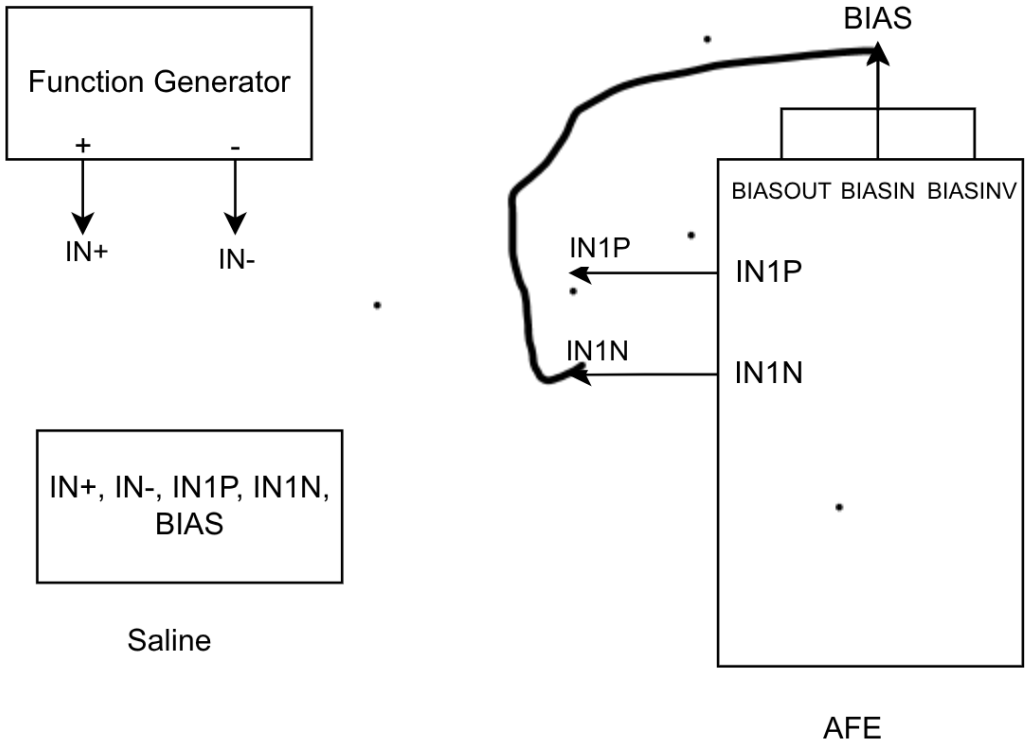
02/25/2026

Dmytro Stavskyi

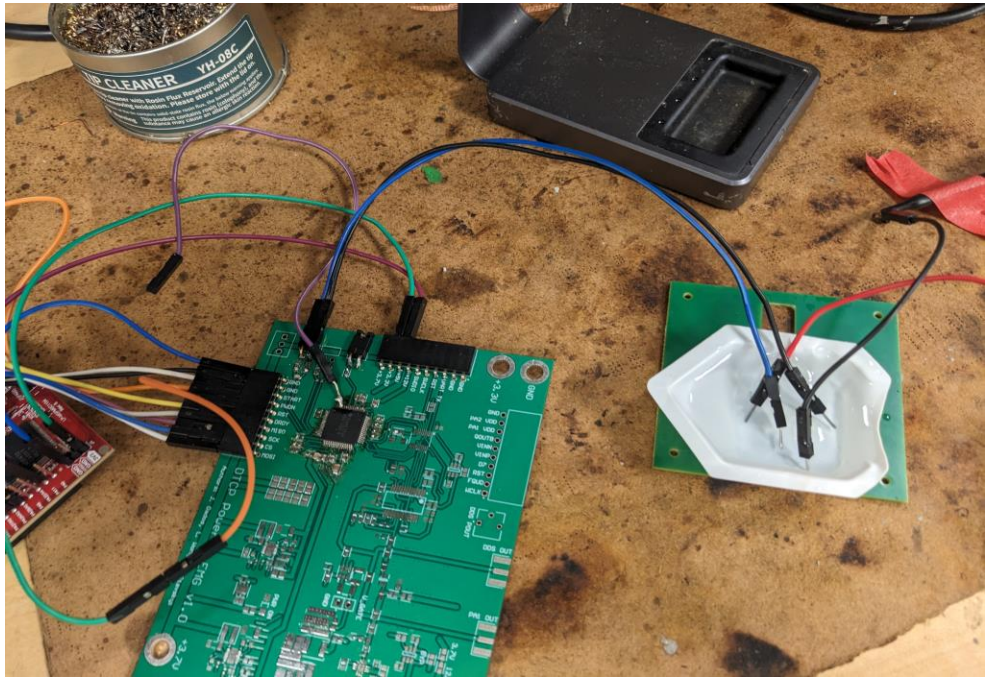
Objective

- Test the setup and the AFE measurements with saline

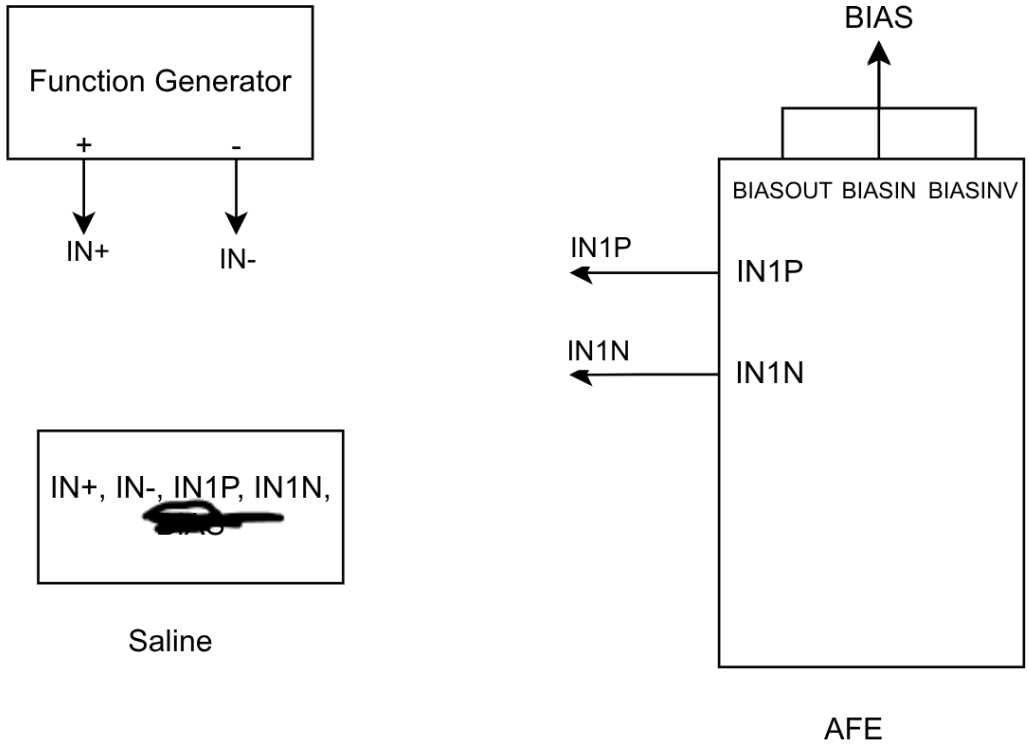
Methods



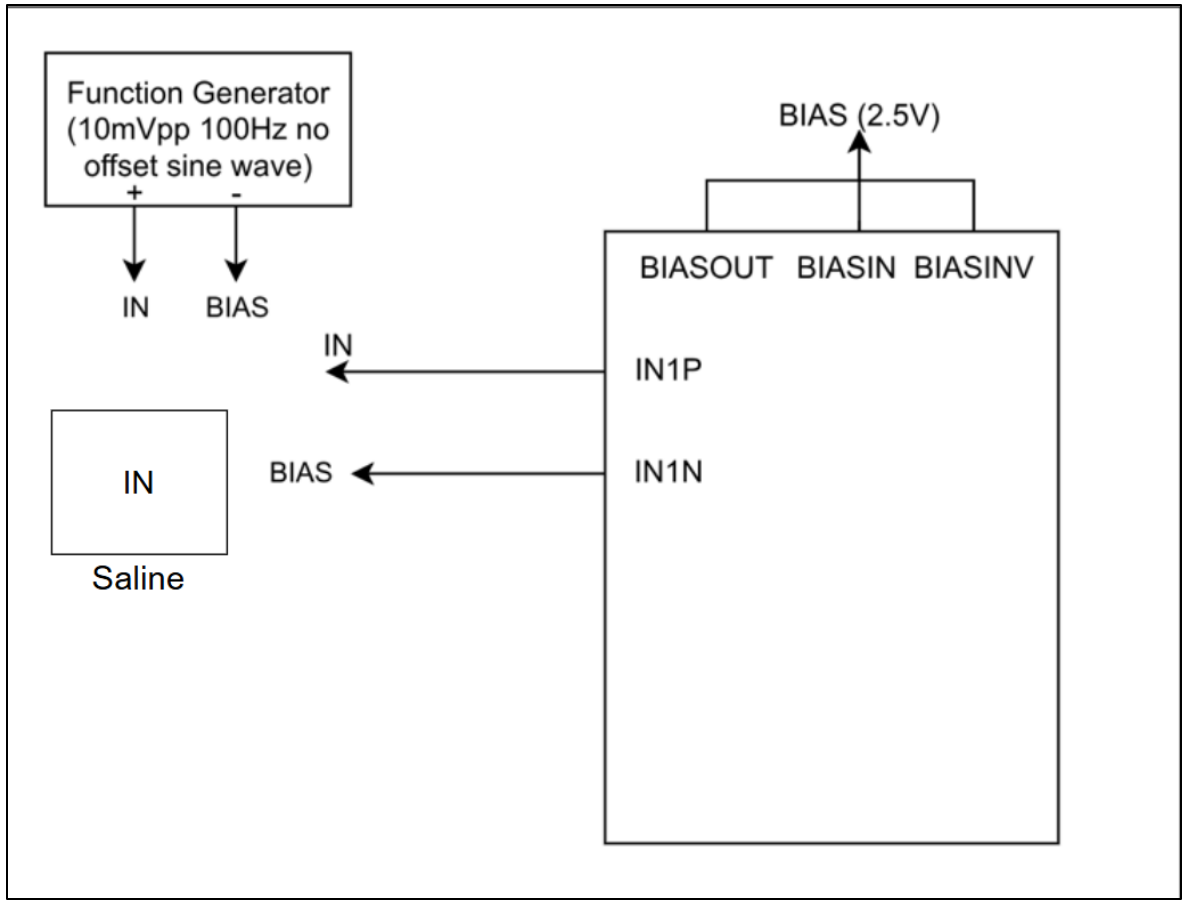
- Power is provided from the LaunchPad board, not the separate power supply

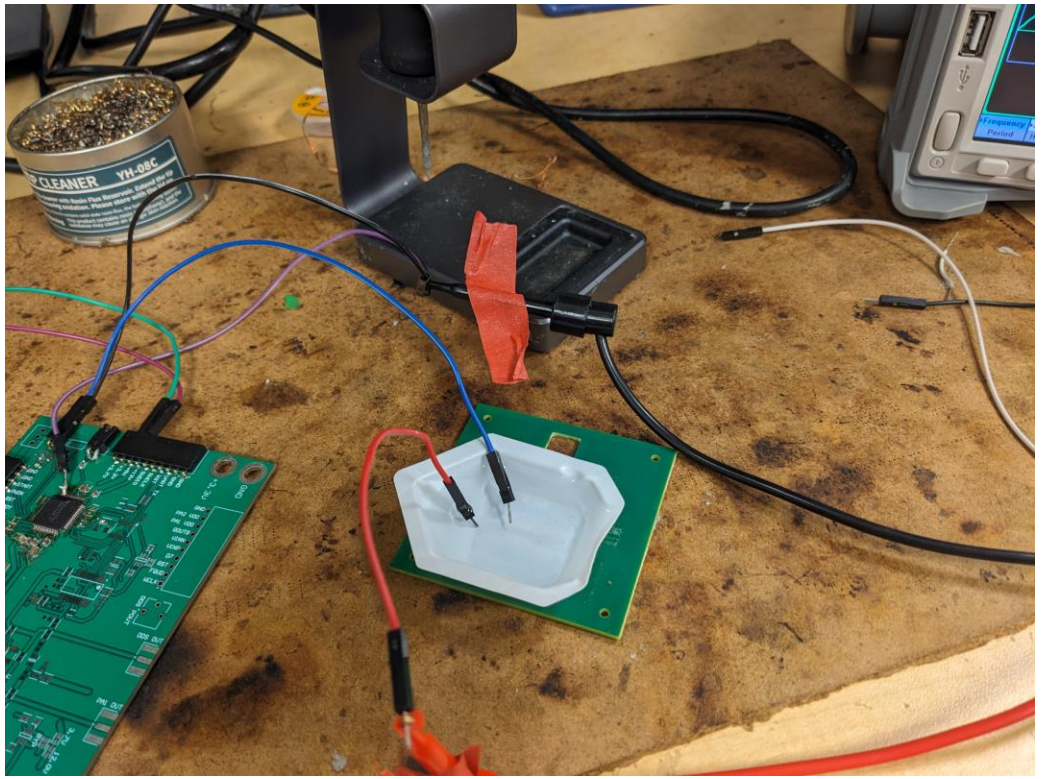


Methods (First Setup, No Bias)

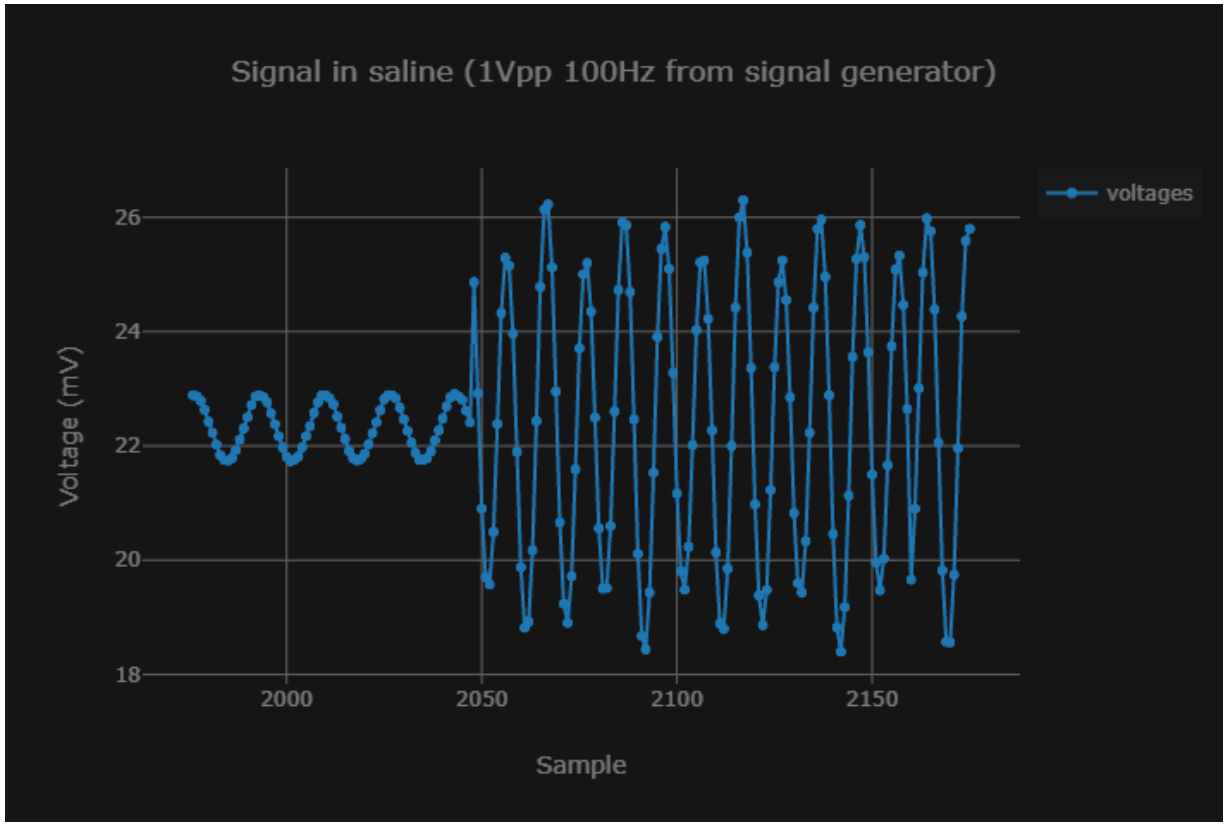


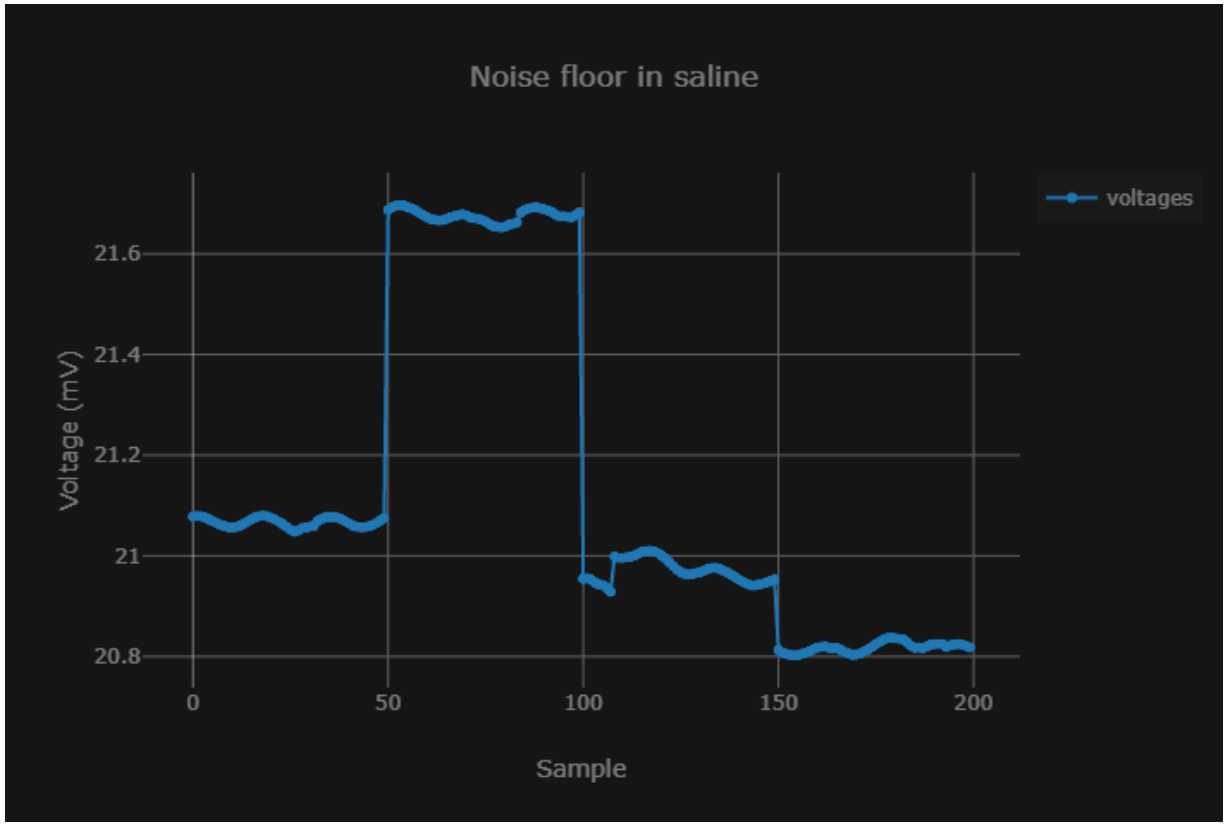
Methods


Function Generator
(10mVpp 100Hz no
offset sine wave)
+
-
IN
BIAS
BIAS (2.5V)
BIASOUT
BIASIN
BIASINV
IN
IN1P
IN
BIAS
IN1N
Saline

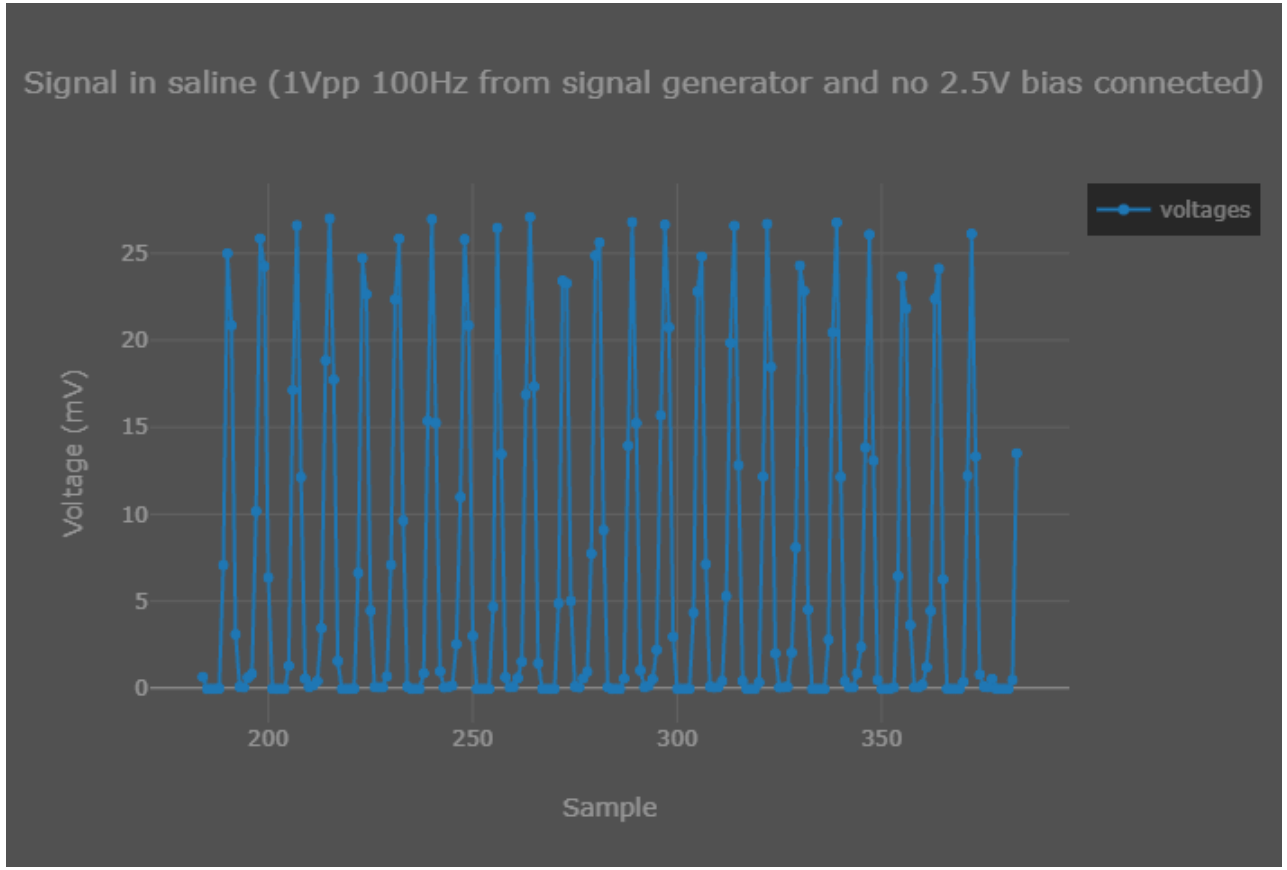


Results (First Setup)

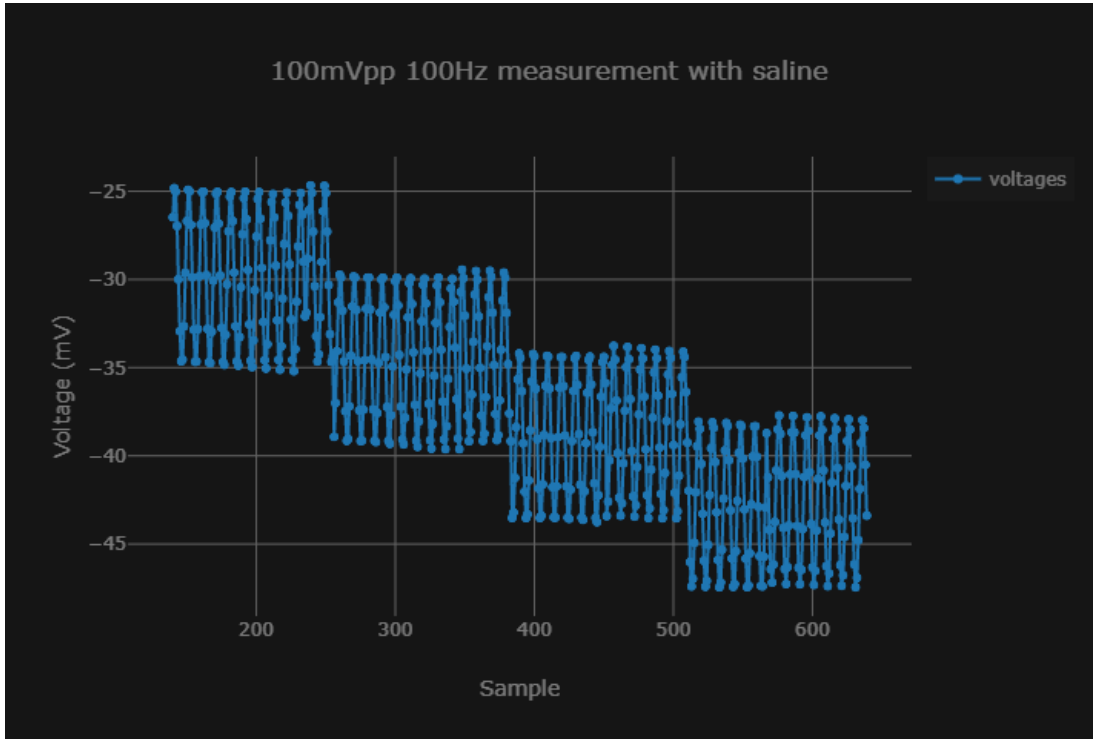


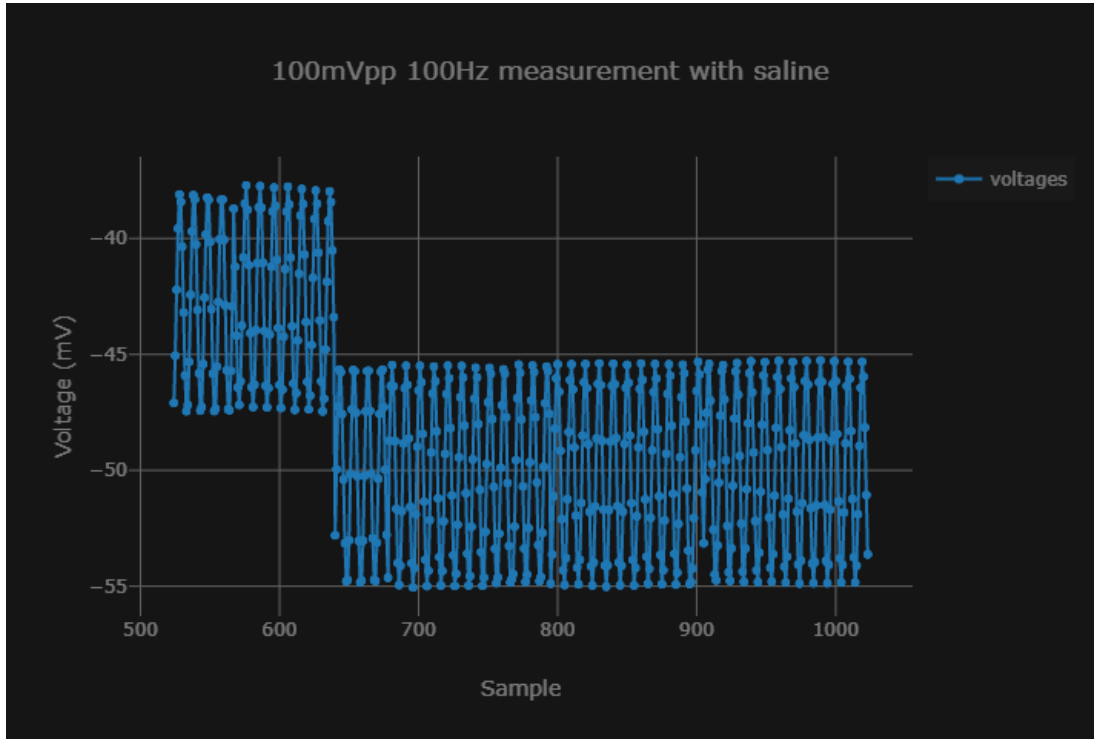


Results (First Setup, No BIAS)



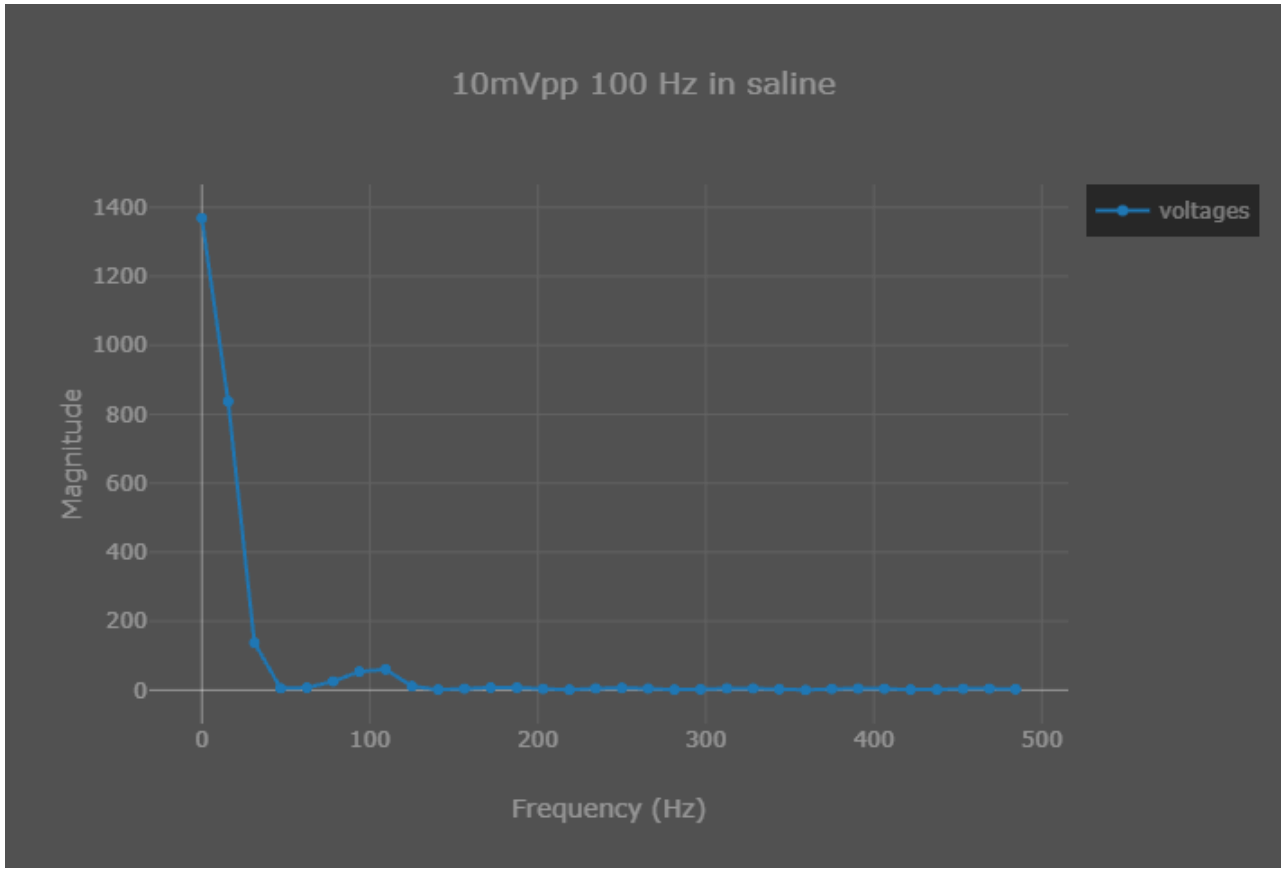
Results (Second Setup; 10mVpp 100Hz)



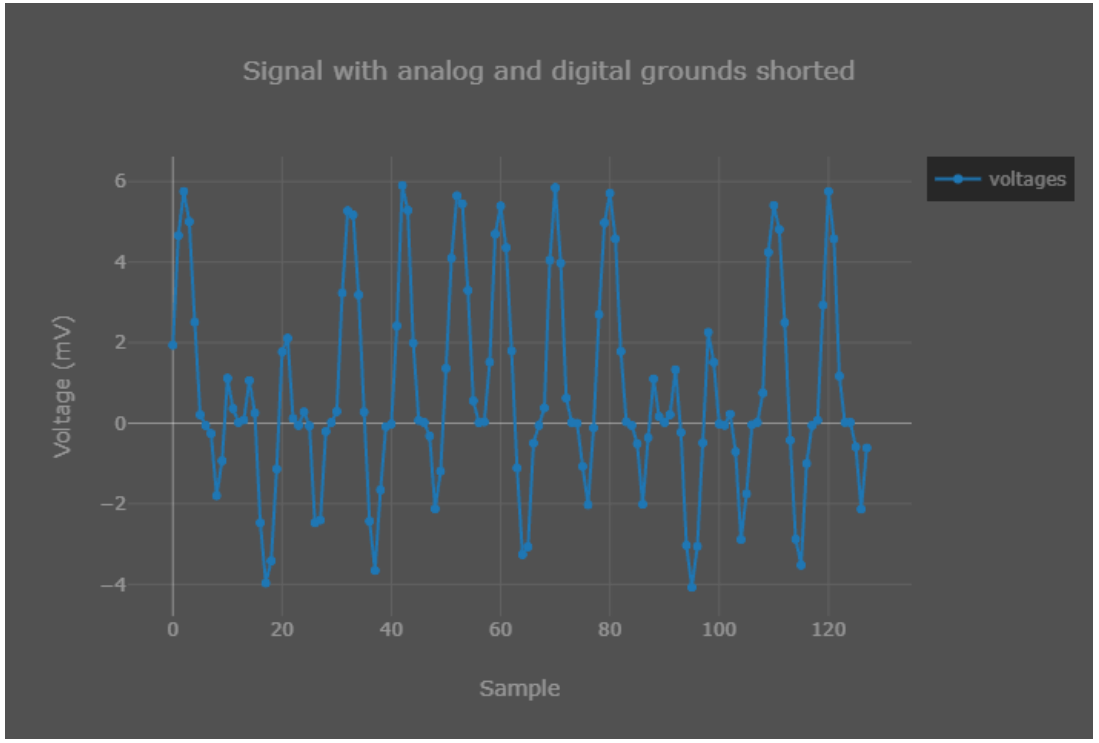


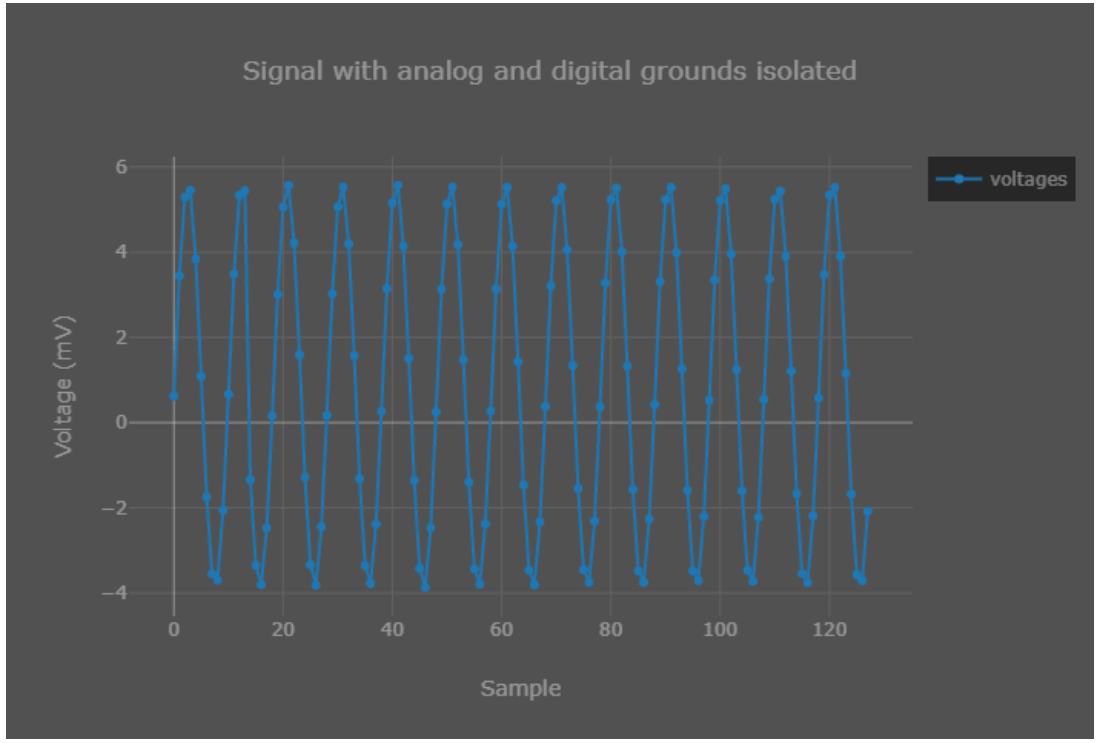
- Measured amplitude 4.794 mV

Observation / Conclusion (FFT; DC component)



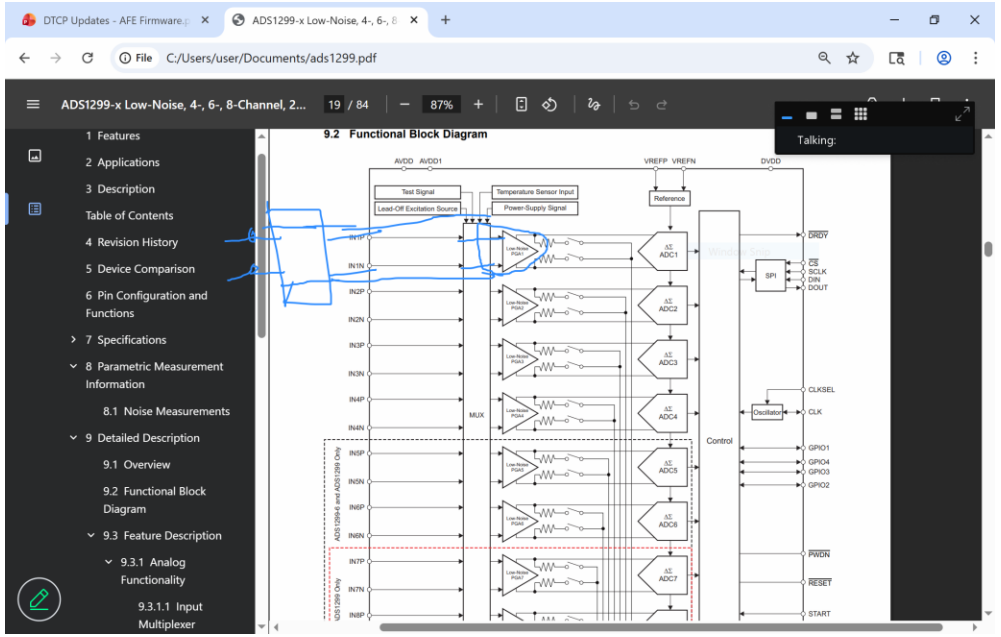
Observation / Conclusion (DGND and AGND)





Next steps

- Review the PCB schematics and note the i
  pin
- Improve the code
- Test the setup more
  - Repeat the tests (with detailed setups: drawings, saline connections, wire connections)
  - Internal power amplifier drawing
  - Filter for the input
  - System diagram
  - Four setups (methodology and results for each setup, and video): everything in saline, no bias in saline but shorted to the negative terminal, no bias, bias in saline but not shorted to the negative terminal



9.3.2.4.5 section of datasheet

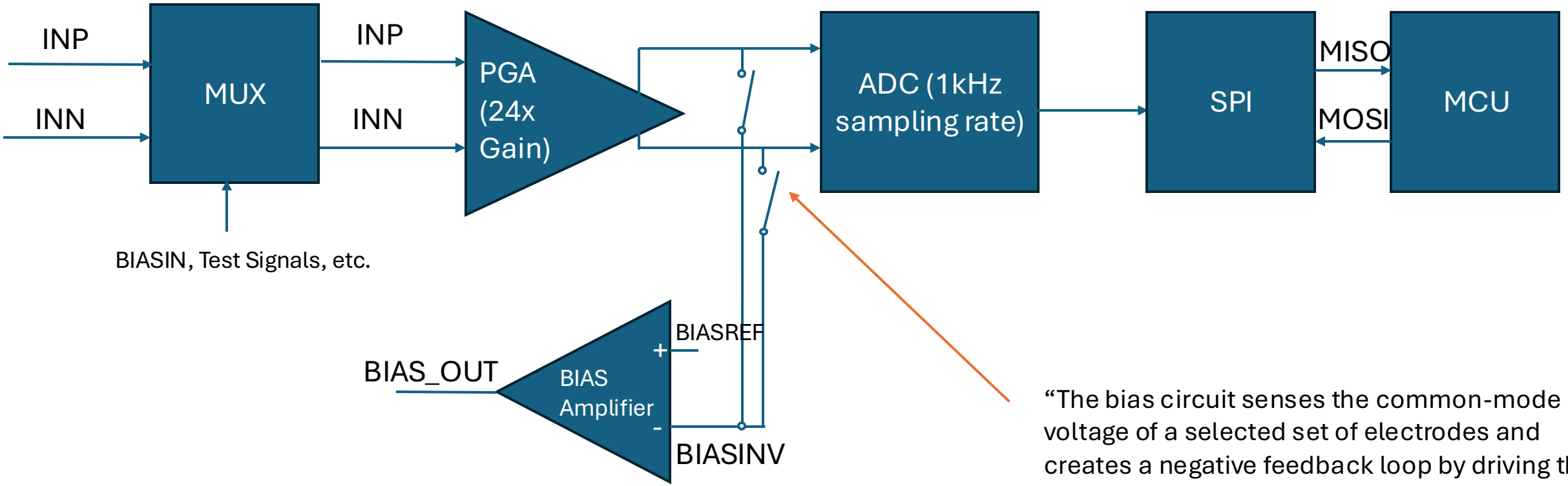
3/4/2026

Dmytro Stavskyi

Objective

- Test the setup and the AFE measurements with saline more

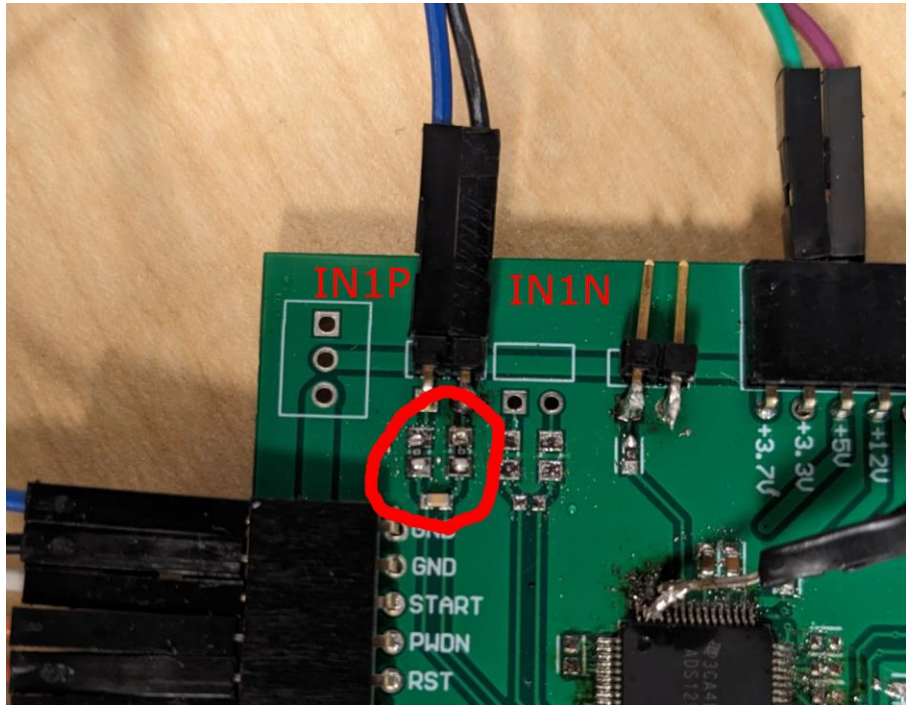
Methods (AFE System Diagram)

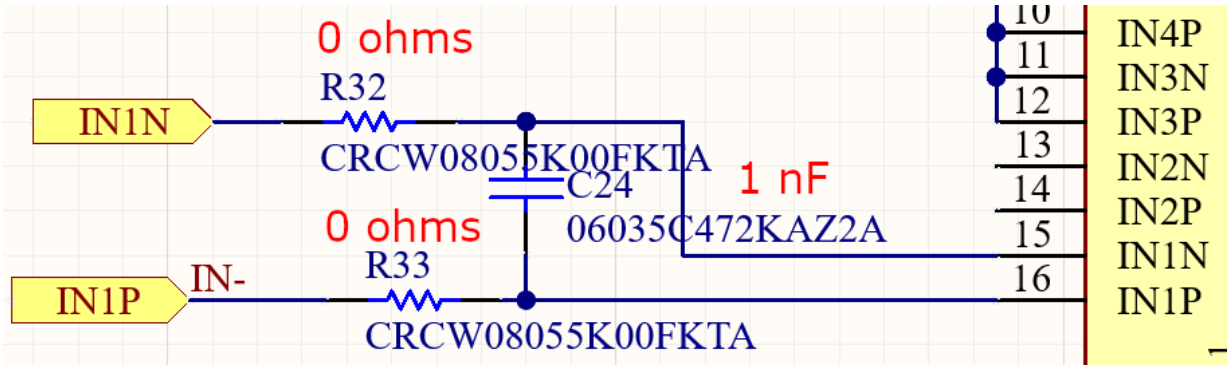


"The bias circuit senses the common-mode voltage of a selected set of electrodes and creates a negative feedback loop by driving the body with an inverted common-mode signal."

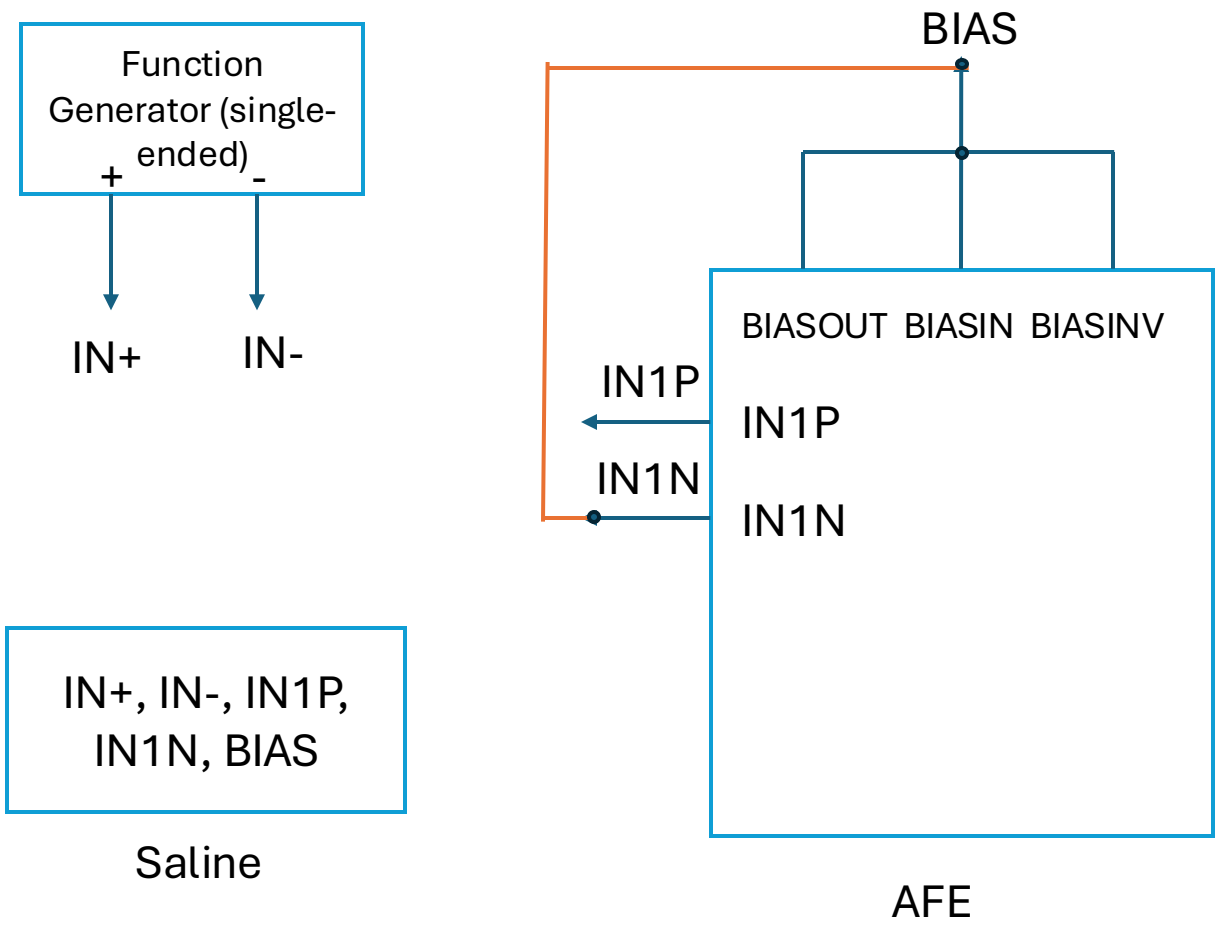
BIASREF = (AVDD + AVSS) / 2
If AVDD = 5V and AVSS = 0V, BIASREF = 2.5V

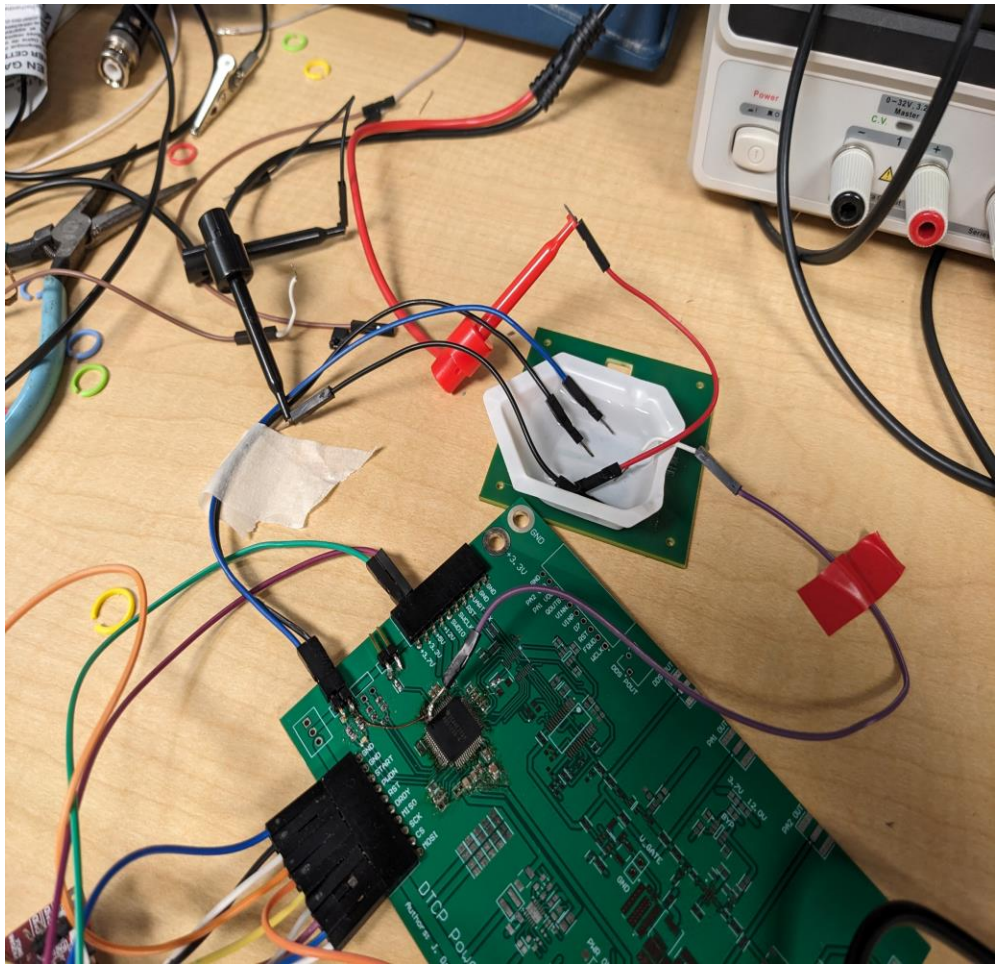
Methods (Filter on the board right now)



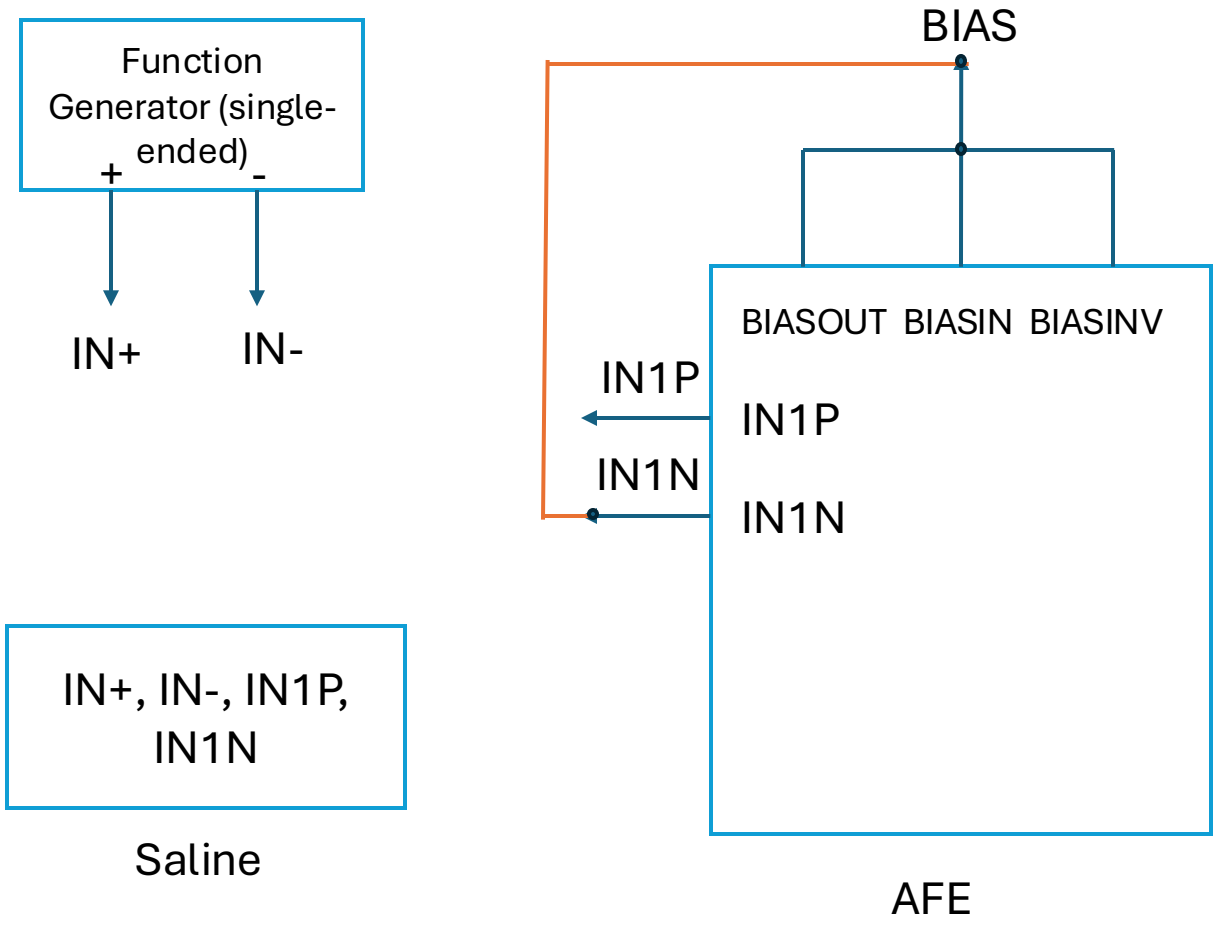


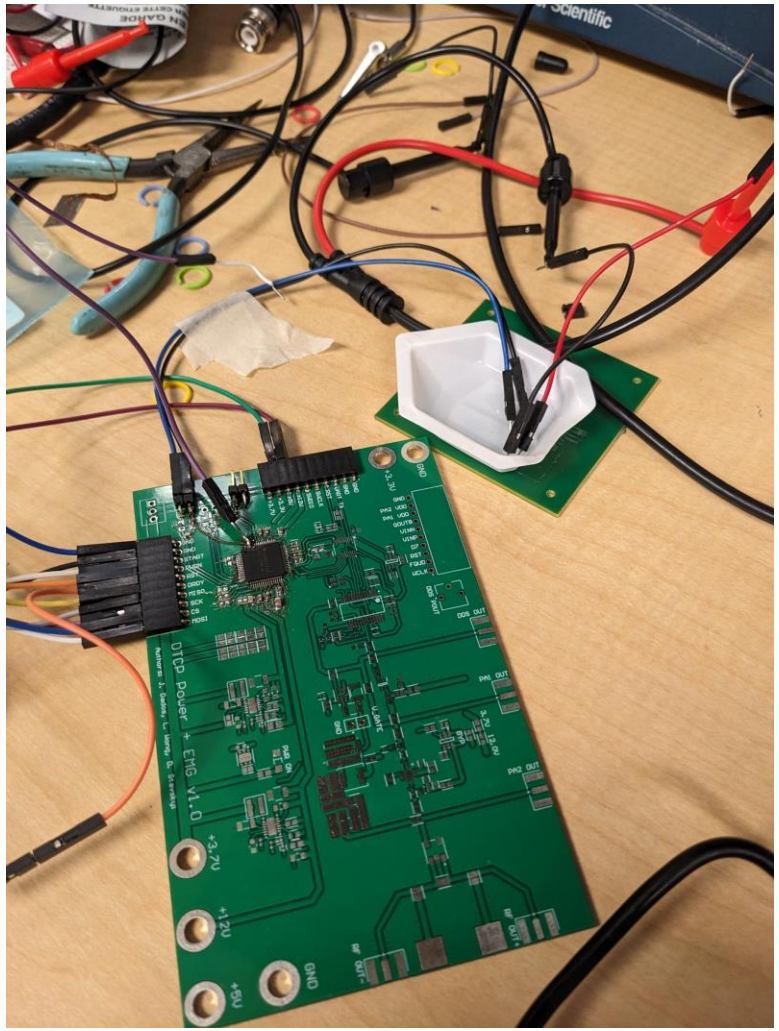
Methods (First Setup; everything in saline)



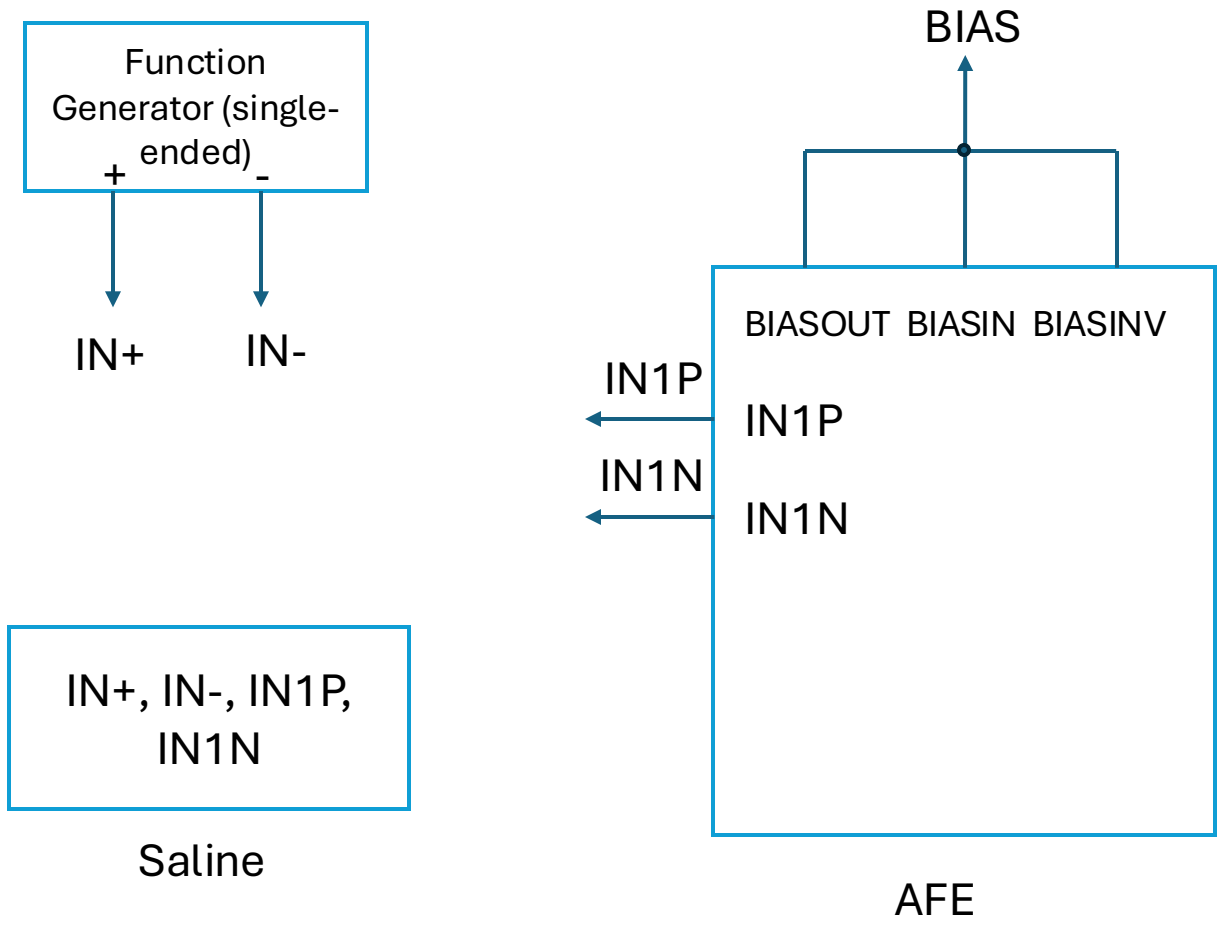


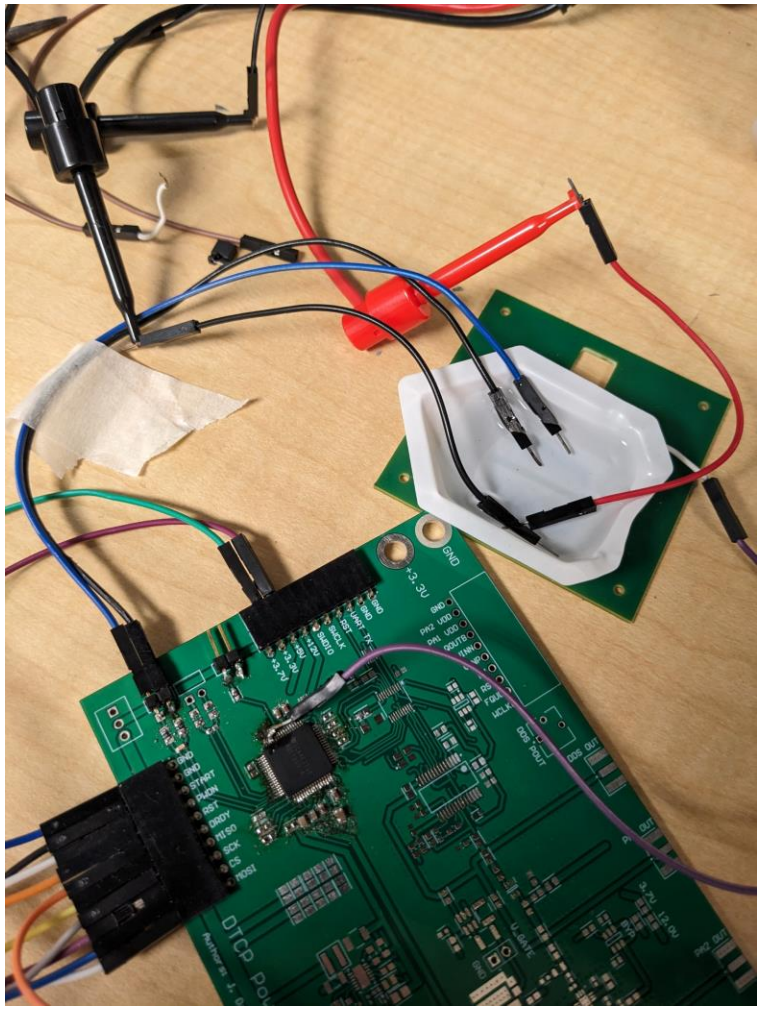
Methods (Second Setup; no bias in saline but shorted to the negative terminal)



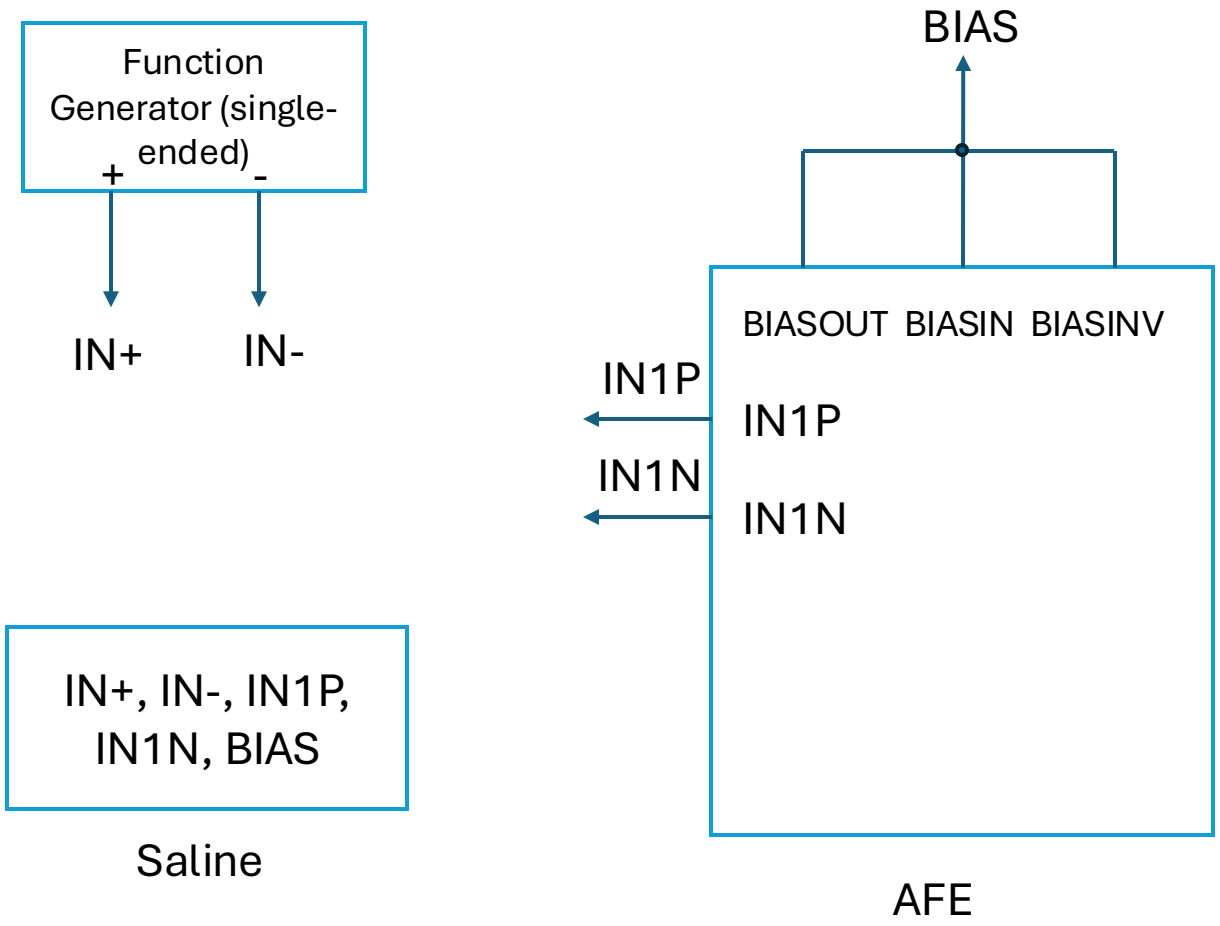


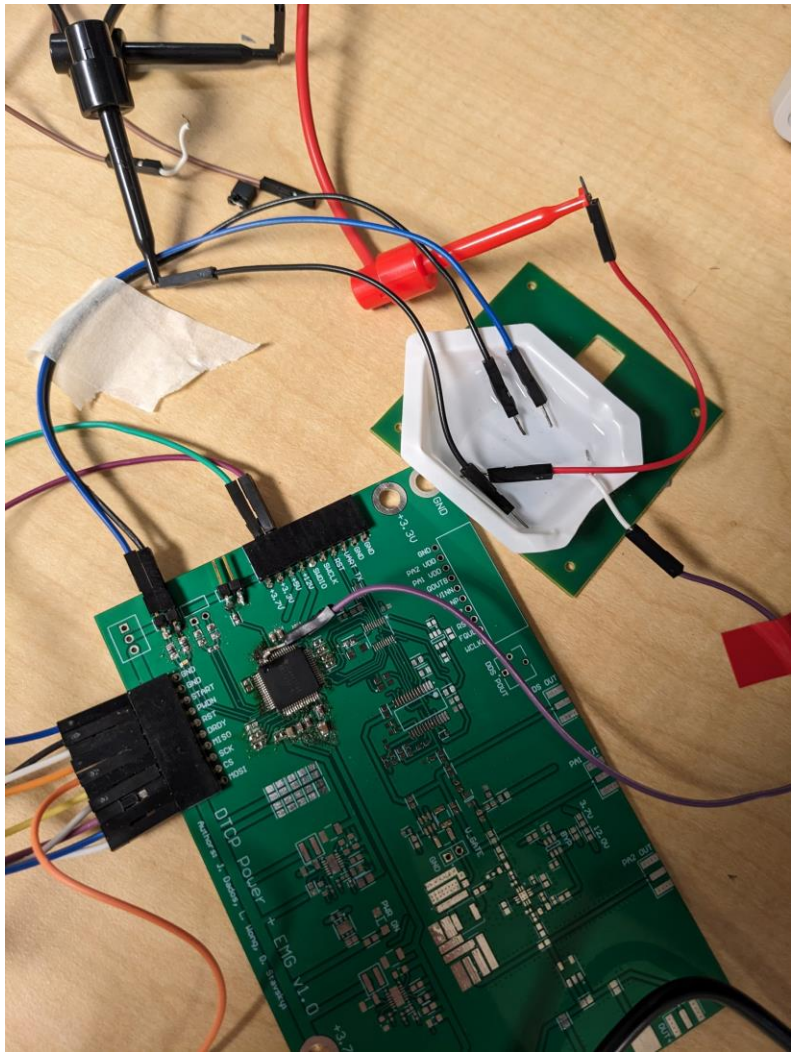
Methods (Third Setup; no bias)





Methods (Fourth Setup; bias in saline but not shorted to the negative terminal)

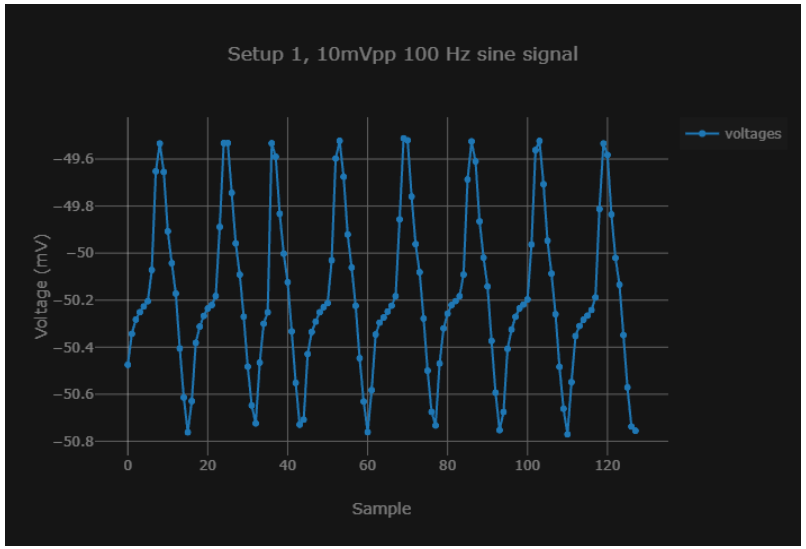


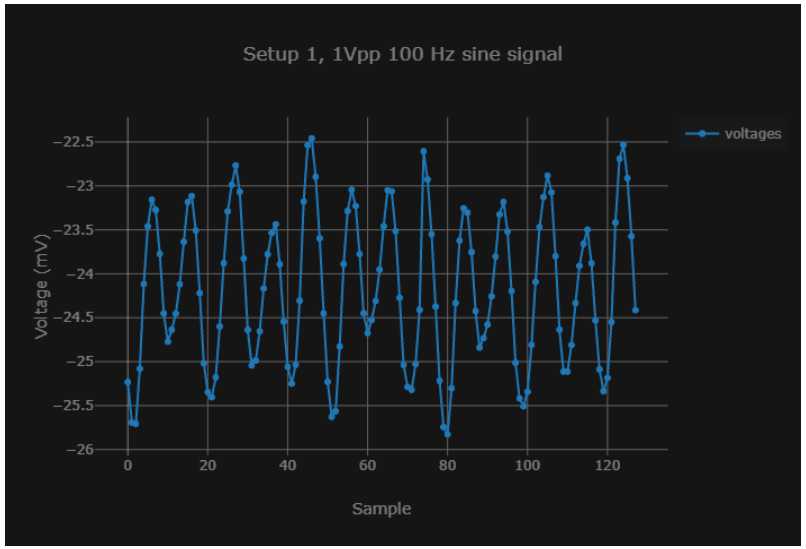


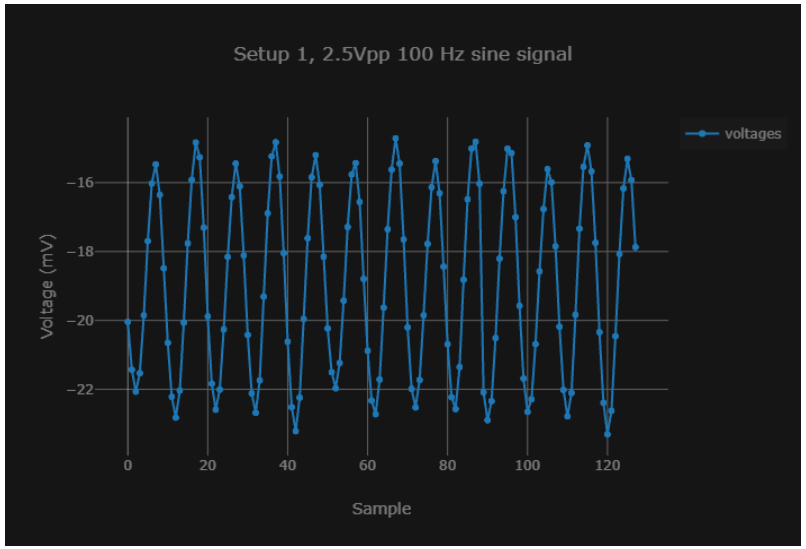
Methodology

- Apply a sinusoidal signal of 100 Hz and of amplitude of 10mVpp, 1Vpp and 2.5Vpp

Results (First Setup; everything in saline)





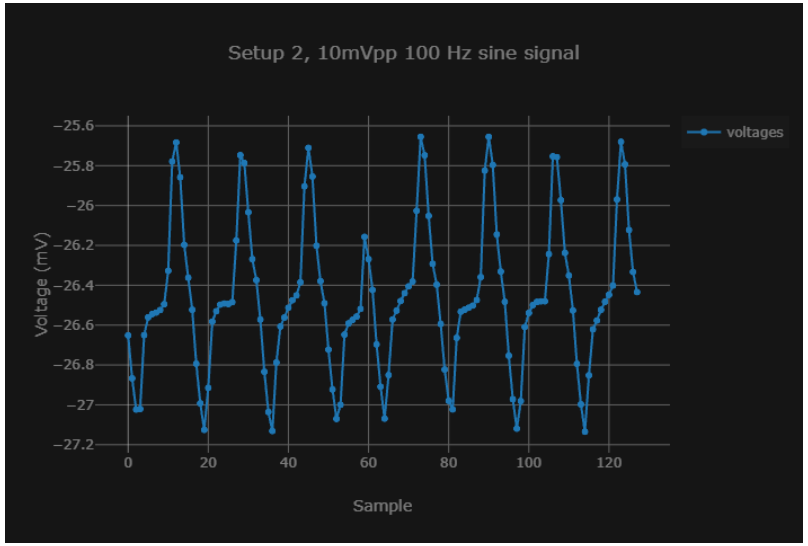


Amplitude: 0.629 mV (noise ?; double check)

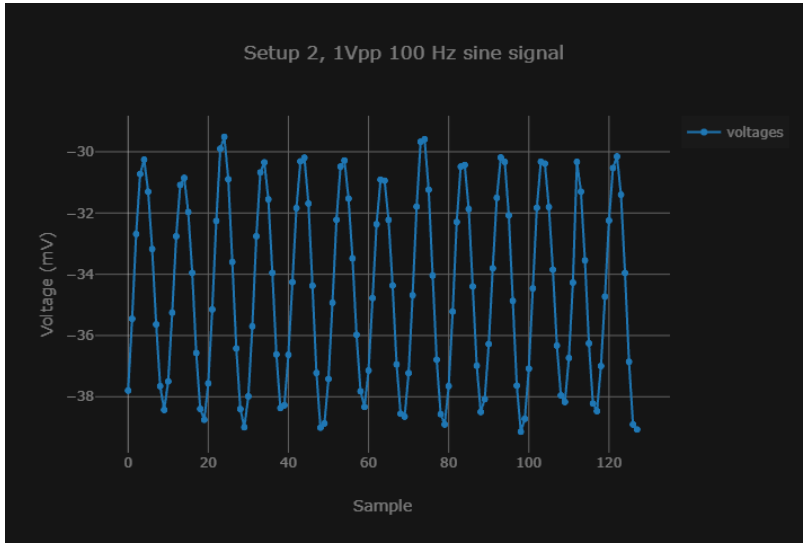
Amplitude: 1.684 mV

Amplitude: 4.296 mV

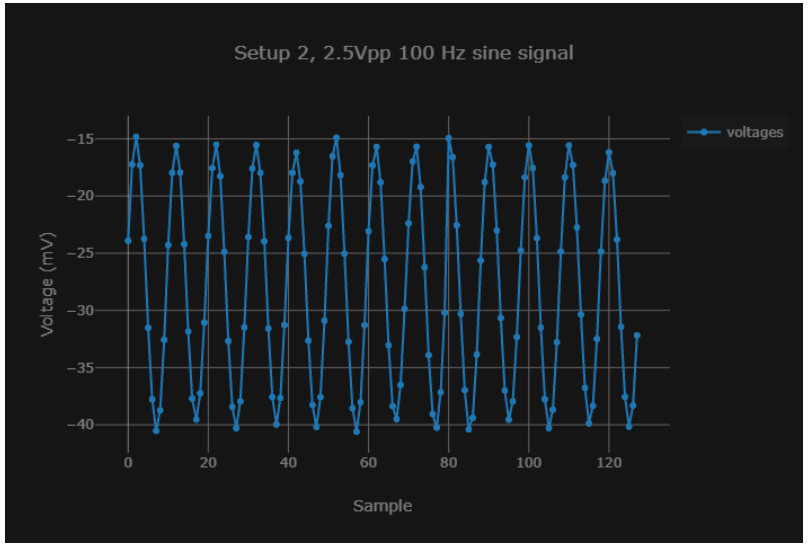
Results (Second Setup; no bias in saline but shorted to the negative terminal)



Amplitude: 0.739 mV (noise ?)

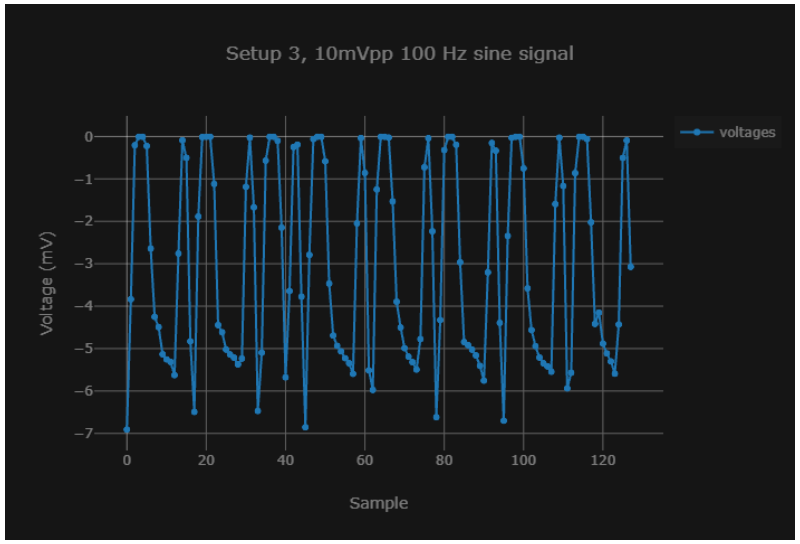


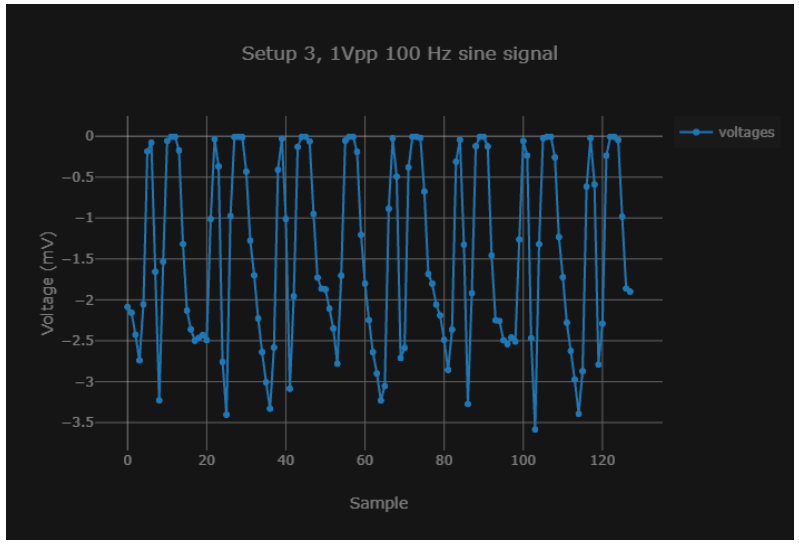
Amplitude: 4.815 mV

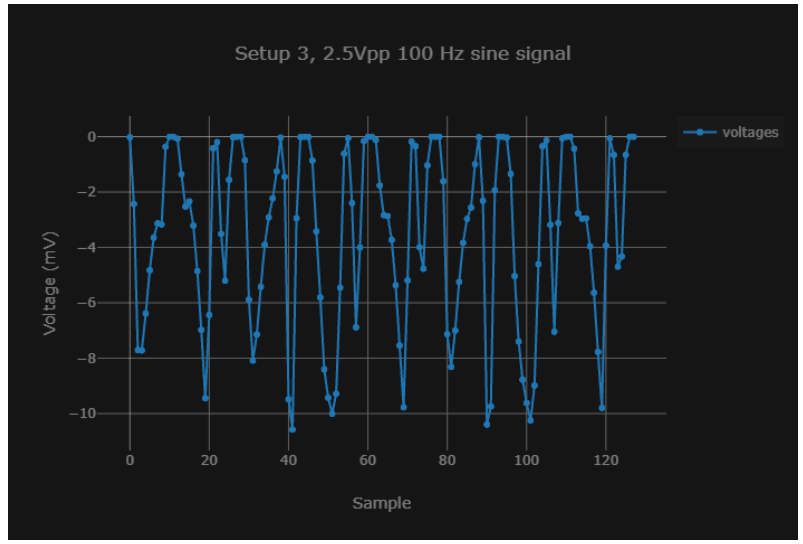


Amplitude: 12.891 mV

Results (Third Setup; no bias)

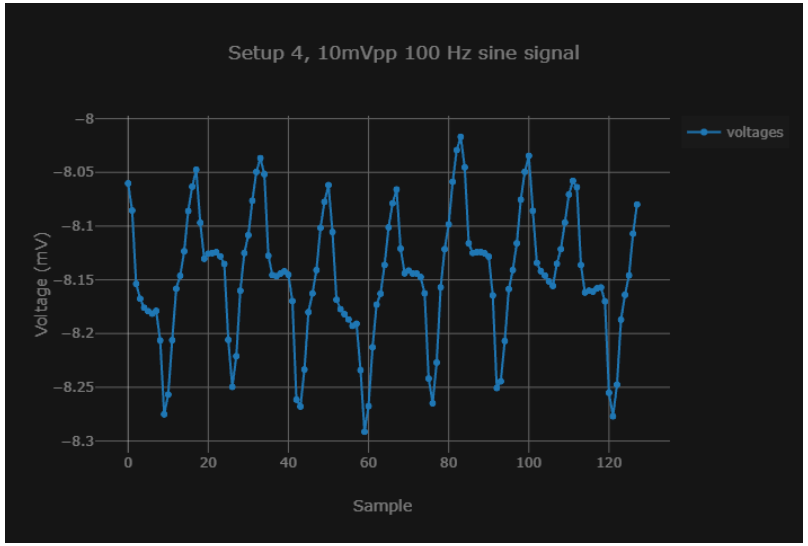




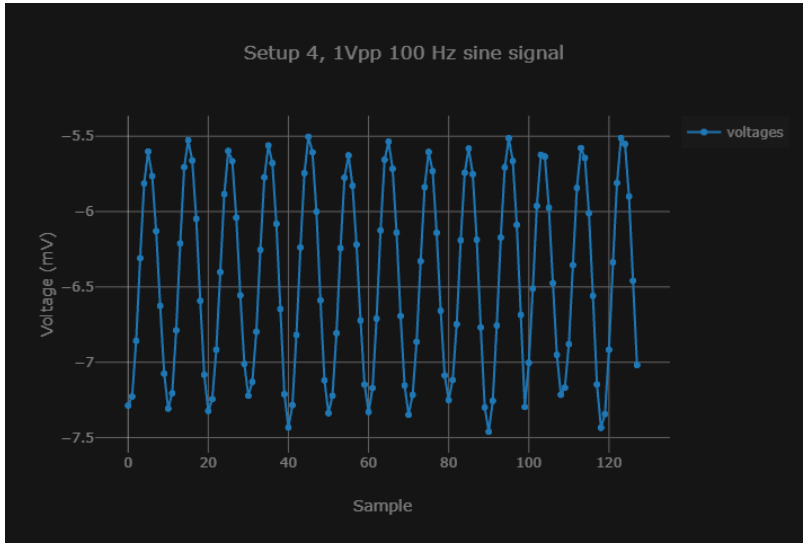


Essentially noise as the shape does not change as the amplitude increases

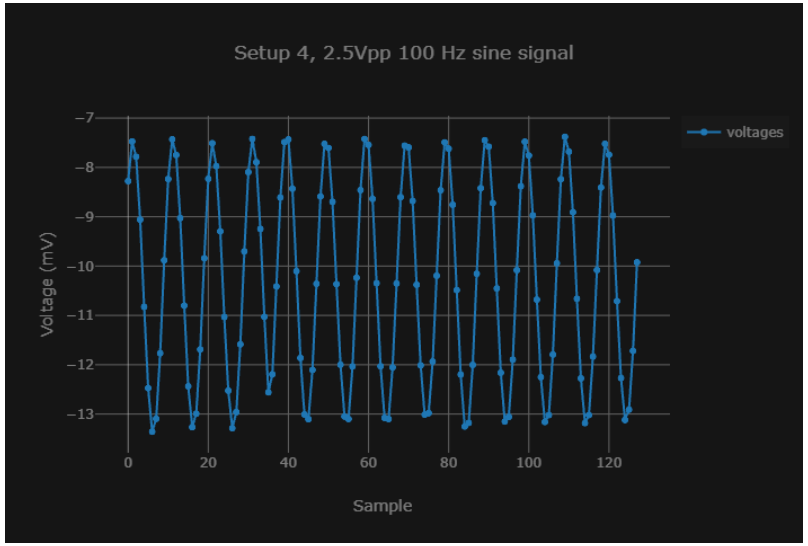
Results (Fourth Setup; bias in saline but not shorted to the negative terminal)



Amplitude: 0.137 mV
(noise ?)



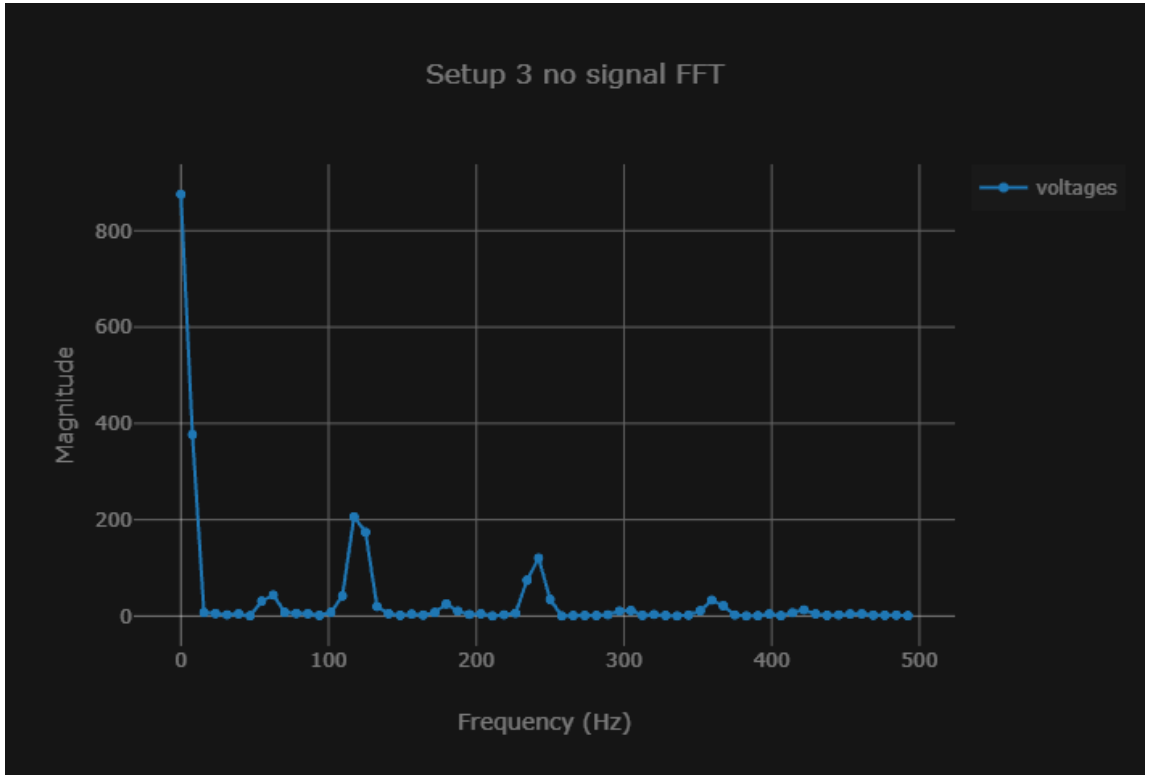
Amplitude: 0.978 mV

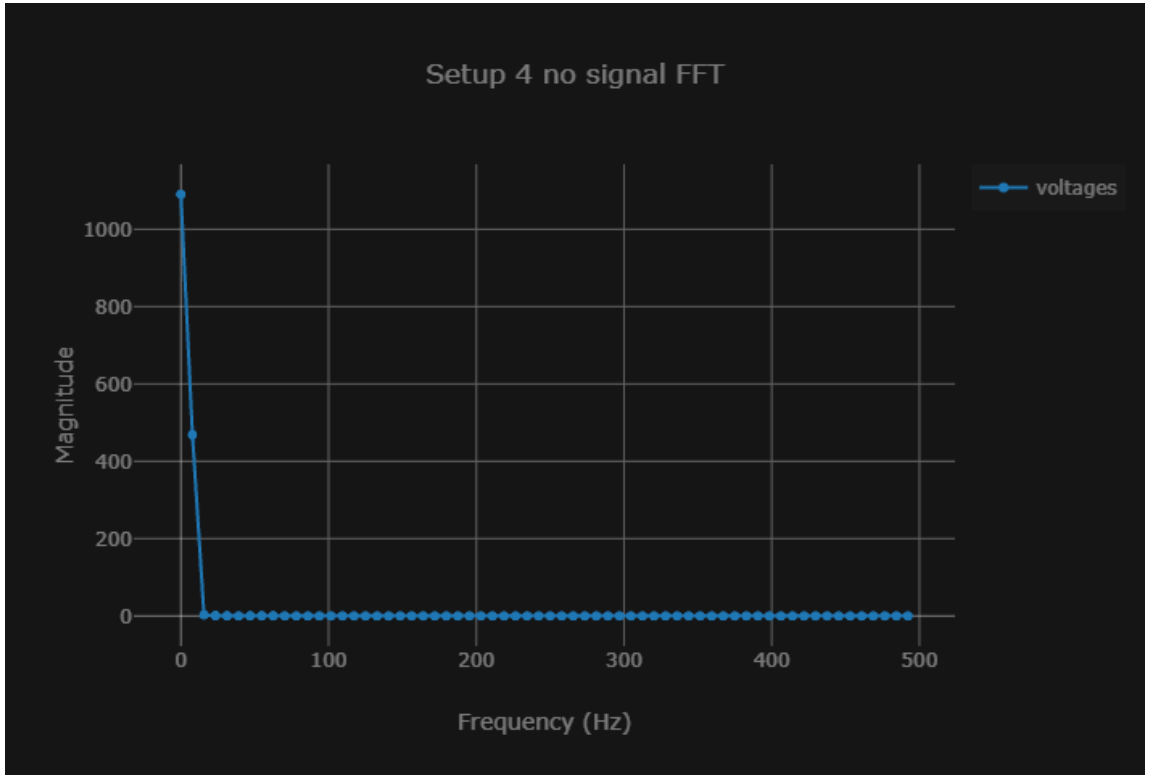


Amplitude: 2.989 mV

Observation / Conclusion

- In terms of amplitude, setup 2 seems to be the best, though all setups with biasing recreate a sine wave at sufficiently large signal
- Based on setup 3, where increasing the signal amplitude does not really do anything, some form of biasing is necessary
    - Biasing does help cancel noise as seen in the graphs below





Next steps

- Work with Luis to get real-time measurements with UART
  - Redo tests 1,2 and 4 with SerialPlot
- PCB Review
- Add one more analog channel and test it
- Connect SRB1 internally for bias
  - Drawing for the setup

3/11/2026

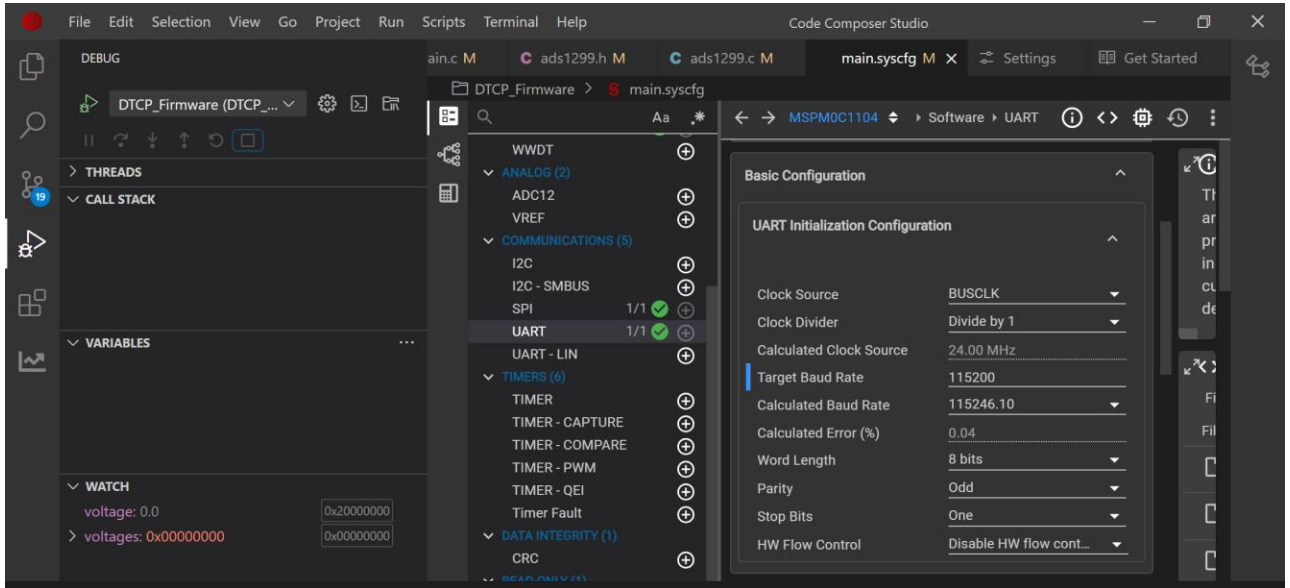
Dmytro Stavskyi, Luis Wong

Objective

- Test the setup and the AFE measurements with saline in real-time with UART.
- Test the setup with channel 2.
- Test SRB1 (“Patient stimulus, reference, and bias signal 1”; description from the datasheet).
- Try to fix the spikes present in UART plotting.

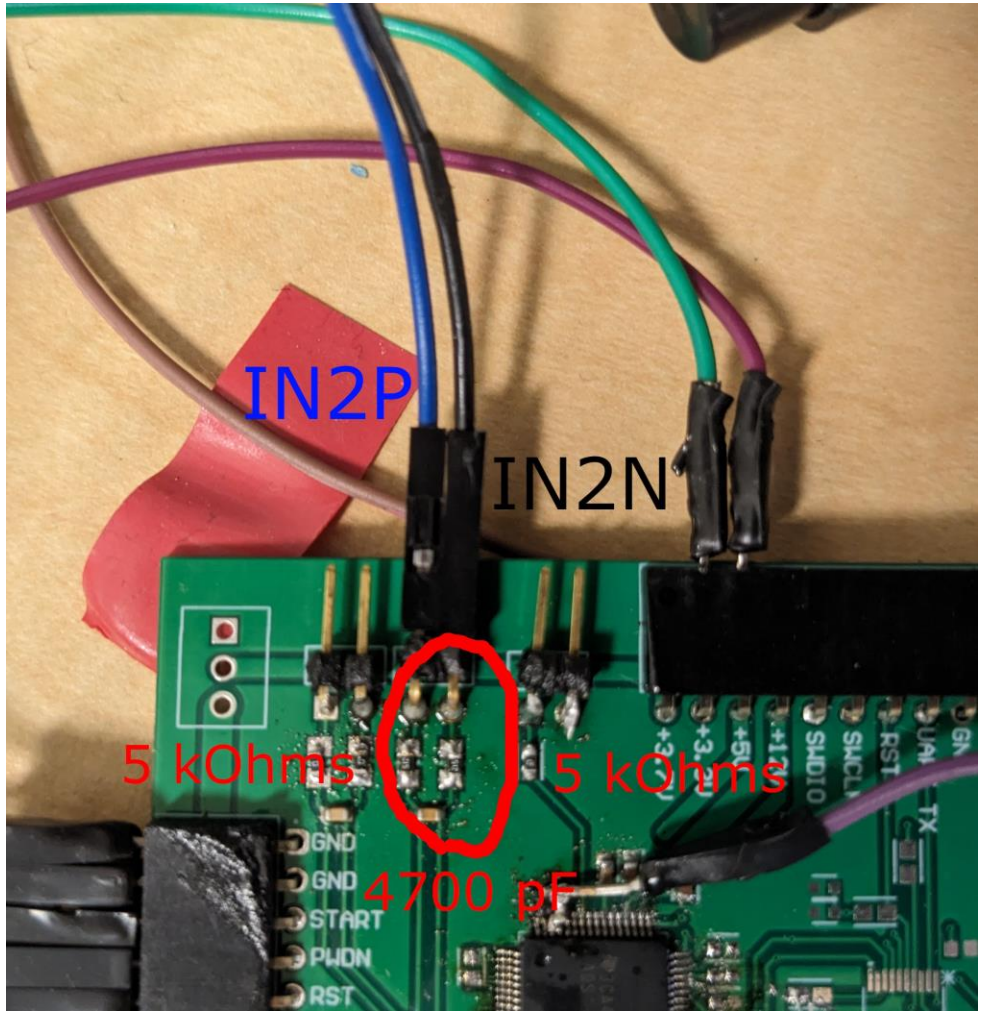
Methods (UART)

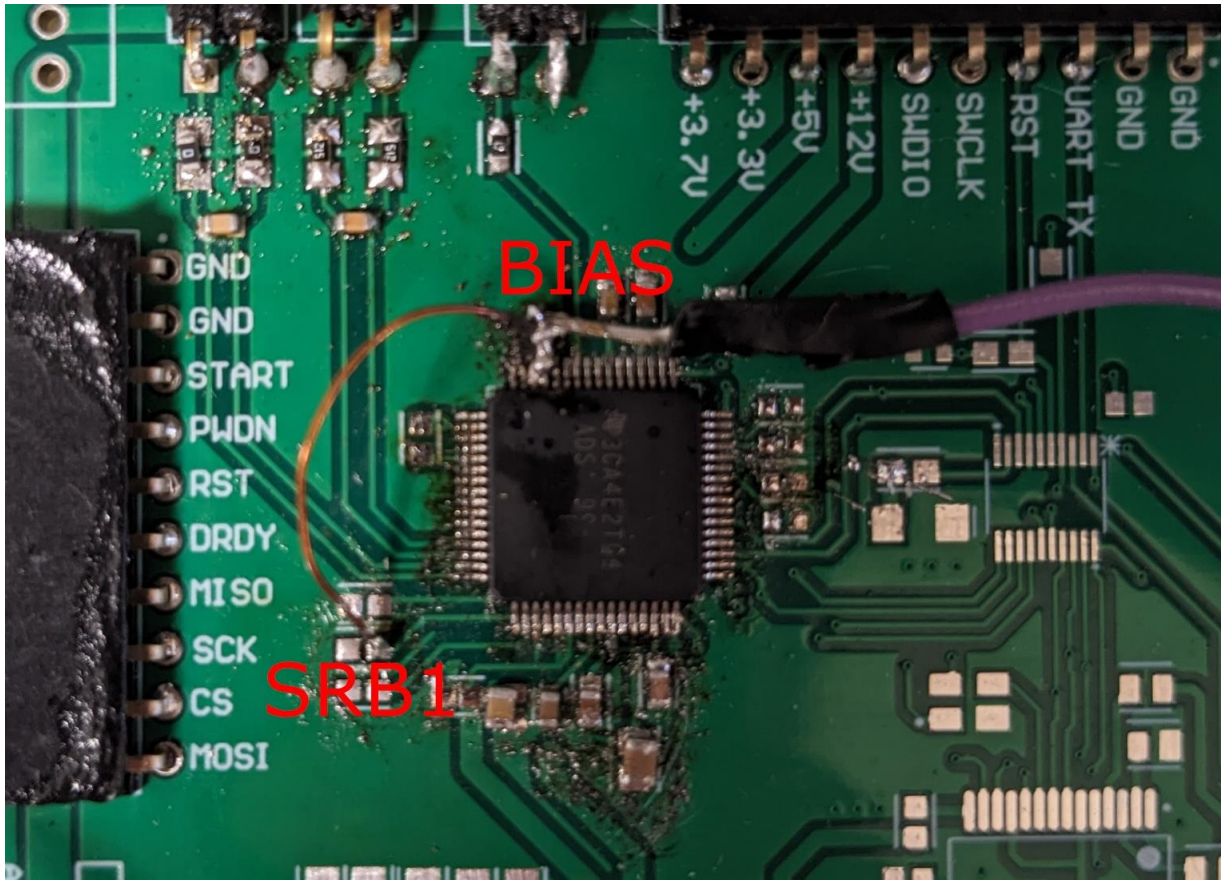
- CCS Sysconfig to change baud rate.
- SerialPlot with ASCII preset for ADS+UART.
- SerialPlot with Binary preset for UART only.
- ADS1299 connected directly to function generator.
  - Similar to setup on slide 37.
- MATLAB to generate test array (100 Hz w/ 2.5 Vpp sampled at 1kHz).



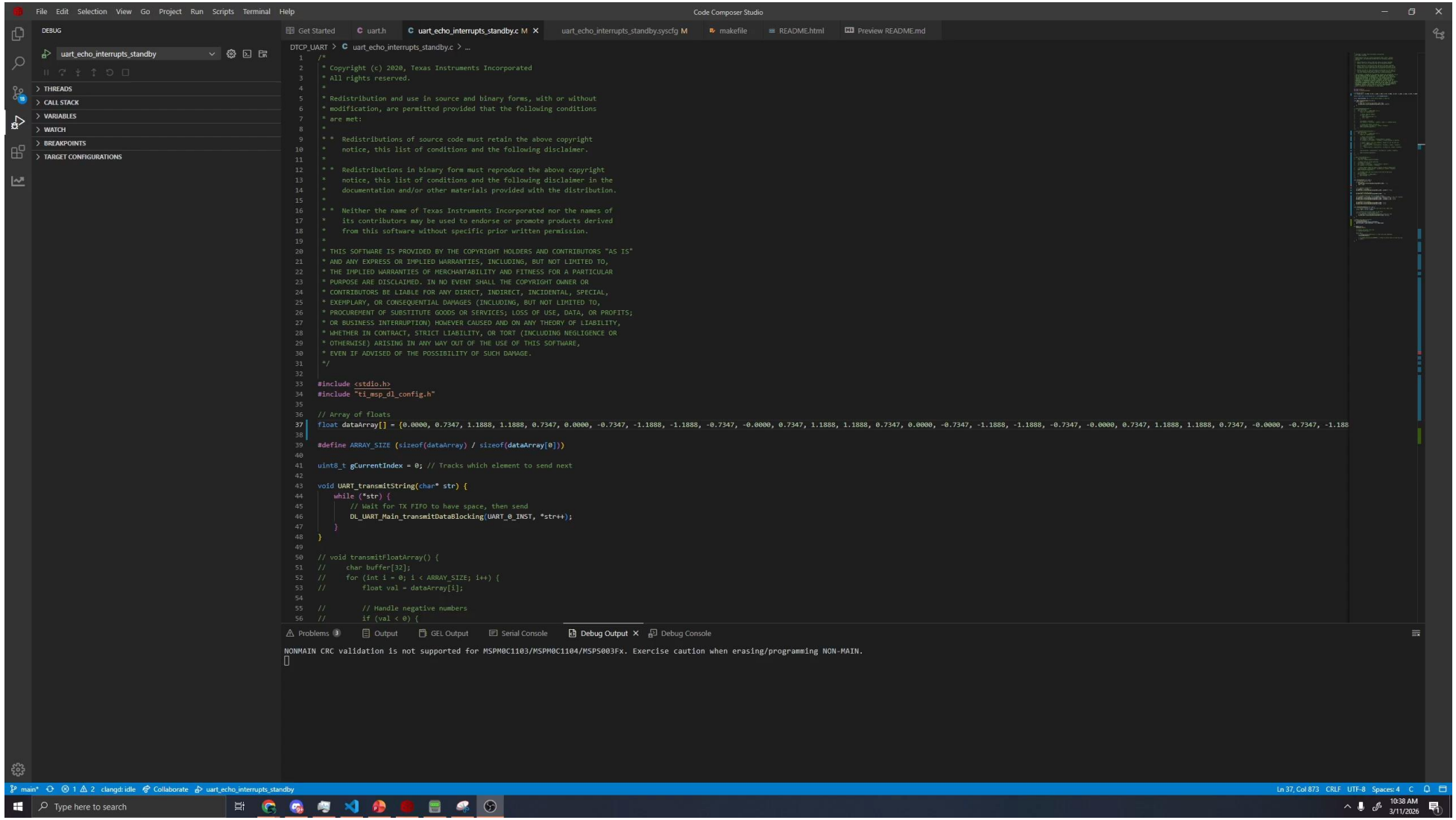

```c
void UART_transmit_voltage(float voltage) {
        /* UART Transmission */
        char buffer[32];
        // Manual float to string conversion
        int intPart = (int)voltage;
        int fracPart = (int)((voltage - (float)intPart) * 100.0f);
        if (fracPart < 0) fracPart = -fracPart;

        // Serial Plotter needs one value + newline to plot a single point
        snprintf(string: buffer, n: sizeof(buffer), format: "%d.%02d\r\n", intPart, fracPart);
        UART_transmitString(str: buffer);
}
```

Methods (Second Channel and SRB1)



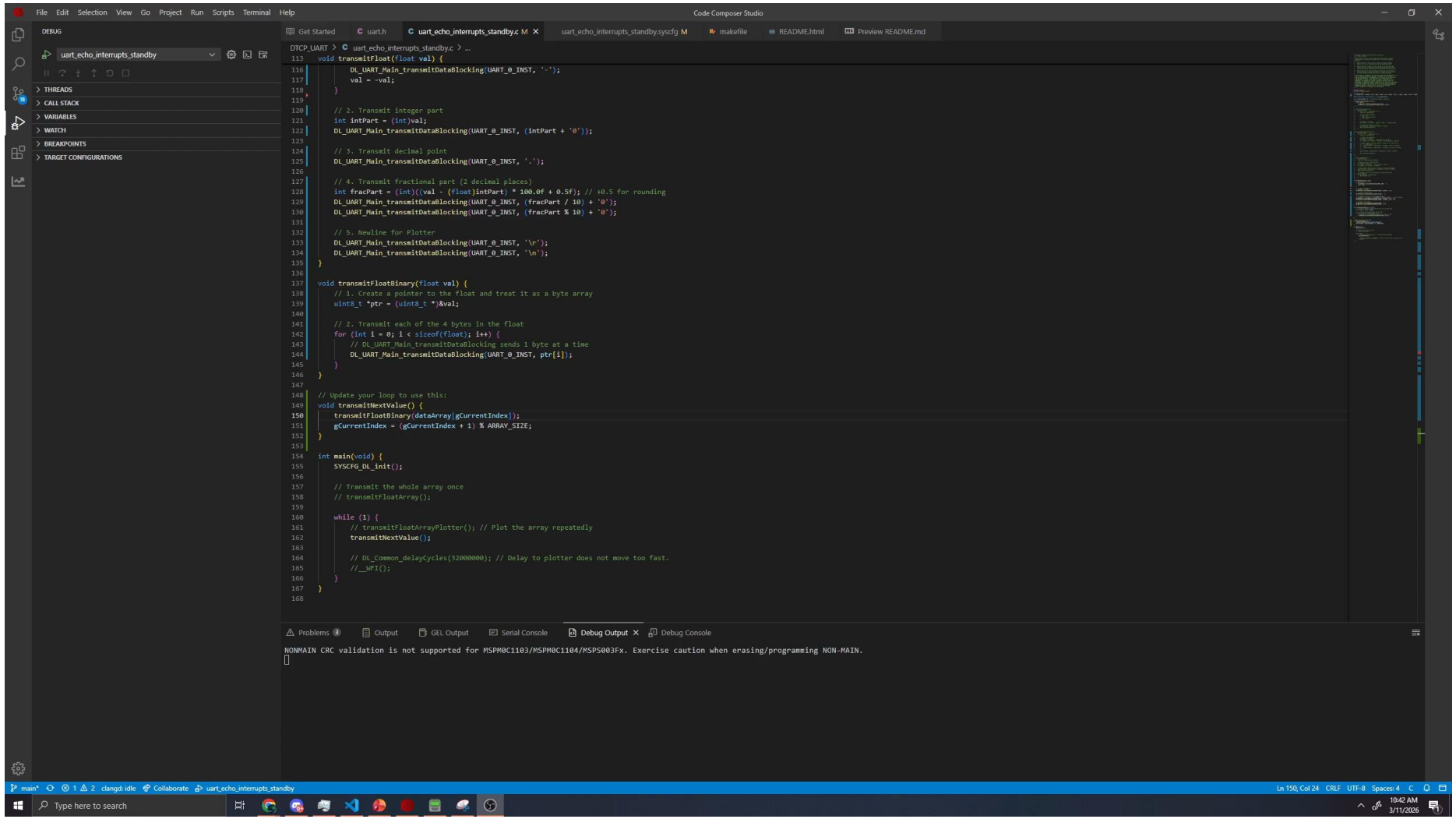


Results (UART w/ Test Array using ASCII)



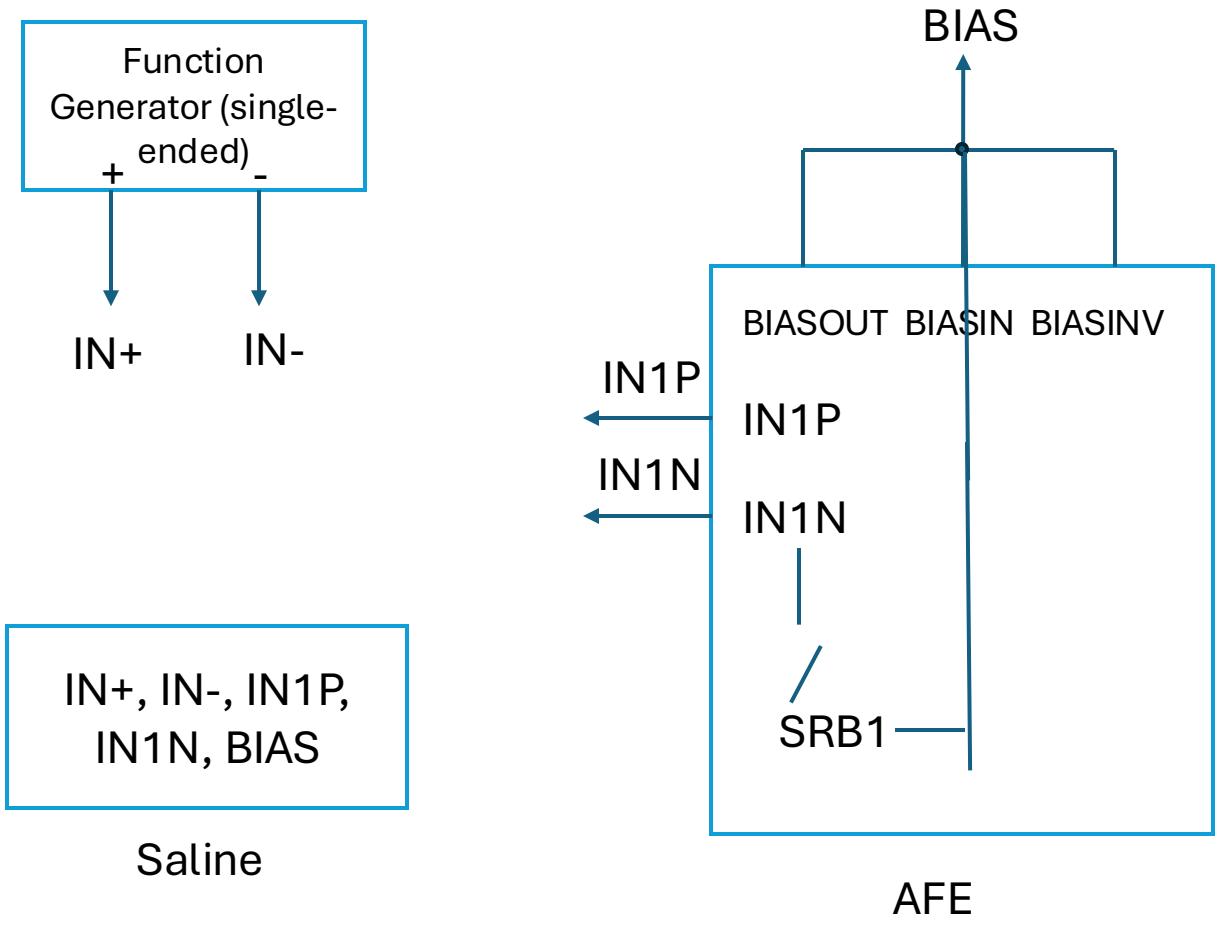
- No UART Spikes when running the UART only code.
- Could be because MCU only needs to worry about one thing unlike the code in the real setup where it needs to manage SPI to communicate with the ADS1299.
- Setup: Just the LaunchPad by itself.

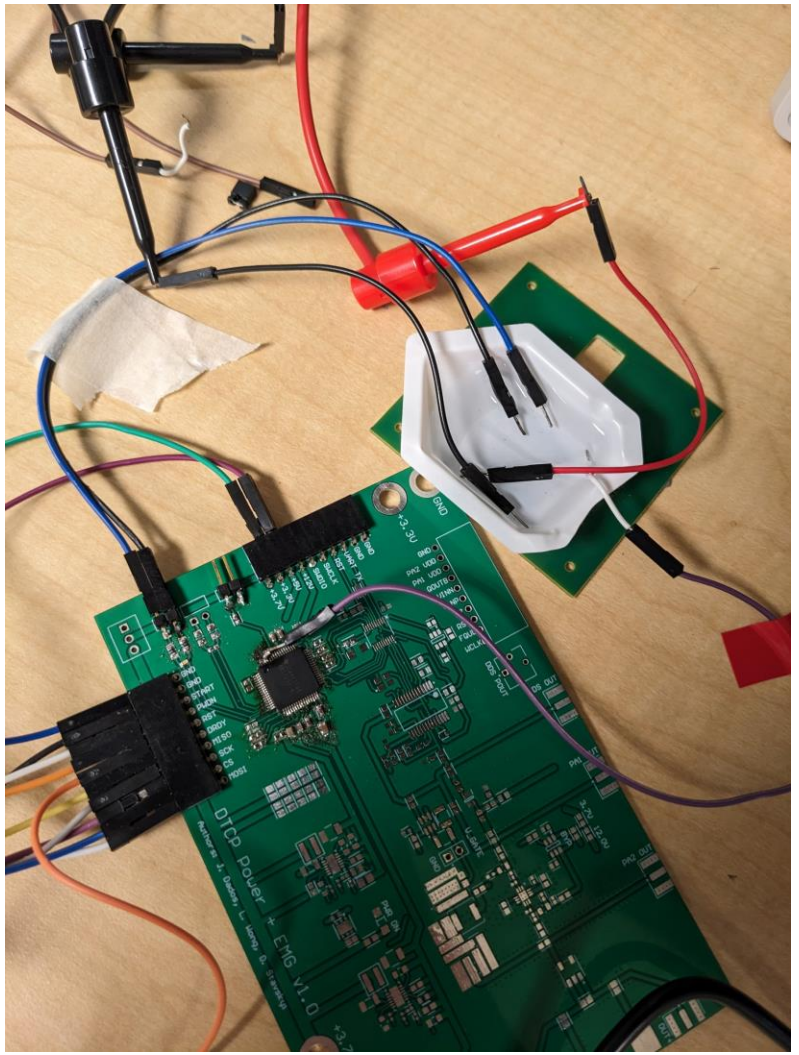
Results (UART w/ Test Array using Binary)



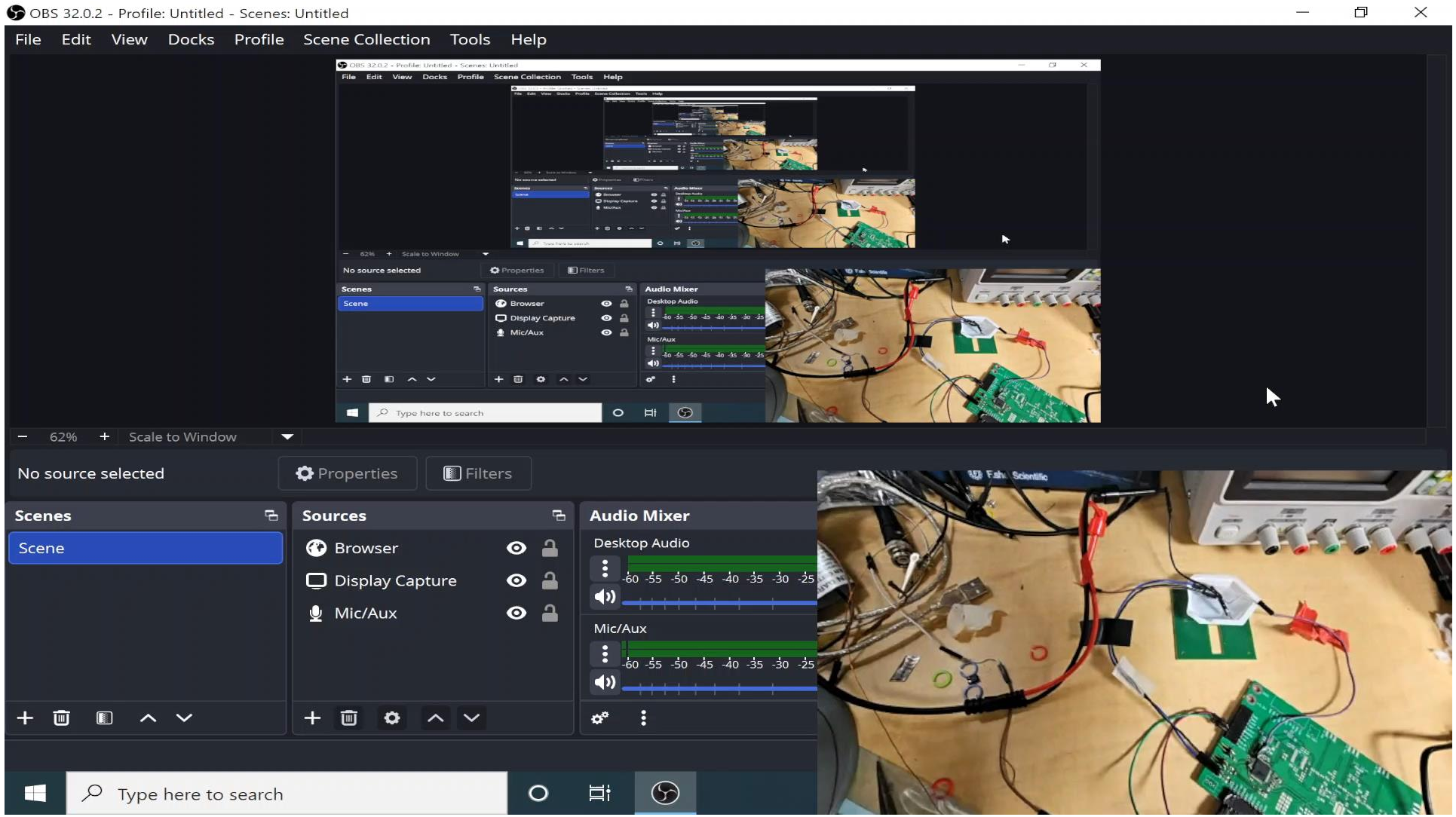
- Next time we test UART+ADS we can try sending raw binary values to see if reducing the strain on the MCU could reduce the spikes we see.
- 4 bytes (w/o header or checksum) vs 4-10+ bytes (delimited w/ \n + conversion).
- Setup: Just the LaunchPad by itself.

Methods (Fifth Setup; bias in saline but not shorted to the negative terminal)



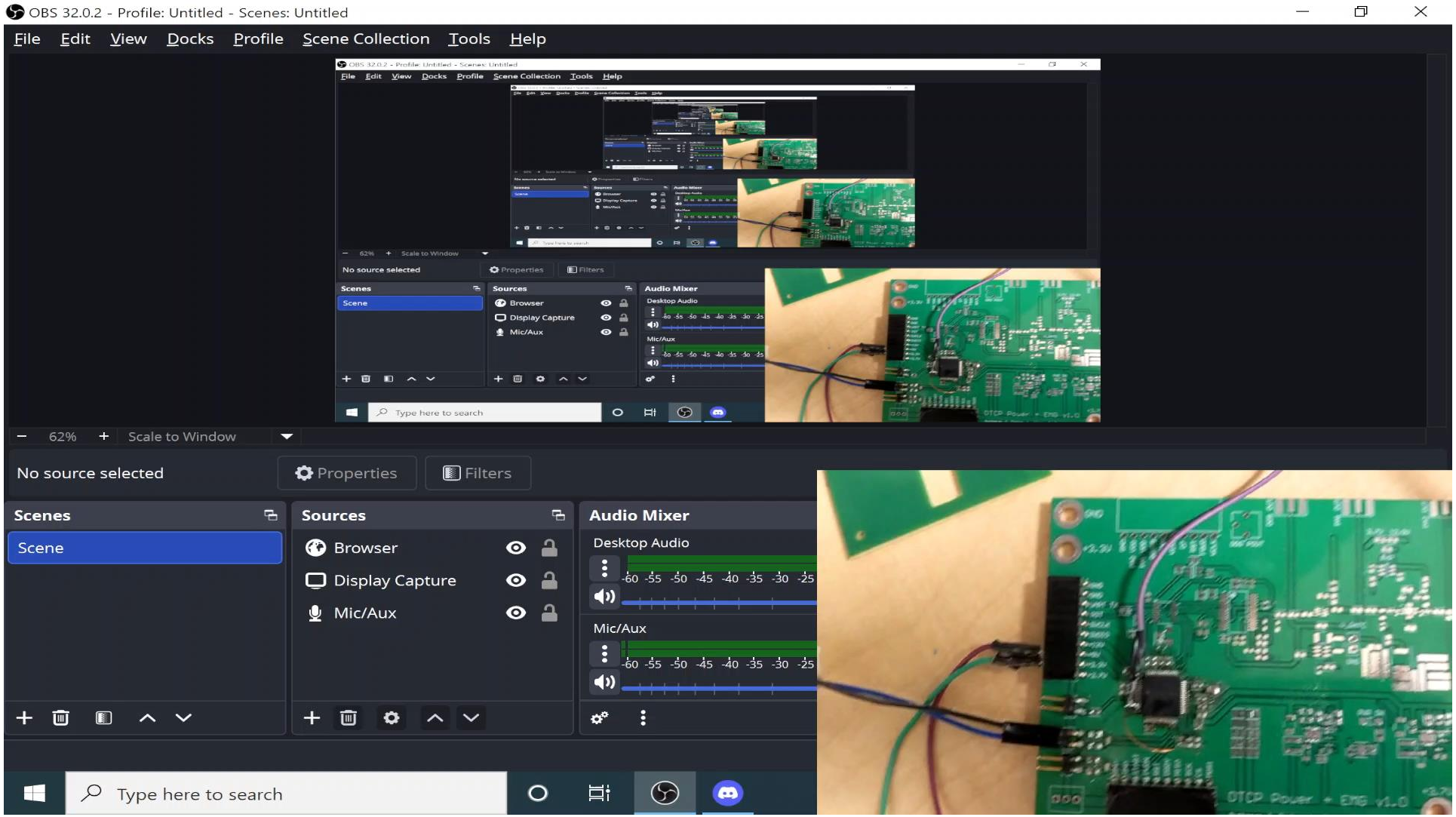


Results (Fifth Setup with channel 1)



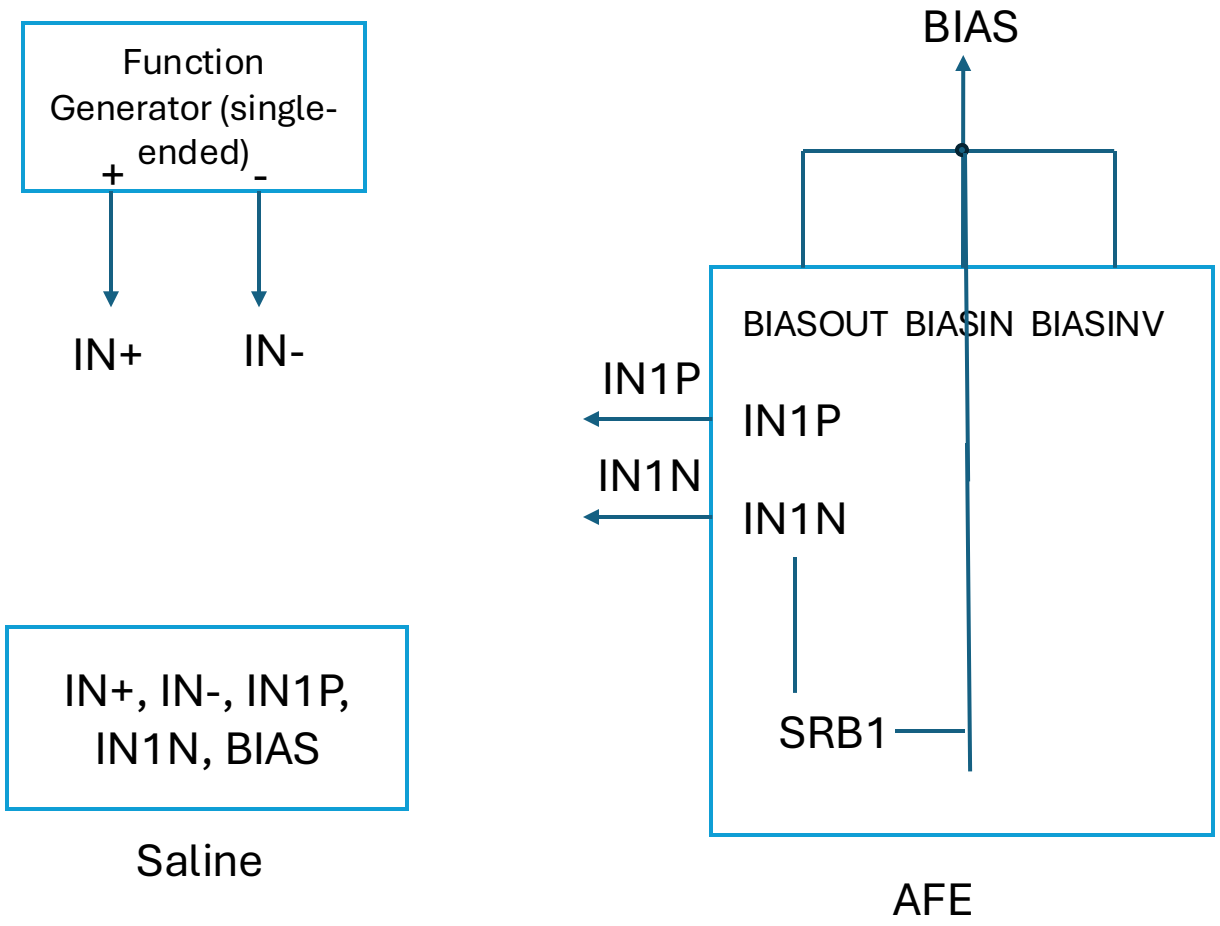
- Methodology: Apply a sinusoidal signal of 100 Hz and of amplitude of 10mVpp, 1Vpp and 2.5Vpp
- The spikes are much bigger than the actual measurement values which increases the range

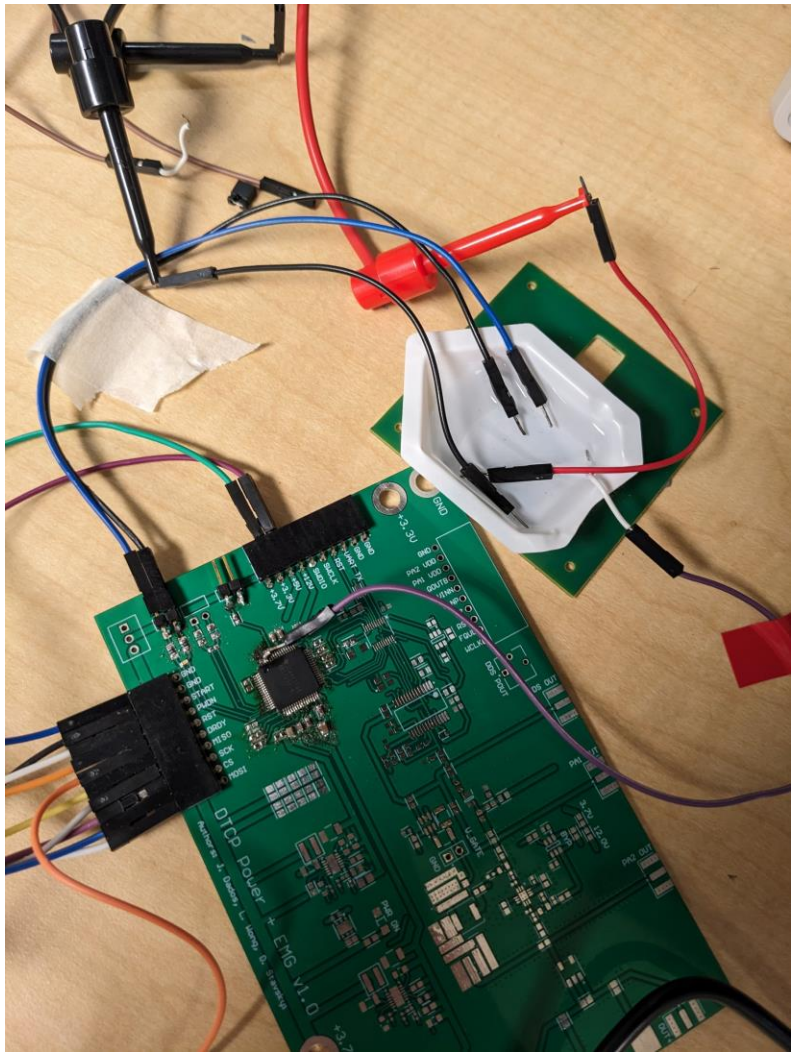
Results (Fifth Setup with channel 2)



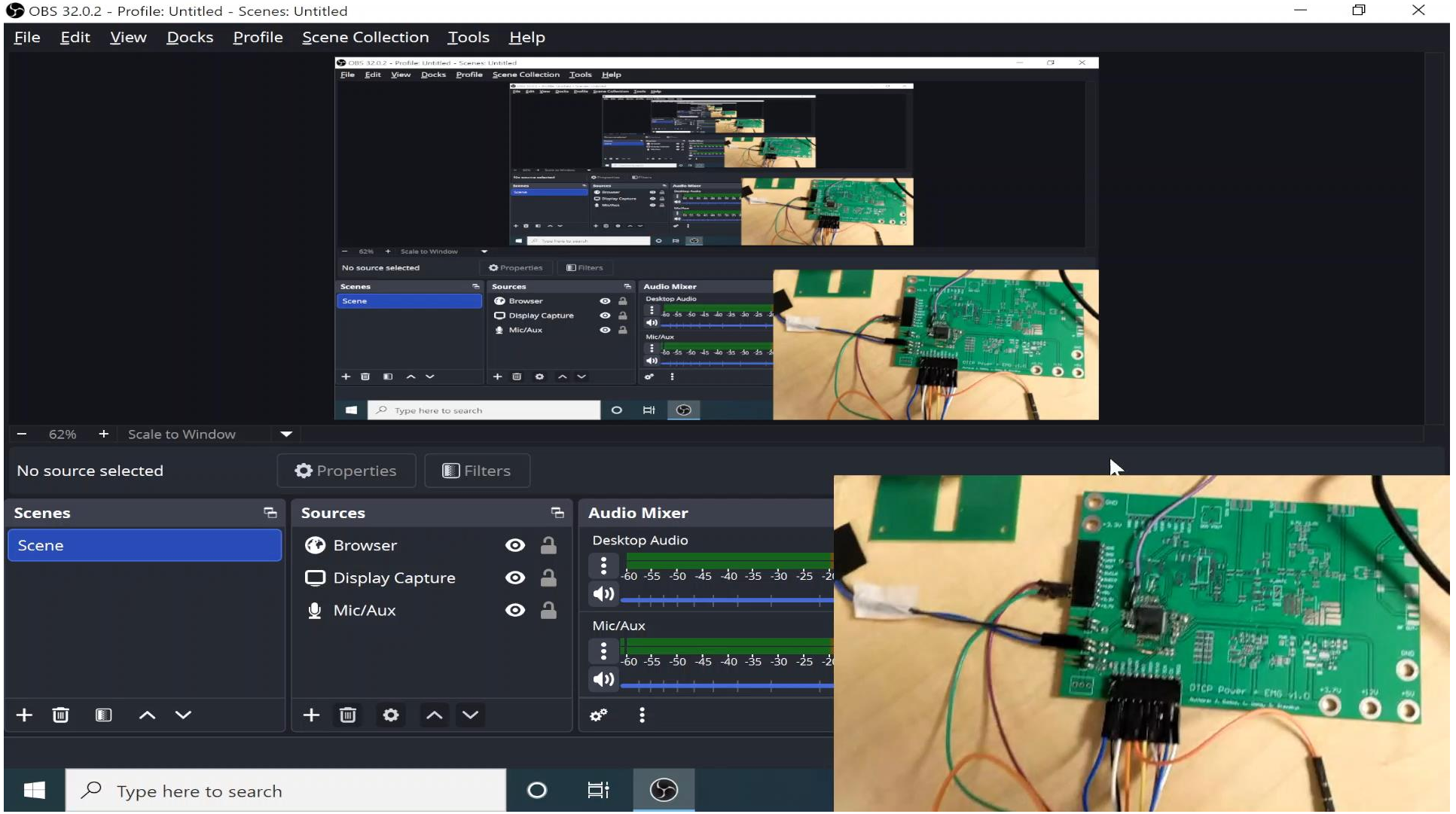
- Methodology: Apply a sinusoidal signal of 100 Hz and of amplitude of 10mVpp, 1Vpp and 2.5Vpp
- Can make measurements with channel 2

Methods (Sixth Setup; bias in saline but not shorted to the negative terminal)





Results (Sixth Setup with channel 2 and SRB1)



- Same setup as in the previous slide but with SRB1 pin enabled in the firmware, meaning that all negative inputs of the channels are connected to SRB1
- Nothing changes as the amplitude of the input signal increases
- Because of the spikes, it is not easy to see, but it saturates to -187.5 mV (tested with CCS graph)

Observation / Conclusion (SRB1)

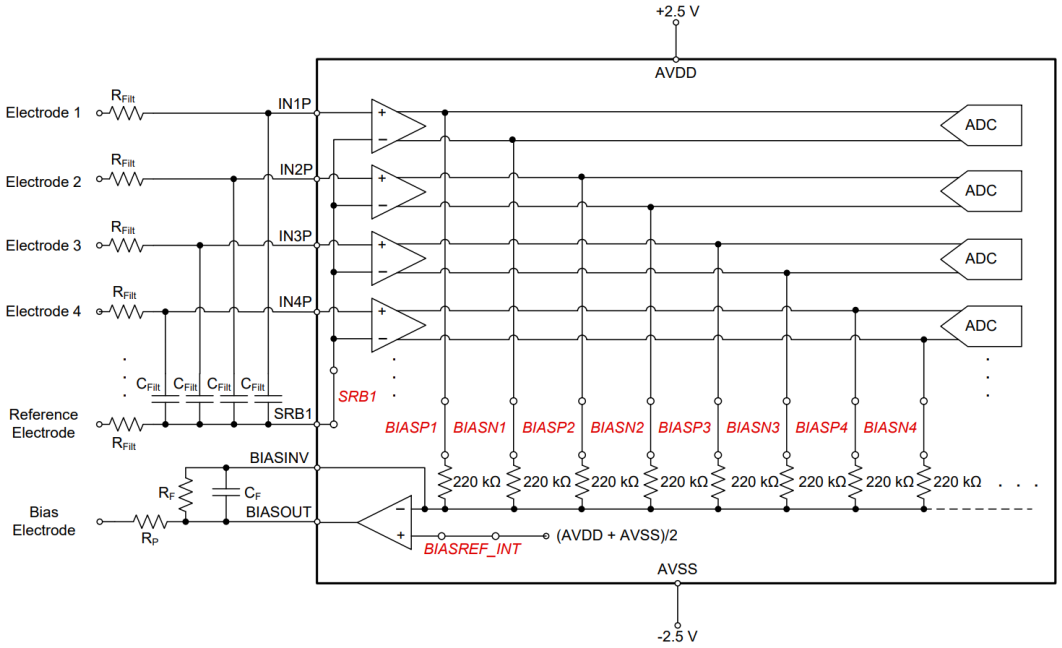
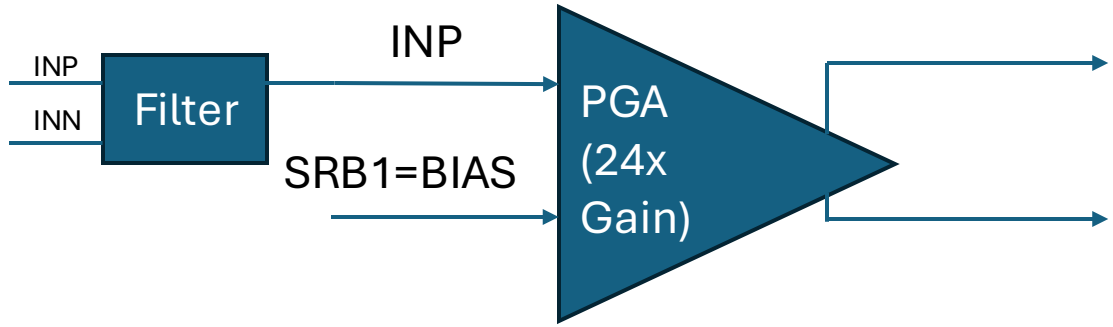


Figure 73. Example Schematic Using the ADS1299 in an EEG Data Acquisition Application, Referential Montage

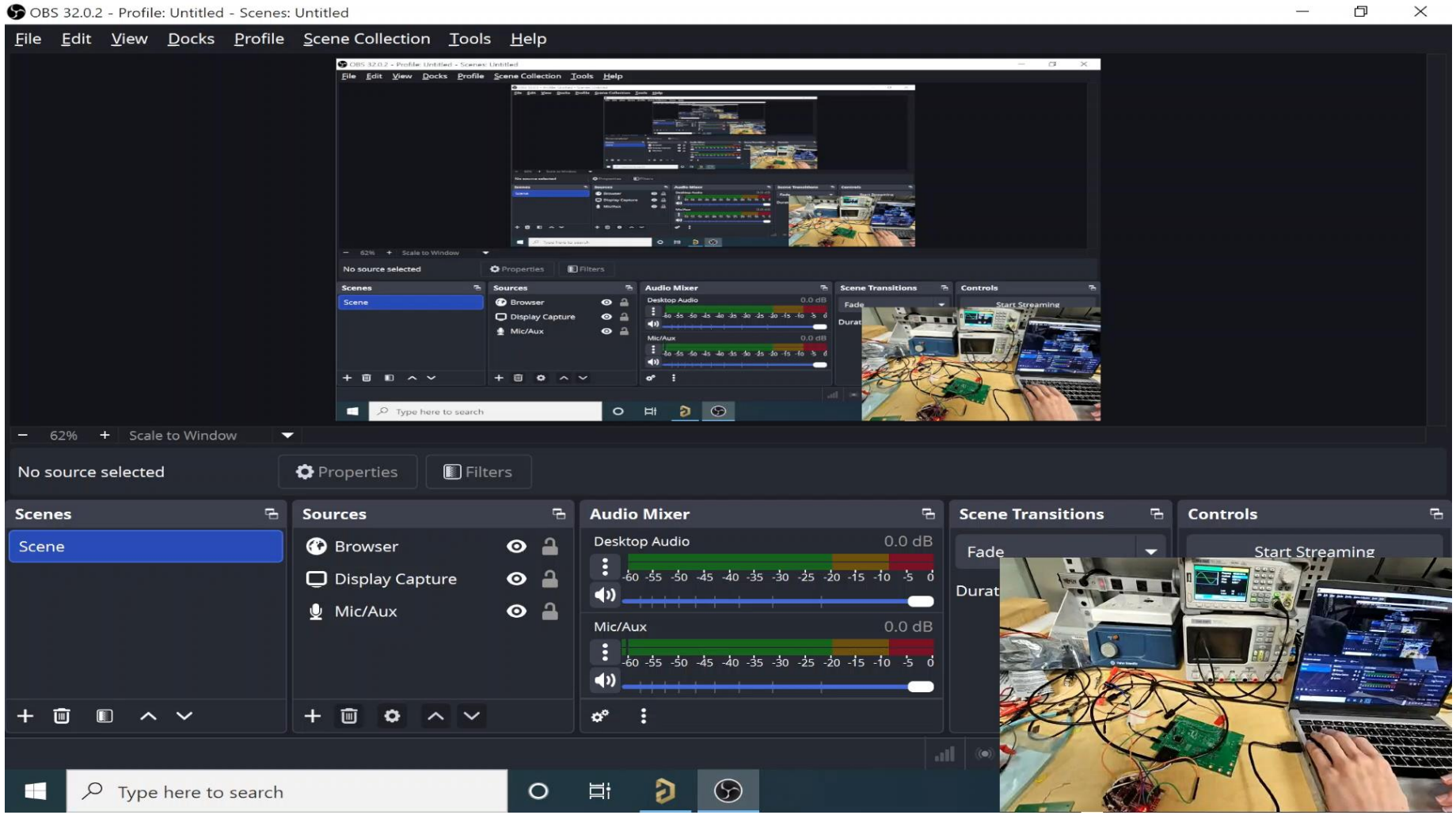
Application example that uses SRB1 from ADS1299 datasheet



What we have if we enable SRB1 on the current board

- The positive input needs to be biased as well

Observation / Conclusion (UART Spikes)



- Setup: no saline, function generator directly connected to analog input channel with bias
- In this particular case, increasing the baud rate to 115200 removed the spikes (but spikes are still present when doing measurements with saline)
- Adding parity or additional stop bit did not get rid of the spikes

Next steps

- Try UART again but send raw binary values of floats instead of sending ASCII values over UART to see if it resolves the spiking as it could be an issue of CPU stalling.
  - Also bypasses the need to manually translate the float to a string which would eliminate potential issues that could occur there.
- Test SRB1 more
  - Remove short between bias and SRB
- UART through Python (PySerial) for data processing
- Load an example EMG signal (from some dataset; with a spike) into the function generator
- Spike detection code (print inside the MCU to determine if its UART or the real data)

3/25/2026

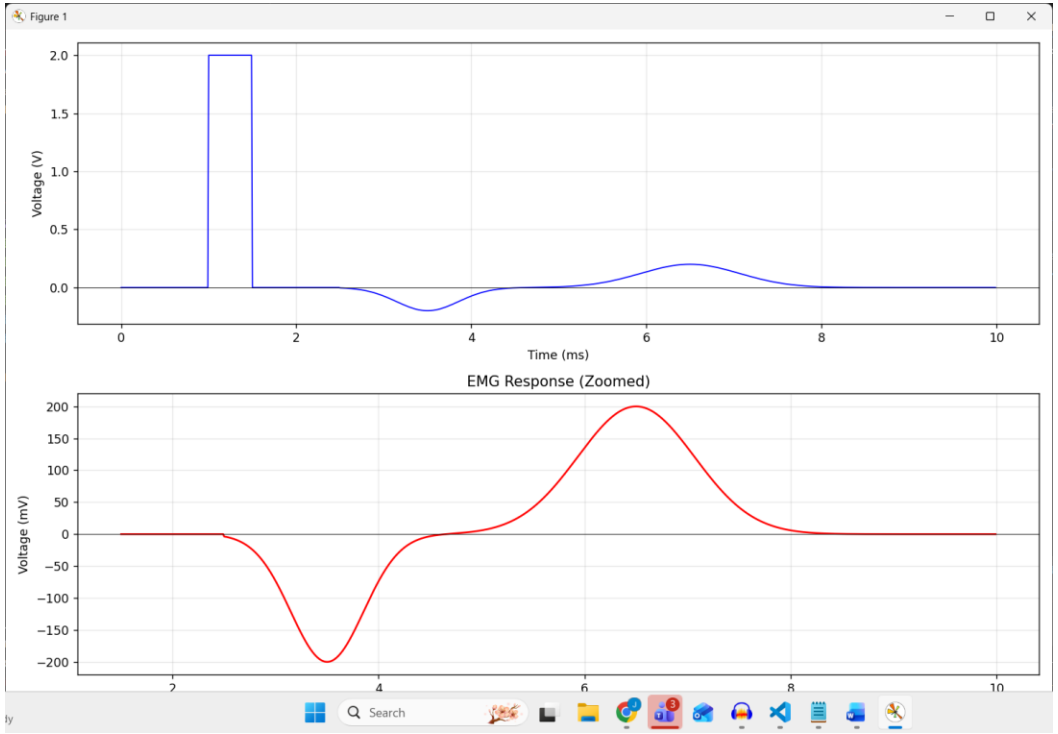
Dmytro Stavskyi, Luis Wong

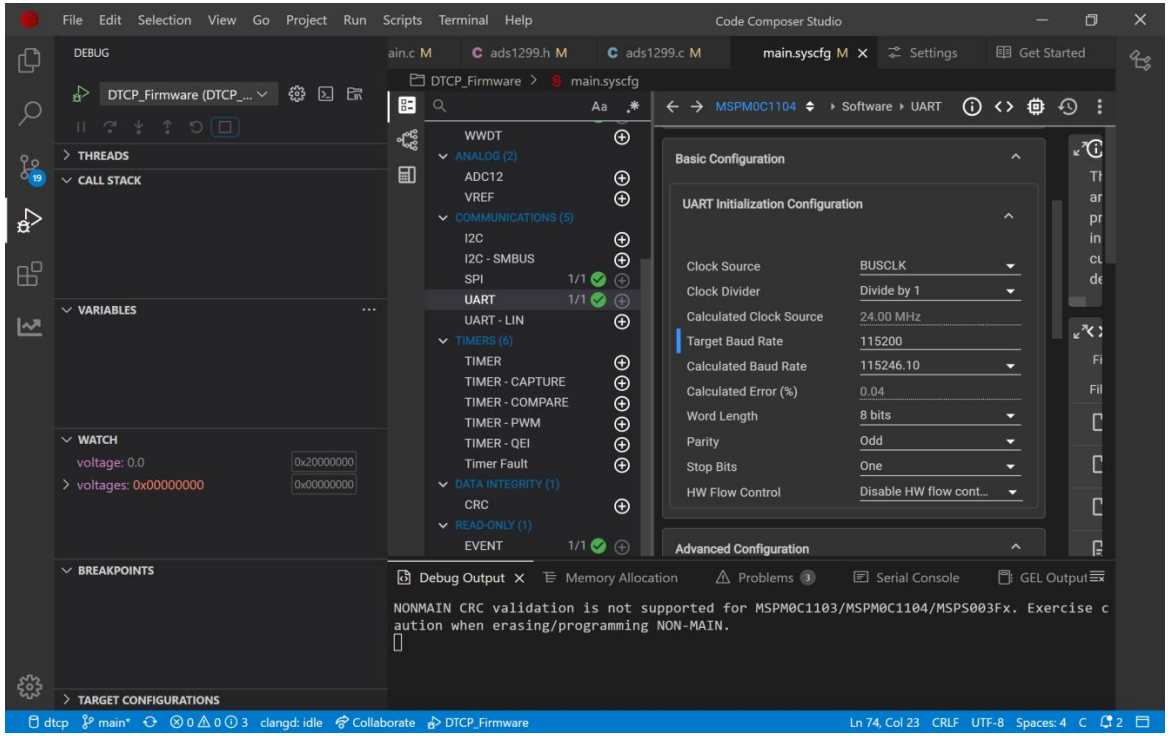
Objective

- Try to fix the spikes present in UART plotting.
  - Test the setup and the AFE measurements with saline in real-time with UART sending raw binary data as opposed to ASCII like last time.
- Load a real EMG signal into the function generator to test the setup's ability to read the signal and begin developing the artifact detection algorithm.

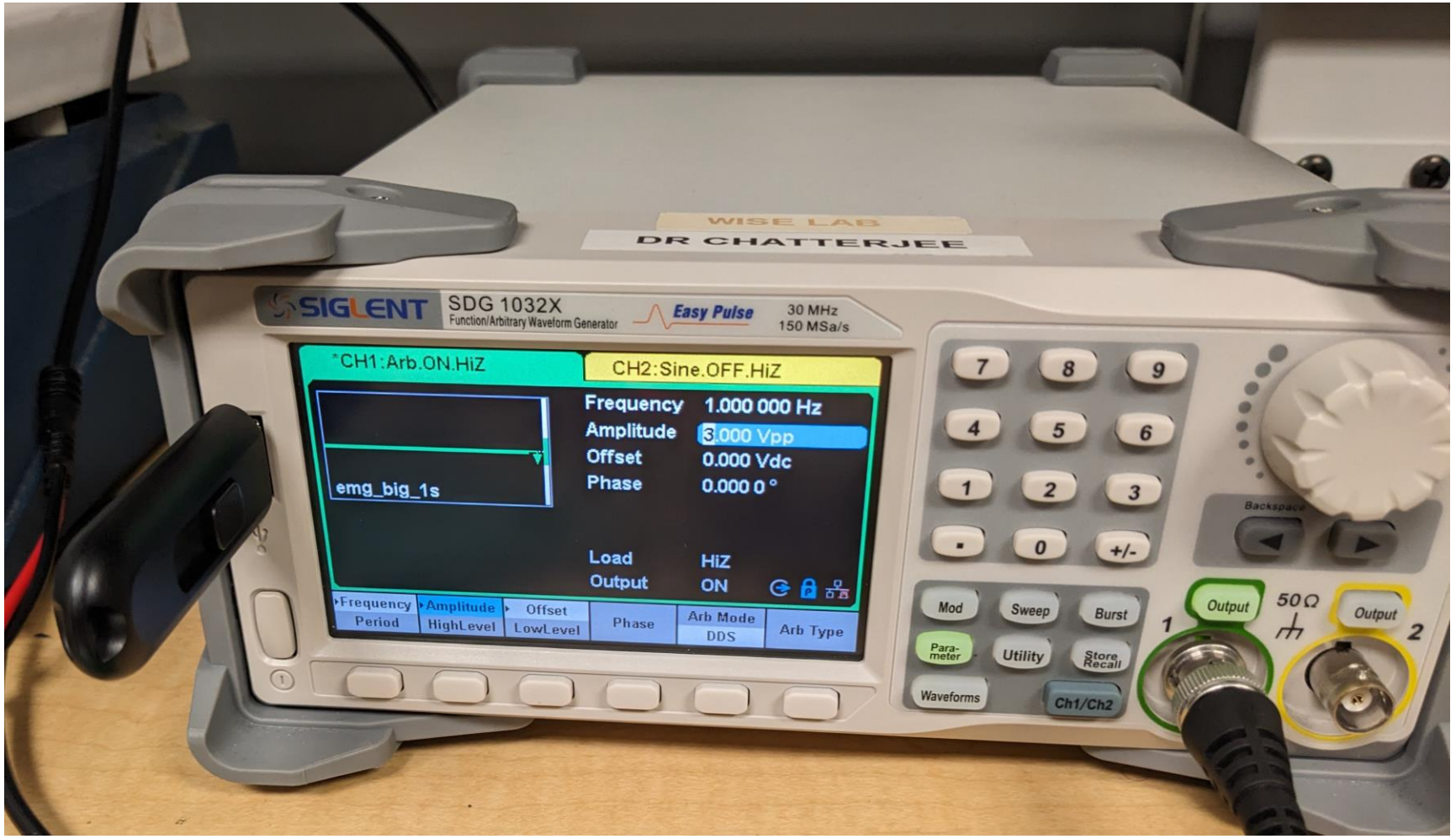
Methods (UART)

- CCS Sysconfig to change baud rate.
- SerialPlot with Binary preset for UART.
- Python code written by Jerimiah to generate EMG signal + artifact.Function generator with generated EMG signal loaded into it.
- CCS C code to change sampling rate of the ADS1299.



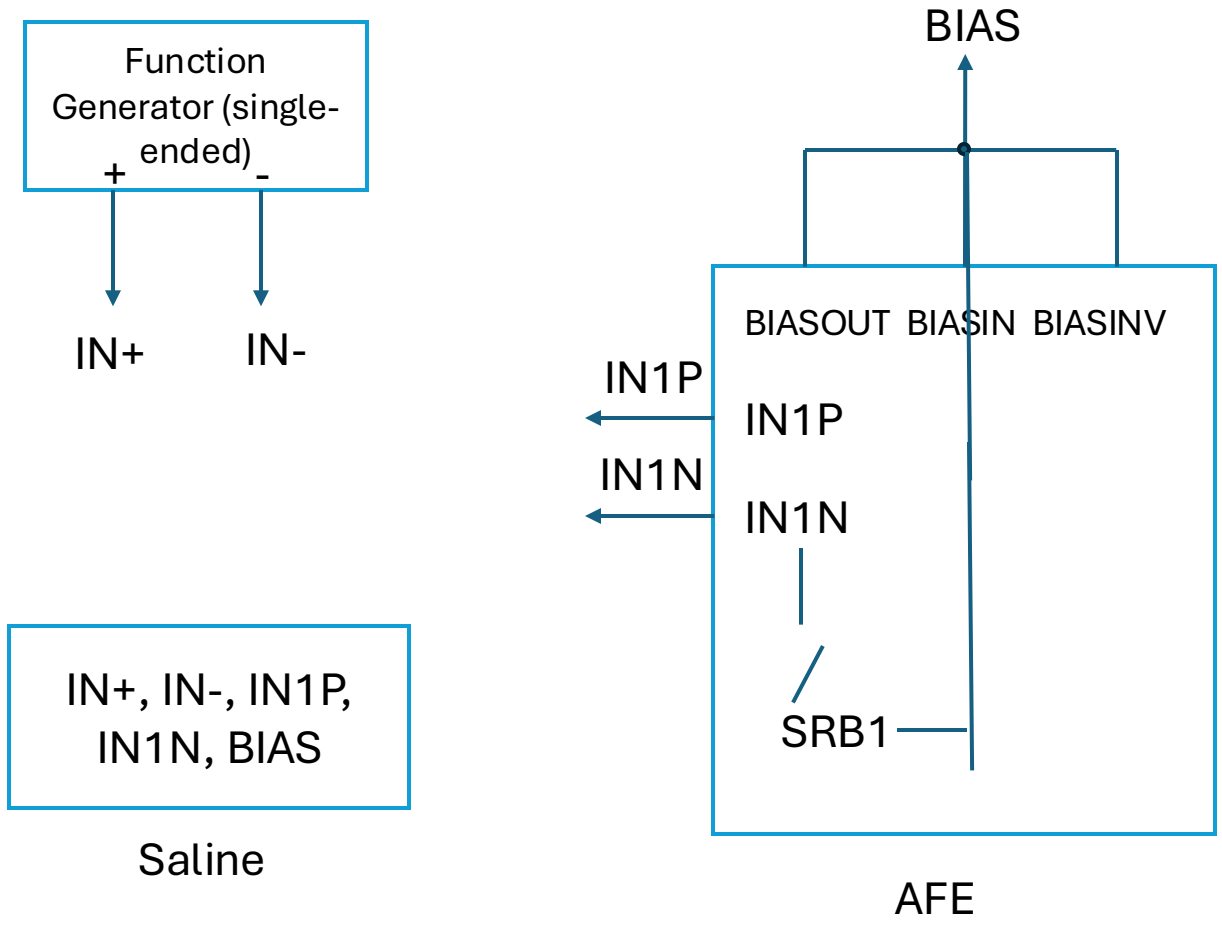


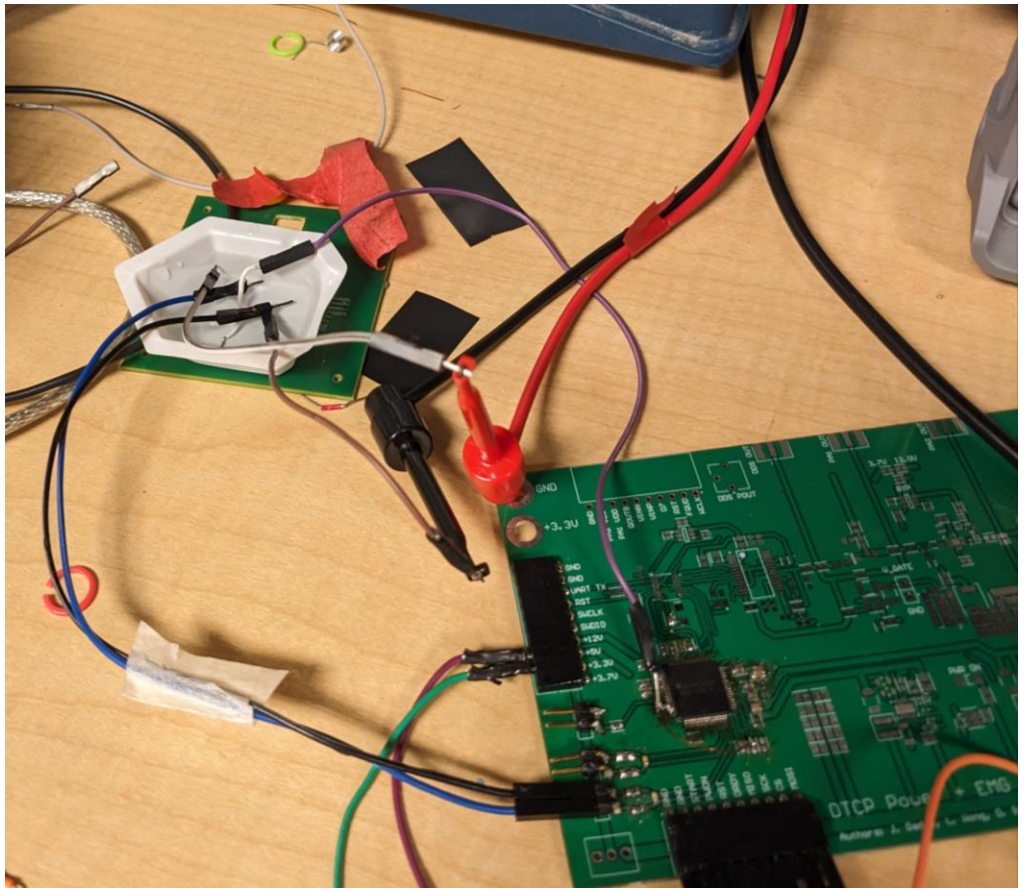
Methods (Function Generator and Methodology)



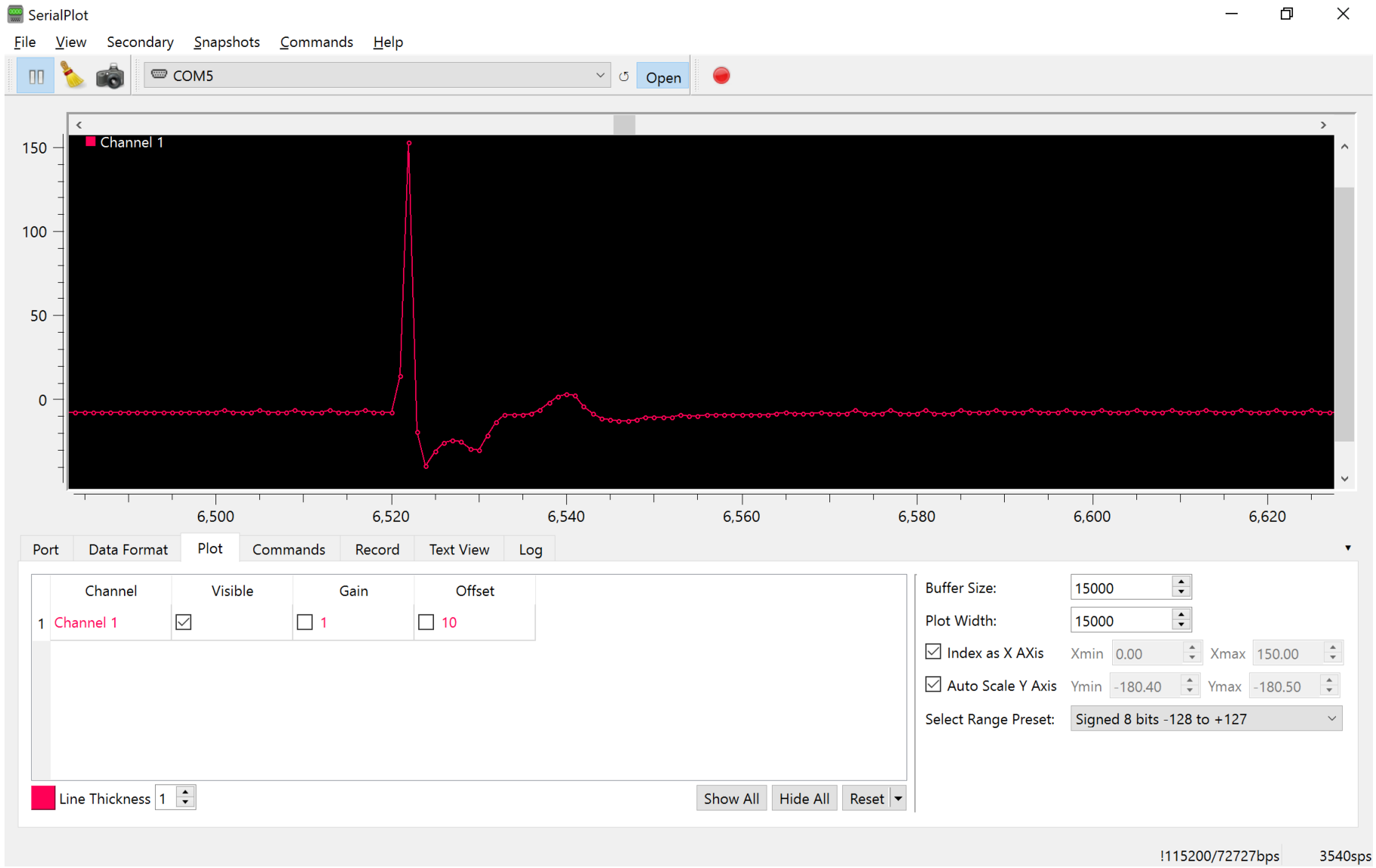
- Methodology: load a test EMG signal represented in a csv file to the function generator and change frequency / amplitude until we get adequate measurements (measured signal replicates the test signal)

Methods (Fifth Setup; bias in saline but not shorted to the negative terminal)





Results (Fifth Setup)



- Configuration: 3Vpp, 1Hz, 8 kHz sampling rate, baud rate of 800 kbps
- The positive peak after the stimulus is around 3 mV
- This peak is measured at sample 6542, while the stimulus peak is measured at sample 6524, so there is 18 sample difference between the two, which with 8kHz sampling rate, corresponds to 2.25 ms ($18 * 1/8000 * 10^3 = 2.25$ ms)

Observation / Conclusion

- Changing how UART sends data from ASCII string to raw binary fixed the spiking issue in the previous configuration (1kHz sample rate, 115200 bps baud)
- With the sample rate of 1kHz (which is what we had before), each sample represents 1 ms ($1 * 1/1000 * 10^3 = 1$ ms) , which is too low for us
  - The maximum sampling rate for ADS1299 is 16kHz, but it can also do 2kHz, 4kHz and 8kHz
- However, as we increased the sampling rate, we saw very frequent spikes again
  - With UART baud rate of 115200 bps and each sample being 32 bits (float), we can only process 3600 samples per second (115200 bits per second / 32 bits per sample = 3600 samples per second), which is not enough for 16kHz
  - Increasing the baud rate does help with the spiking
- After testing different configurations for the sampling rate (AFE) and baud rate (UART), right now we are doing 8kHz sample rate and 800 kbps baud rate, which seems the fastest we can do without any distortions

Next steps

- Currently we have to skip a few bytes in SerialPlot in order to have a properly synced plot.
  - Try fixing this though it may not be possible as we found that it may result in the need for an increased baud rate. Test adding header bytes anyways just to make sure.
- Create the artifact detection algorithm and light up an LED when we detect an EMG signal being recorded.
  - With real signal
  - Long wire as an electrode cable (1 meter; crocodile clips)
  - Test setups 2 and 4

4/1/2026

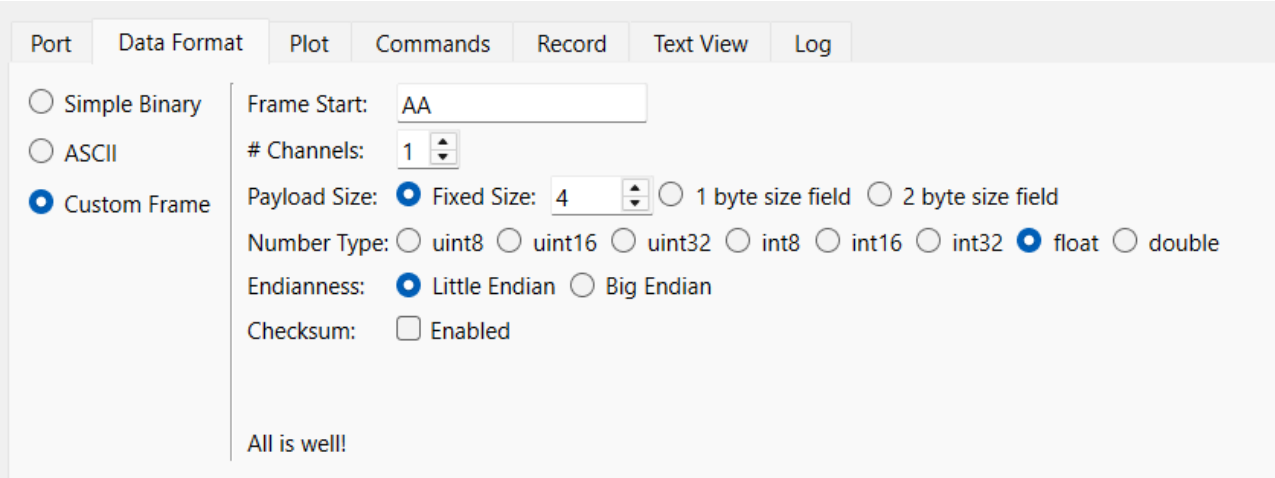
Dmytro Stavskyi, Luis Wong

Objective

- Try adding a header byte to help synchronize with SerialPlot

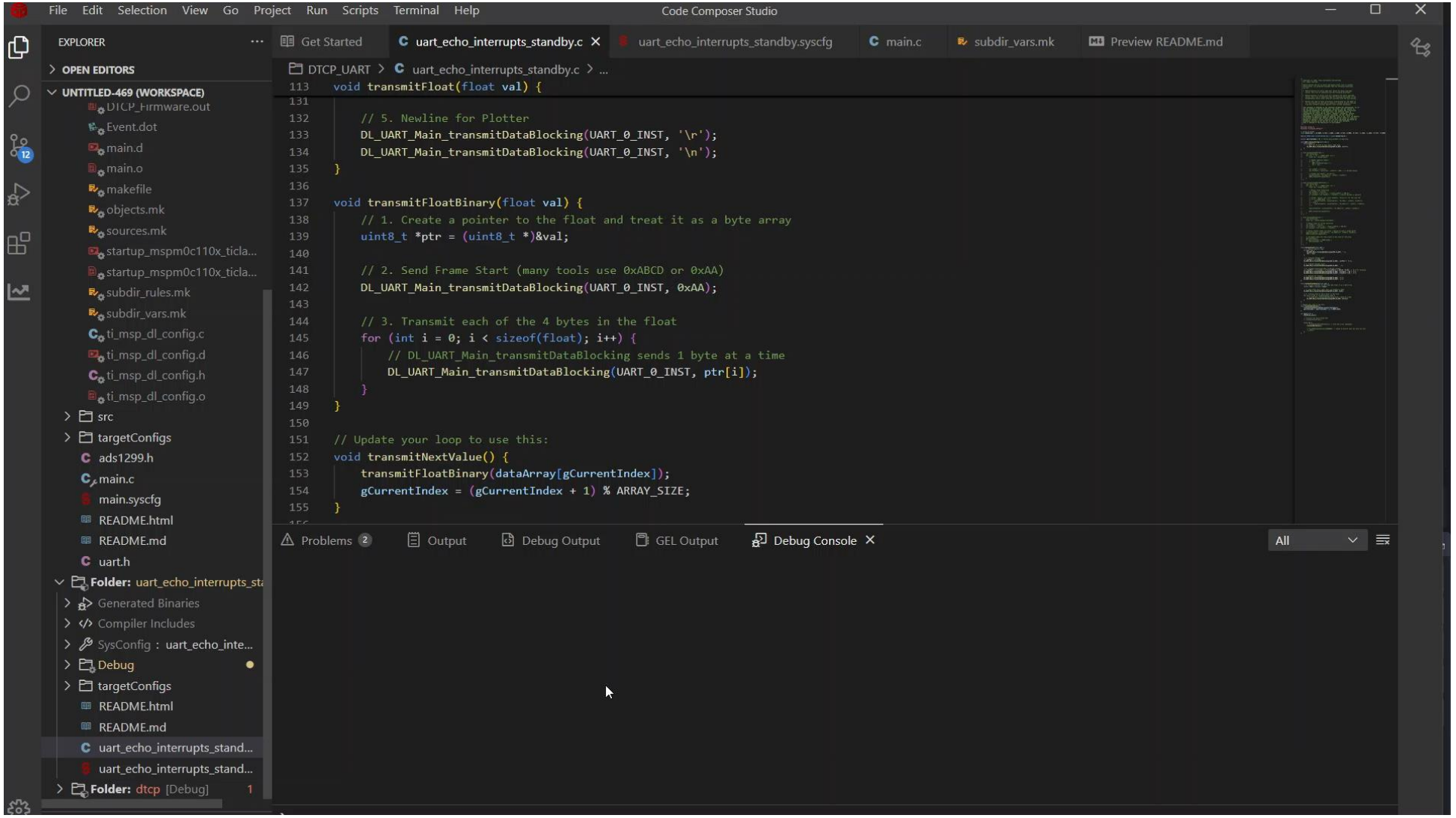
Methods

- CCS to hardcode header byte to each data frame.
- SerialPlot with Custom Frame preset.
  - Used setting in the screenshot.
- LaunchPad with UART only code.
- MATLAB to generate test array (100 Hz w/ 2.5 Vpp sampled at 1kHz).




```c
void transmitFloatBinary(float val) {
    // 1. Create a pointer to the float and treat it as a byte array
    uint8_t *ptr = (uint8_t *)&val;

    // 2. Send Frame Start (many tools use 0xABCD or 0xAA)
    DL_UART_Main_transmitDataBlocking(UART_0_INST, 0xAA);

    // 3. Transmit each of the 4 bytes in the float
    for (int i = 0; i < sizeof(float); i++) {
        // DL_UART_Main_transmitDataBlocking sends 1 byte at a time
        DL_UART_Main_transmitDataBlocking(UART_0_INST, ptr[i]);
    }
}
```

Results


File Edit Selection View Go Project Run Scripts Terminal Help
Code Composer Studio
EXPLORER
OPEN EDITORS
UNTITLED-469 (WORKSPACE)
Get Started
uart_echo_interrupts_standby.c
uart_echo_interrupts_standby.syscfg
main.c
subdir_vars.mk
Preview README.md
DTCP_UART > uart_echo_interrupts_standby.c > ...
113 void transmitFloat(float val) {
131
132 // 5. Newline for Plotter
133 DL_UART_Main_transmitDataBlocking(UART_0_INST, '\r');
134 DL_UART_Main_transmitDataBlocking(UART_0_INST, '\n');
135 }
136
137 void transmitFloatBinary(float val) {
138 // 1. Create a pointer to the float and treat it as a byte array
139 uint8_t *ptr = (uint8_t *)&val;
140
141 // 2. Send Frame Start (many tools use 0xABCD or 0xAA)
142 DL_UART_Main_transmitDataBlocking(UART_0_INST, 0xAA);
143
144 // 3. Transmit each of the 4 bytes in the float
145 for (int i = 0; i < sizeof(float); i++) {
146 // DL_UART_Main_transmitDataBlocking sends 1 byte at a time
147 DL_UART_Main_transmitDataBlocking(UART_0_INST, ptr[i]);
148 }
149 }
150
151 // Update your loop to use this:
152 void transmitNextValue() {
153 transmitFloatBinary(dataArray[gCurrentIndex]);
154 gCurrentIndex = (gCurrentIndex + 1) % ARRAY_SIZE;
155 }
DTCP_Firmware.out
Event.dot
main.d
main.o
makefile
objects.mk
sources.mk
startup_mspm0c110x_ticla...
startup_mspm0c110x_ticla...
subdir_rules.mk
subdir_vars.mk
ti_msp_dl_config.c
ti_msp_dl_config.d
ti_msp_dl_config.h
ti_msp_dl_config.o
src
targetConfigs
ads1299.h
main.c
main.syscfg
README.html
README.md
uart.h
Folder: uart_echo_interrupts_sta
Generated Binaries
Compiler Includes
SysConfig : uart_echo_inte...
Debug
targetConfigs
README.html
README.md
uart_echo_interrupts_stand...
uart_echo_interrupts_stand...
Folder: dtcp [Debug] 1
Problems 2
Output
Debug Output
GEL Output
Debug Console
All

Observation / Conclusion

- We were able to add the header byte to the UART transmission code and read a signal.
- Some initial testing in the lab showed lack of spikes when signal was supplied to the AFE through our DAD board (for function generation) though we need to test more thoroughly with the saline setup to be sure.
  - Should this testing be successful this will provide greater usability as we no longer have to skip bytes to ensure proper synchronization.

Next steps

- Create the artifact detection algorithm and light up an LED when we detect an EMG signal being recorded.
  - With real signal (using header bytes should we detect no spiking occurs).
  - Long wire as an electrode cable (1 meter; crocodile clips) and
  - Use a larger container with PBS instead of saline (container size being similar to size of a rat; create distance between positive / negative electrodes and the bias electrode)
  - Units for real data EMG CSV file: left column (x) is in , right column (y) is in Volts (time different between each stimulus peak is 1s; tune based on the setup)
  - Test setups 2 and 4
  - Maybe code in a threshold (large value) to detect stimulation artifact and wait (some time) for another (smaller value) for EMG signal, then LED lights up?
- Validate the new PCB works after assembly and test combined firmware for power transmission and AFE.

4/8/2026

Dmytro Stavskyi, Luis Wong

Objective

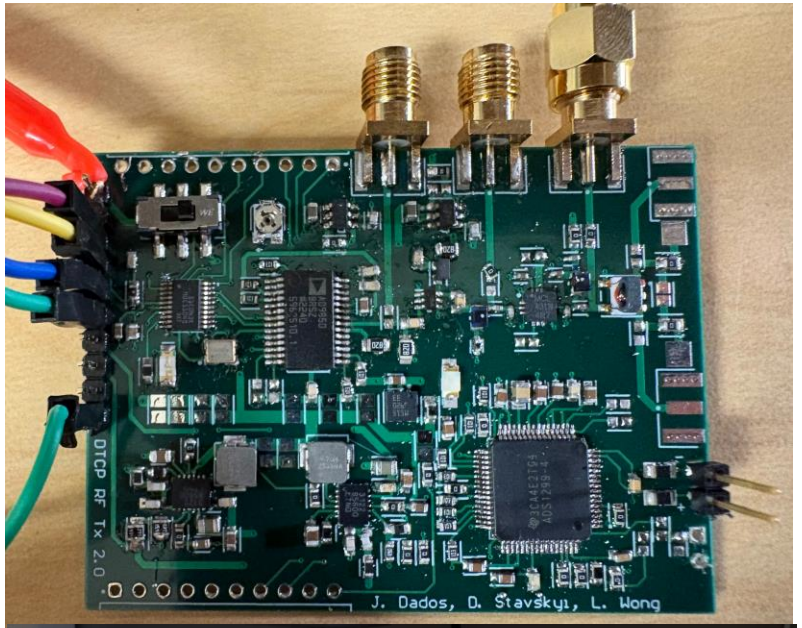
- Validate the AFE functionality on the new PCB
- Test combined firmware for power transmission and AFE measurements

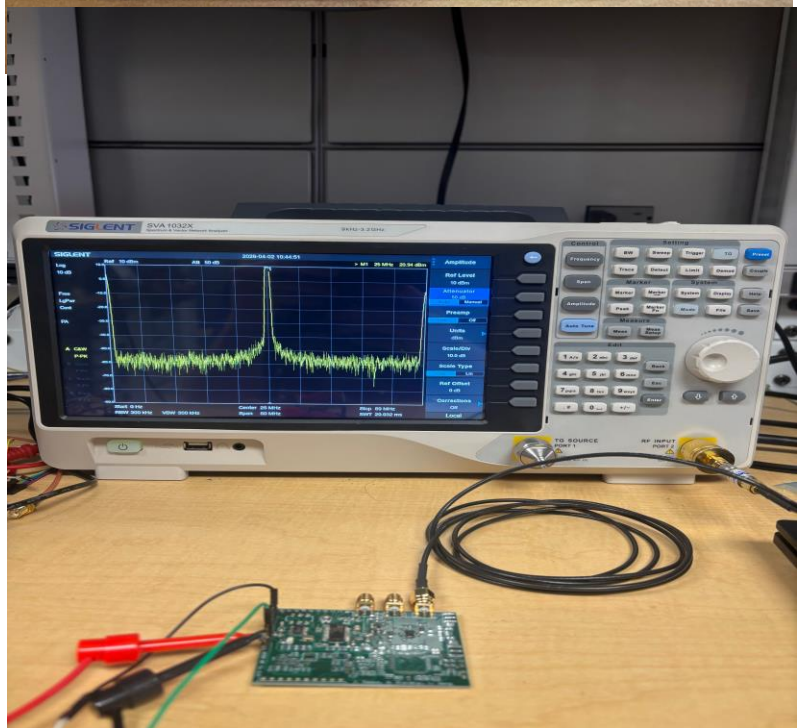
Methods (AFE)

- For the firmware, we copied the code from the projects used to program AFE and UART to the project that programs the power transmission.
  - Initially, after copying everything with no changes, we had memory overflow and so could not build a project, but we managed to solve this after deleting some things.
- Test the AFE incrementally: first check whether we can read/write registers (determine if SPI working and the chip works correctly), then whether we can measure the internal test signal (to determine if UART and the chip works correctly), and then do the normal external input connected directly to function generator, and then normal input in the PBS solution

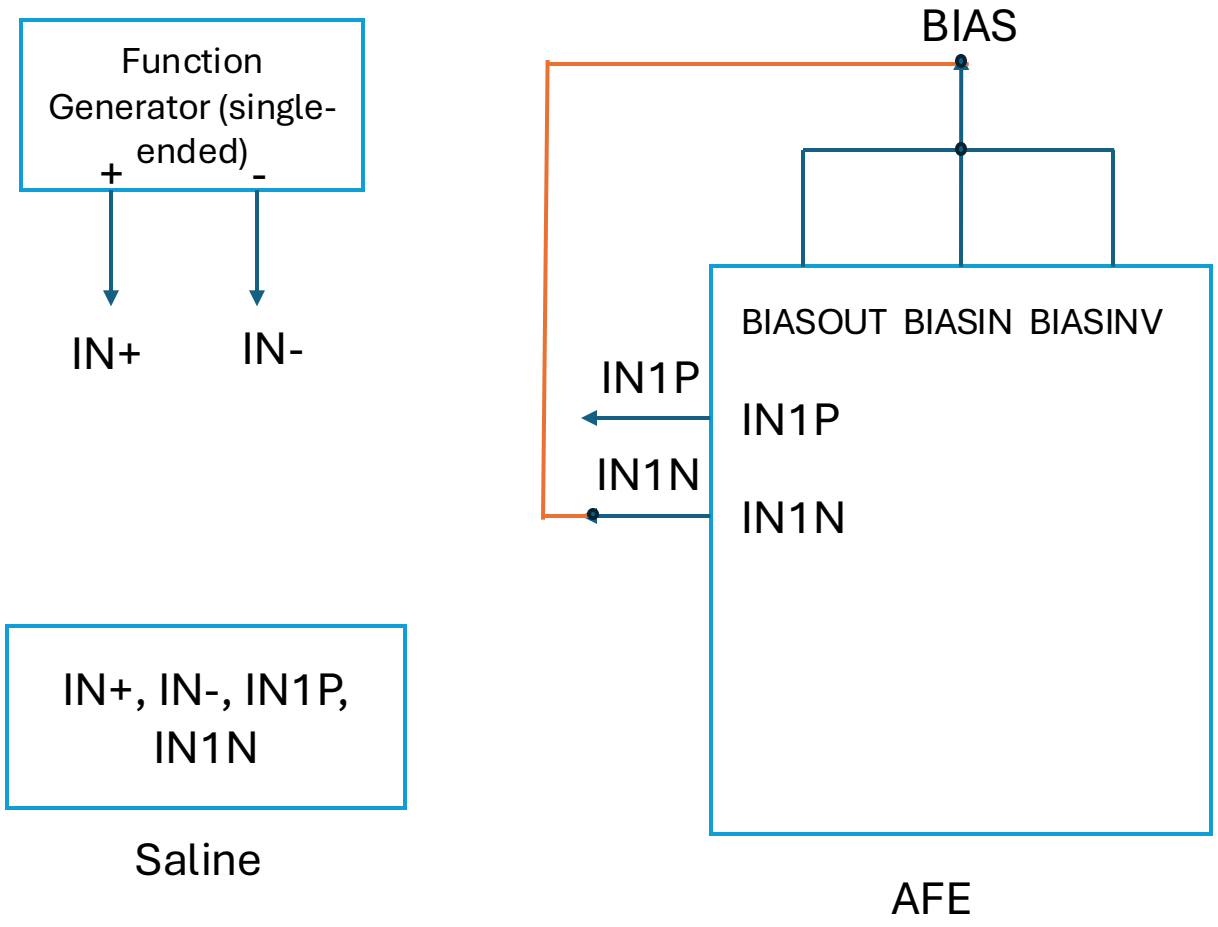
Methods (PA)

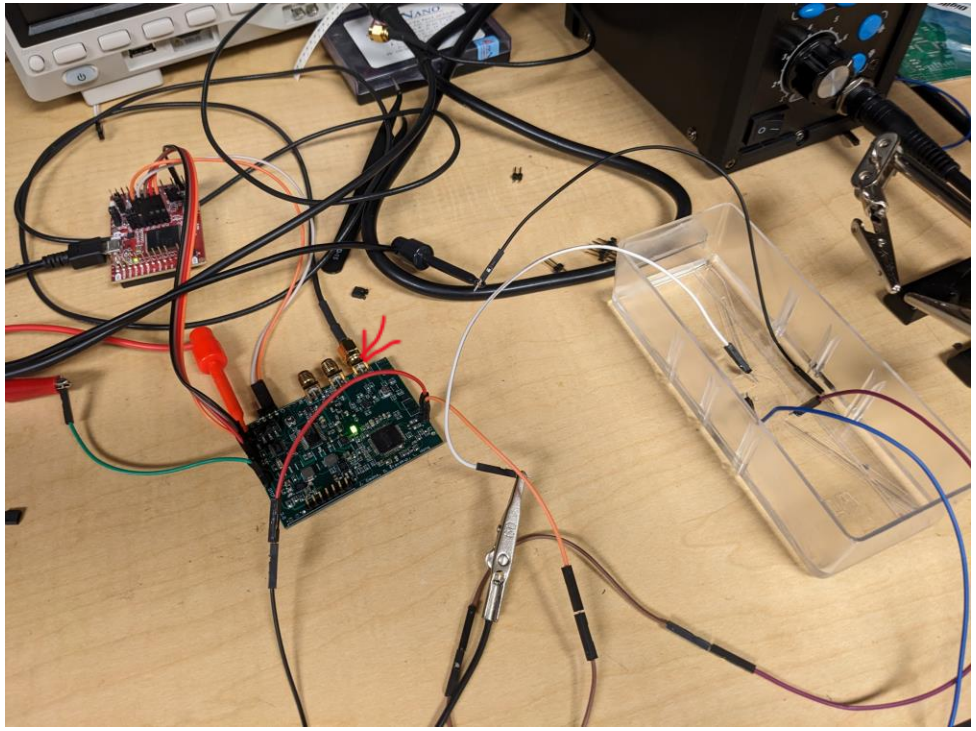
- For the Power Amplifier we connected a 10dBm attenuator to the spectrum analyzer. Centered the spectrum analyzer to 25Mhz and gave a range from 0 –50Mhz. Other setting are Attenuator: Auto, PreAmp: On, Ref Level: 10 dBm. We connect the third SMA connector found on the board to the RF input port of the spectrum analyzer and turn on our device.
- In CCS we would just comment out the code responsible for controlling the power switching when not in use and undo when we wished to test it again.



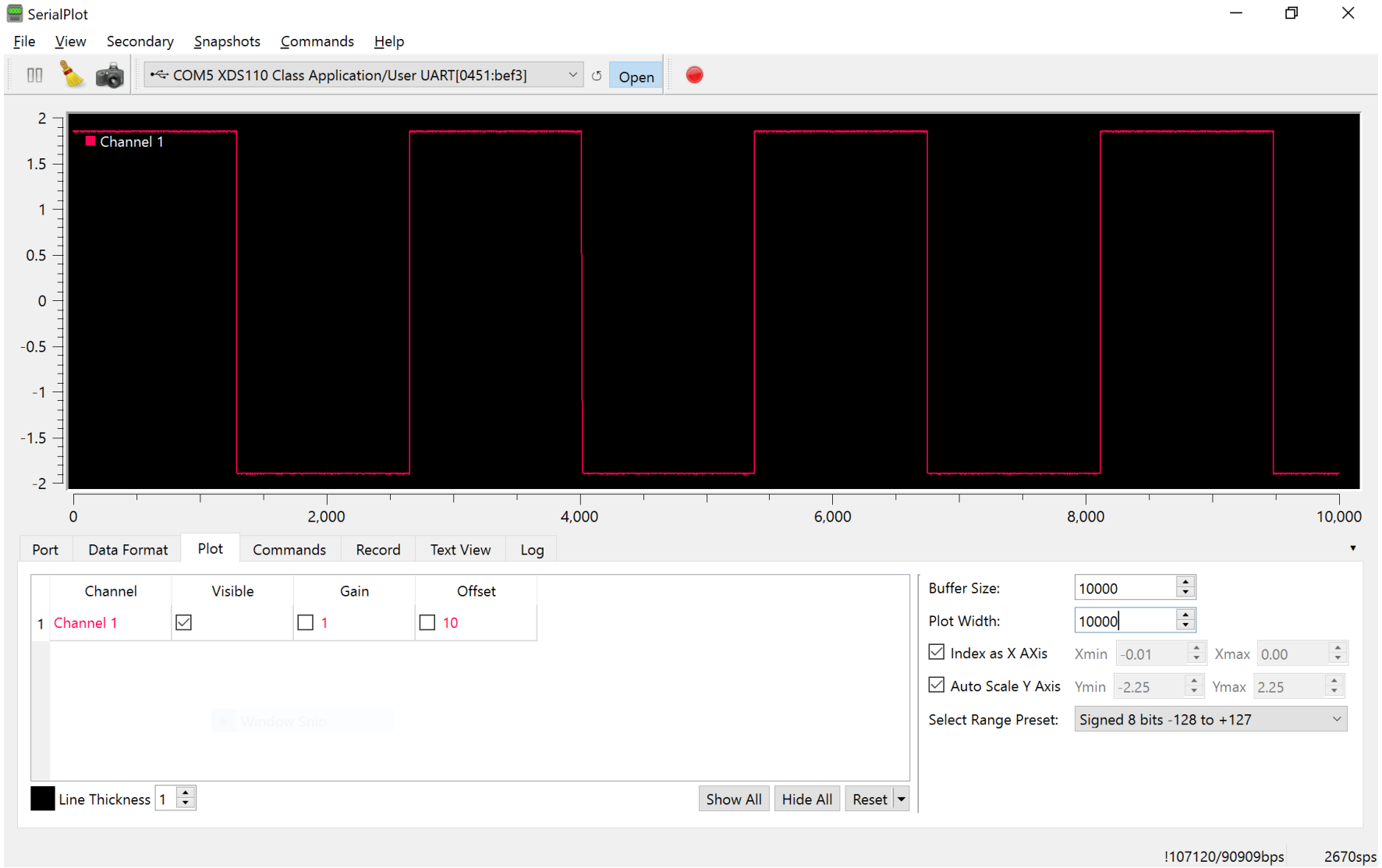


Methods (Second Setup; no bias in PBS but shorted to the negative terminal)



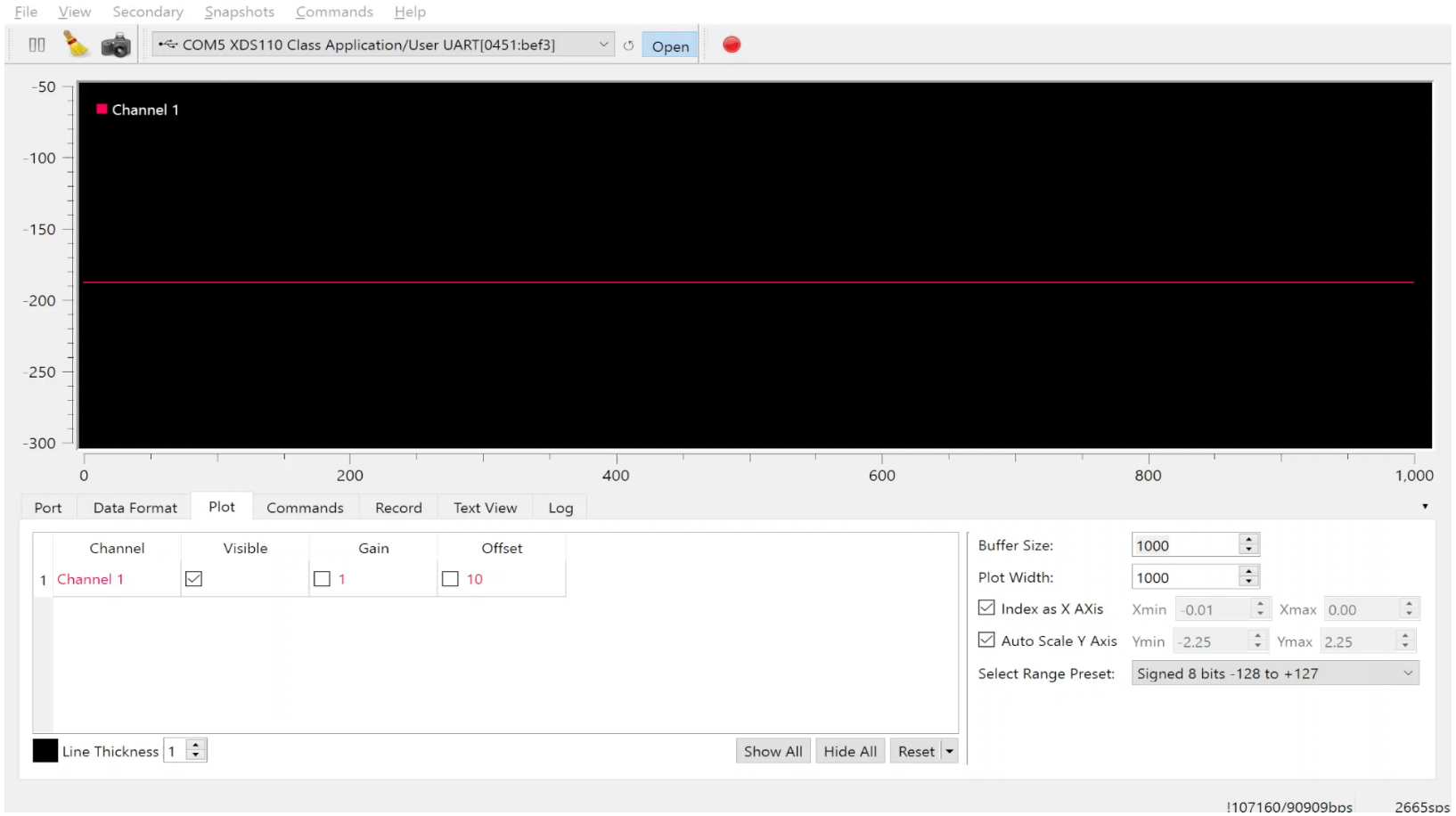


Results (Internal Test Signal)



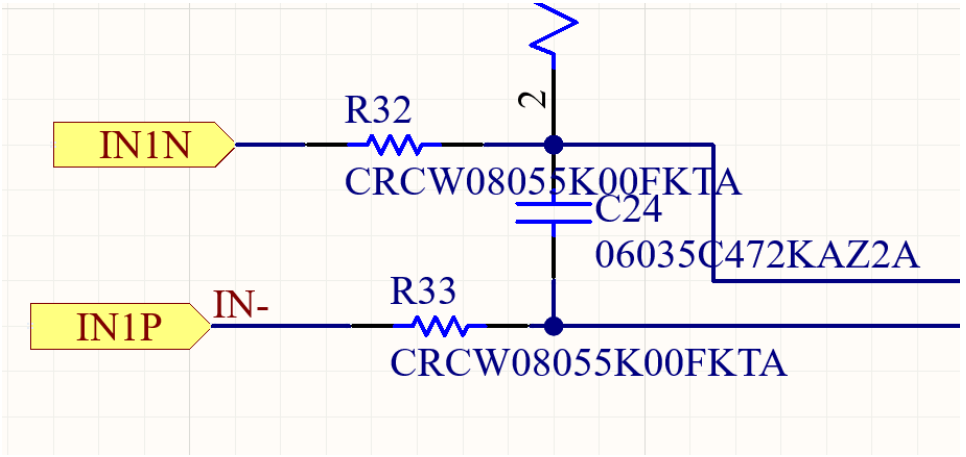
- After resoldering a couple of times and using the ADS1299 chip from the previous board in this new board, we managed to record the test signal properly (square wave)

Results (Measurements directly connected to function generator)

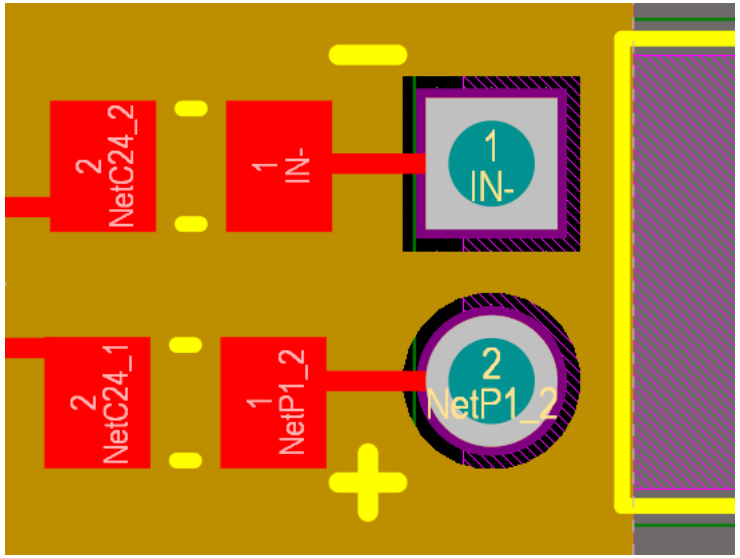


- First, no signal, then 10 mVpp, then 50 mVpp and lastly, 100 mVpp (all 100 Hz sine waves, no offset).
- At 10 mVpp amplitude level, which on the previous board was measured with no problem, there are significant distortions, which disappear as the amplitude increases.
- This implies that something is wrong with the biasing and/or input terminals

Results (Input Terminals)



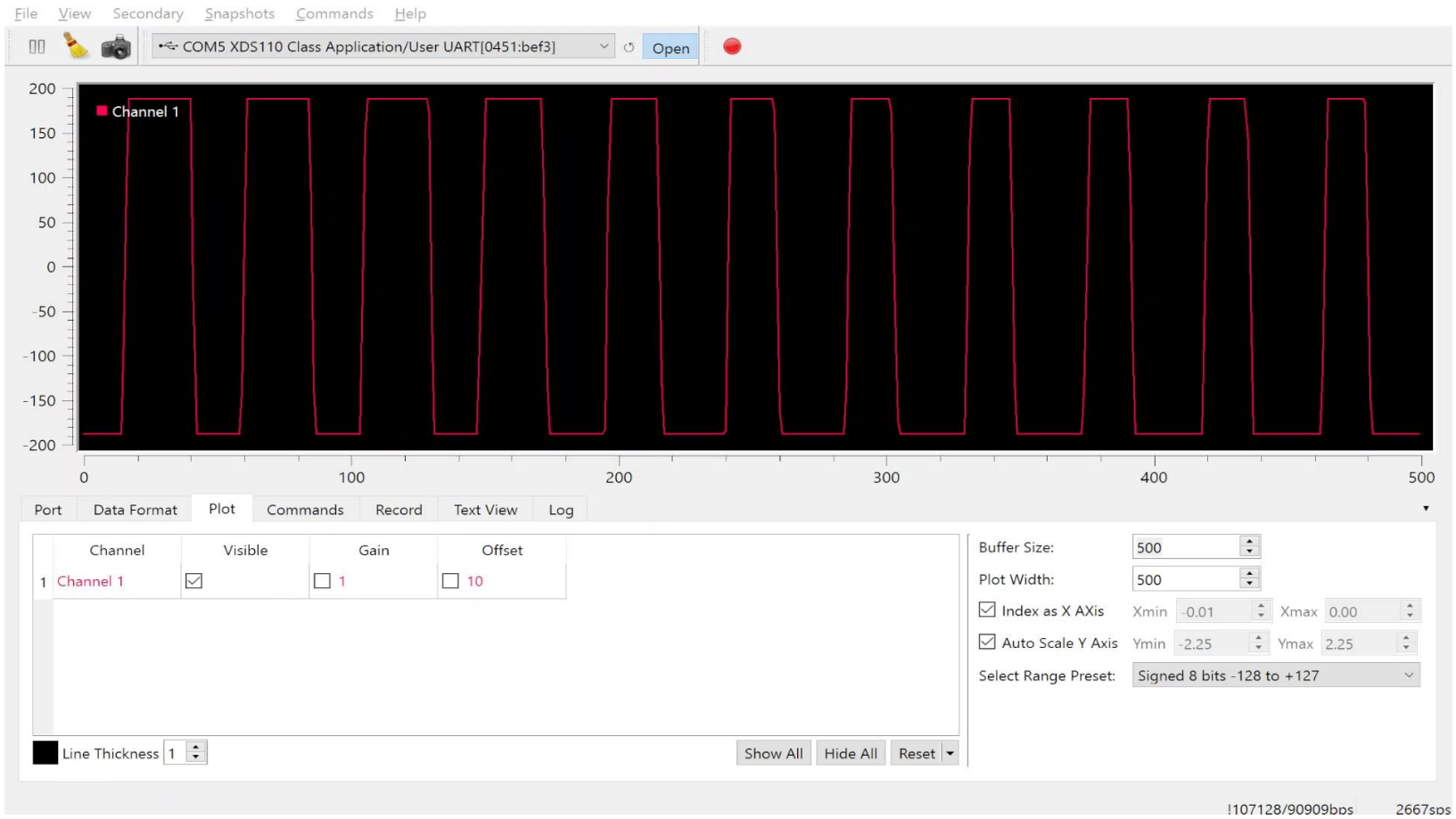
AFE Schematic



PCB Layout

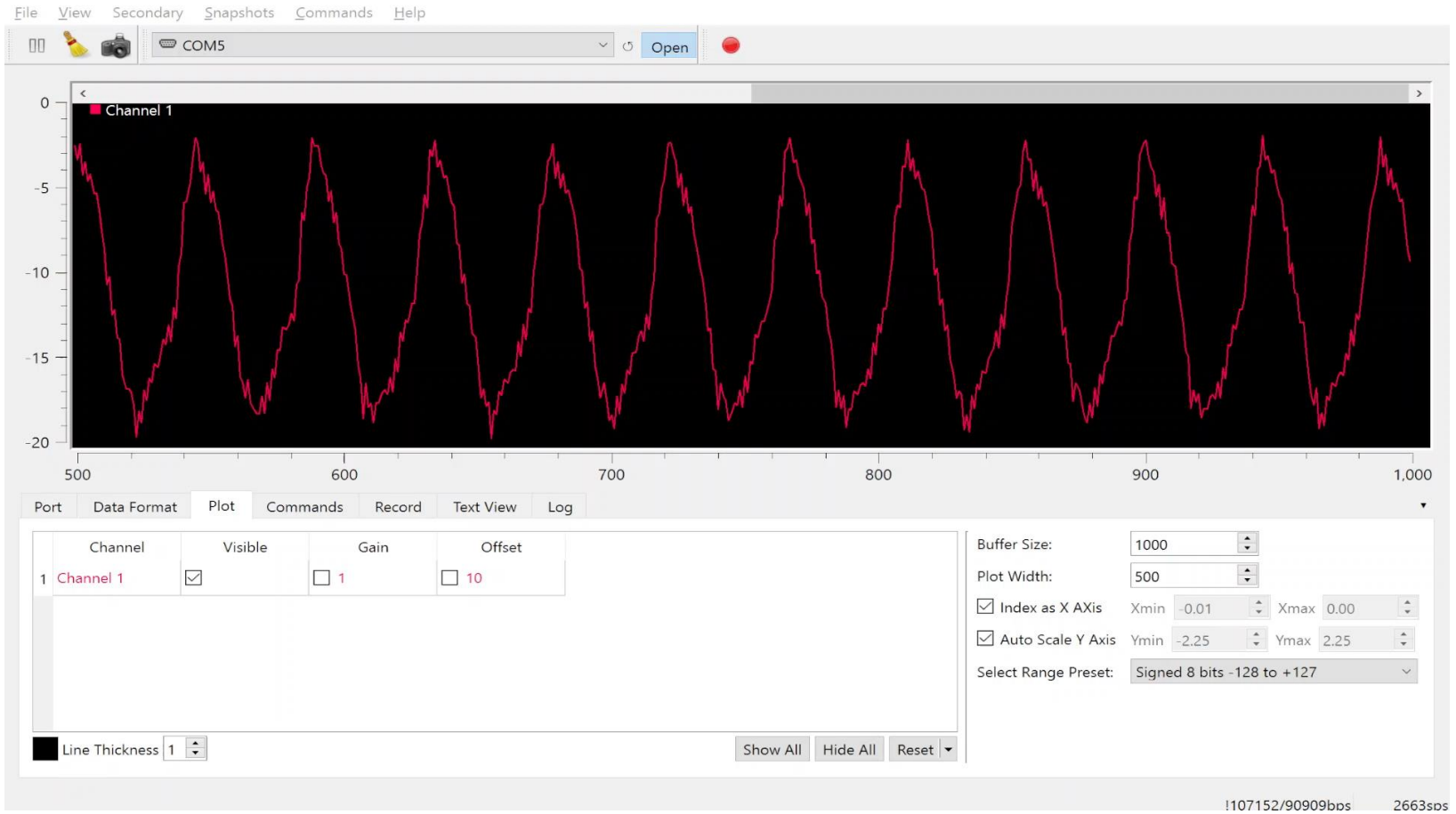
The IN1P was labeled as IN-, so the input terminals are inverted, and bias is connected to the “positive” terminal instead of “negative”.

Results (Measurements directly connected to function generator with “inverted” terminals)



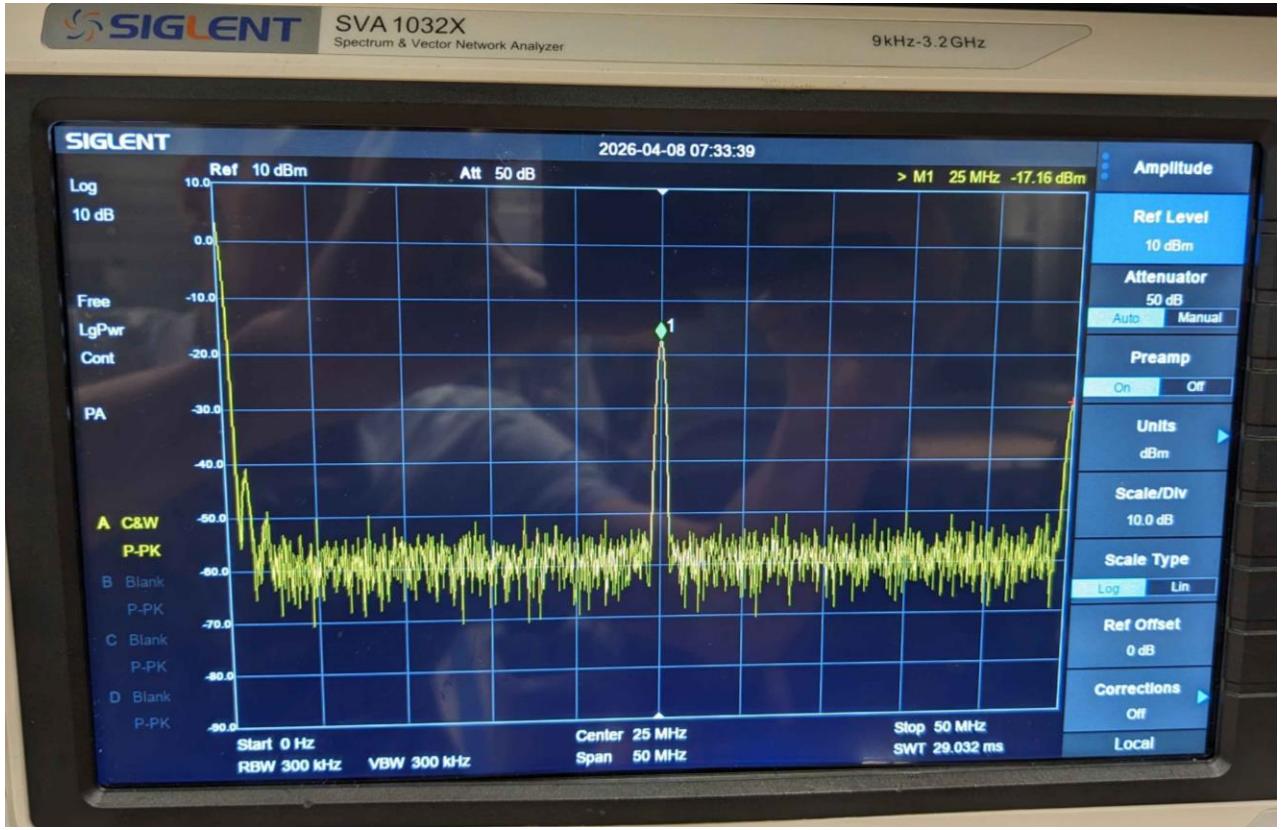
- So, the positive terminal of the function generator is connected to the “negative” (as denoted by the board) input terminal and same with the other terminals.
- Same increments as in the previous slide

Results (Measurements in PBS)



- First no signal and then the test EMG signal increasing from 4Vpp to 12Vpp
- There is a great amount of noise that is not cancelled by the bias.
- Even as we changed the placement of electrodes, we did not manage to eliminate the noise and get proper measurements

Results (Combined firmware running)



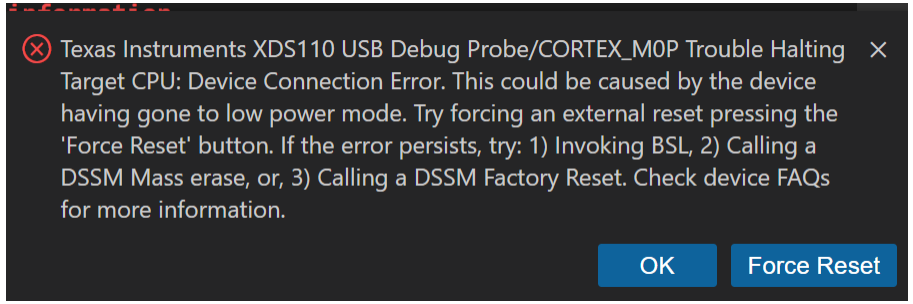
- When running the code for both AFE and RF transmitter, we did not get correct power, even when commented out the code for the AFE, so it may be a hardware issue due to resoldering the AFE parts.
- Need to test this more, but it seems that the presence of the attenuator destroys the measurements

Observation / Conclusion (Measurements in PBS)

- The noise is probably from the power supply, so maybe we can attempt to power the board from the launchpad or try using a battery?
- The issue may also be with biasing, so we can try to copy the last used setup, setup 4, that is, bias as a separate electrode and not shorted to the negative terminal?

Observation / Conclusion (CCS errors)

```
NONMAIN CRC validation is not supported for MSPM0C1103/MSPM0C1104/MSPS003Fx. Exercise caution when erasing/programming NON-MAIN.
JTAG Communication Error: (Error -1001 @ 0x0) Requested operation is not supported on this device. (Emulation package 20.3.0.3656)
Failed to remove the debug state from the target before disconnecting.  There may still be breakpoint op-codes embedded in program memory.  It is rec
ommended that you reset the emulator before you connect and reload your program before you continue debugging
```




```
Error connecting to the target: (Error -6310) PRSC module failed to read a register. (Emulation pa
ckage 20.3.0.3656)
```

- When attempting to program the device or during the debug session, a lot of times we get errors like these.
- Usually, they are fixed by restarting the CCS and/or repowering the board.

Next steps

1. Power the board from the launchpad and try using a battery
    1. See if that solves the noise issue
2. Check the code for the RF and AFE and sent it to Jeremiah
    - RF by itself -> AFE by itself -> AFE and RF working together but not at the same time -> AFE and RF working together
    - Focus on the AFE for now
3. Create the artifact detection algorithm and light up an LED when we detect an EMG signal being recorded.
    1. Maybe code in a threshold (large value) to detect stimulation artifact and wait (some time) for another (smaller value) for EMG signal, then LED lights up?
4. Replicate the last used setup, setup 4, that is, bias as a separate electrode and not shorted to the negative terminal?

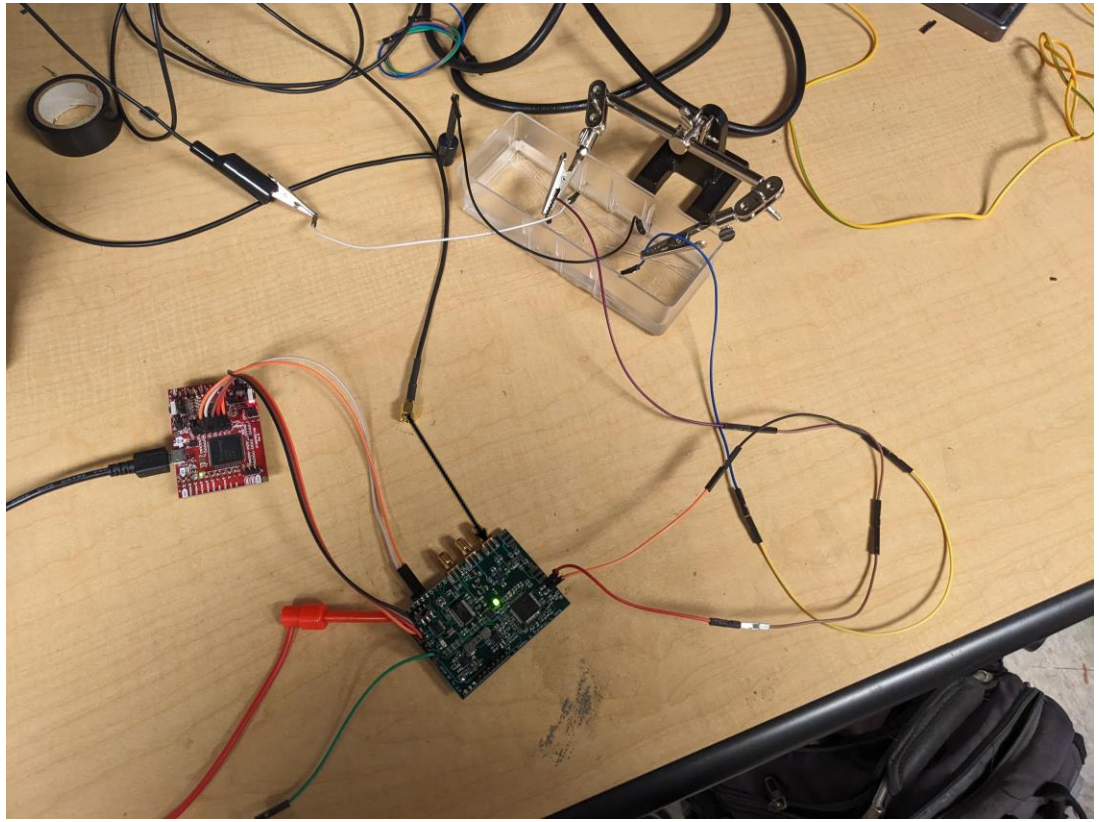
4/15/2026

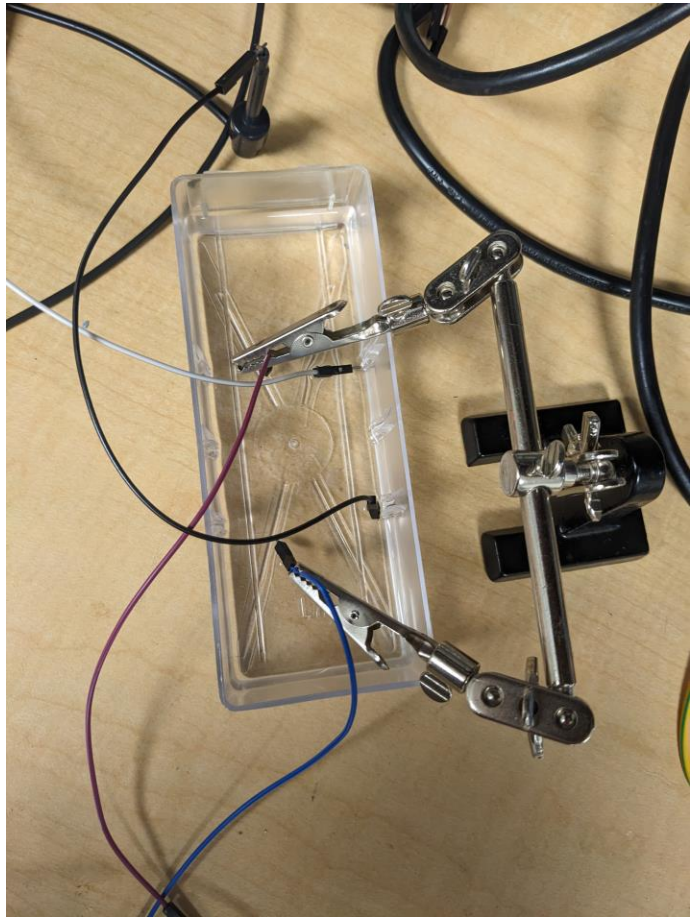
Dmytro Stavskyi, Luis Wong

Objective

- Power the board from the launchpad and see if that will solve the AFE measurements noise issue
- Check the bias and grounds setup

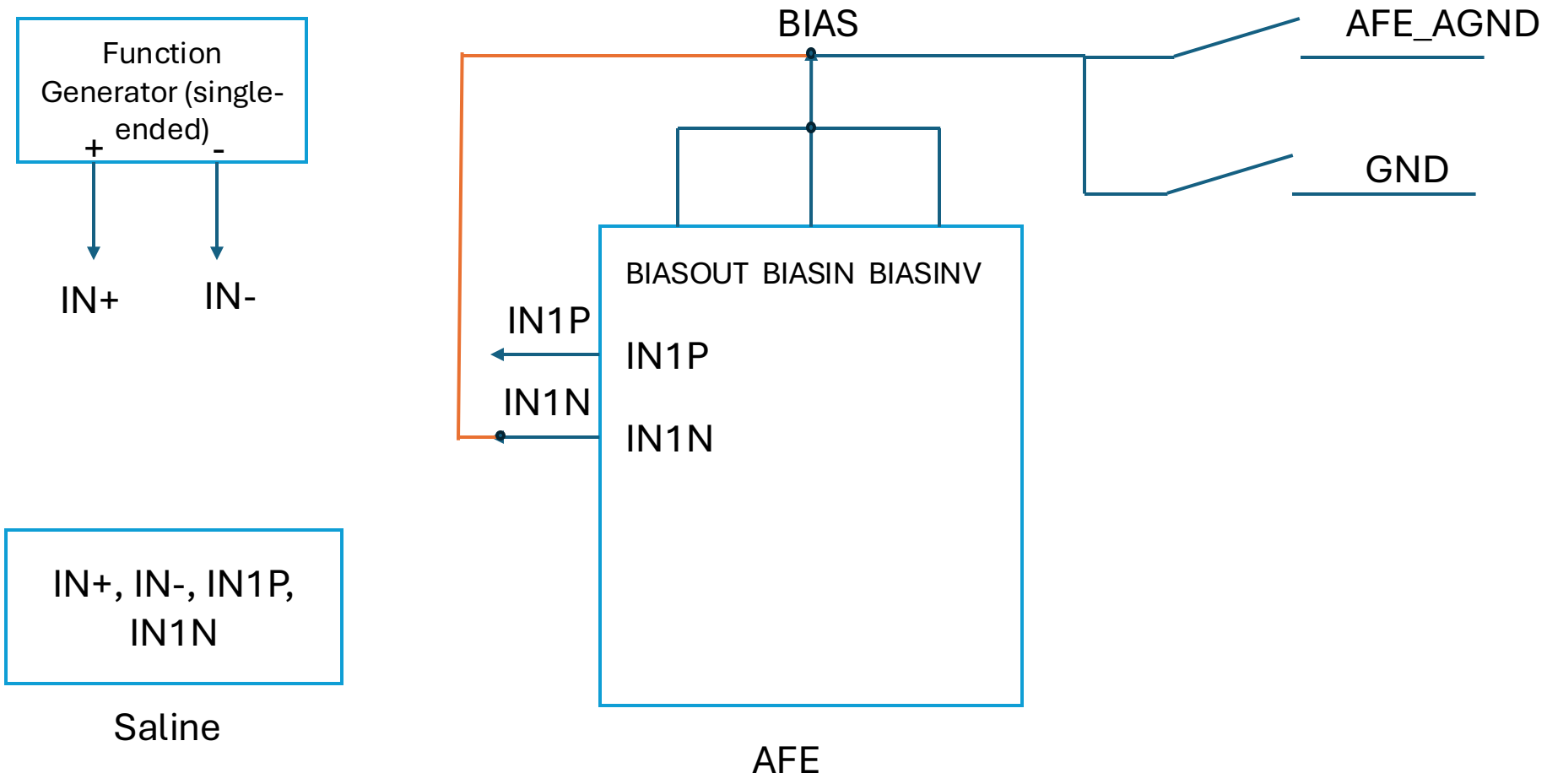
Methods (Setup for slides 116-117)





- White: IN+
- Black: IN-
- Purple:
- INP1
- Blue: INN1

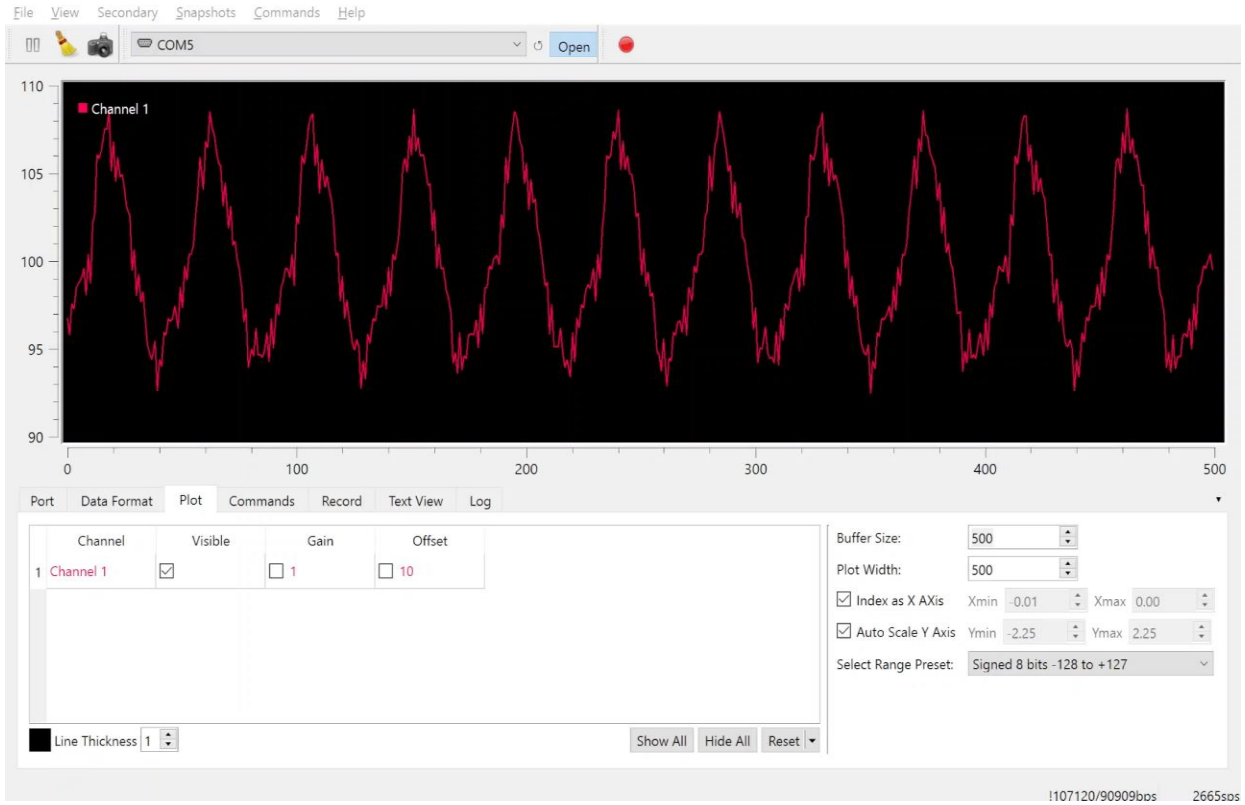
Methods (no bias in PBS but shorted to the negative terminal)



Results (Launchpad and Power supply measurements in PBS)

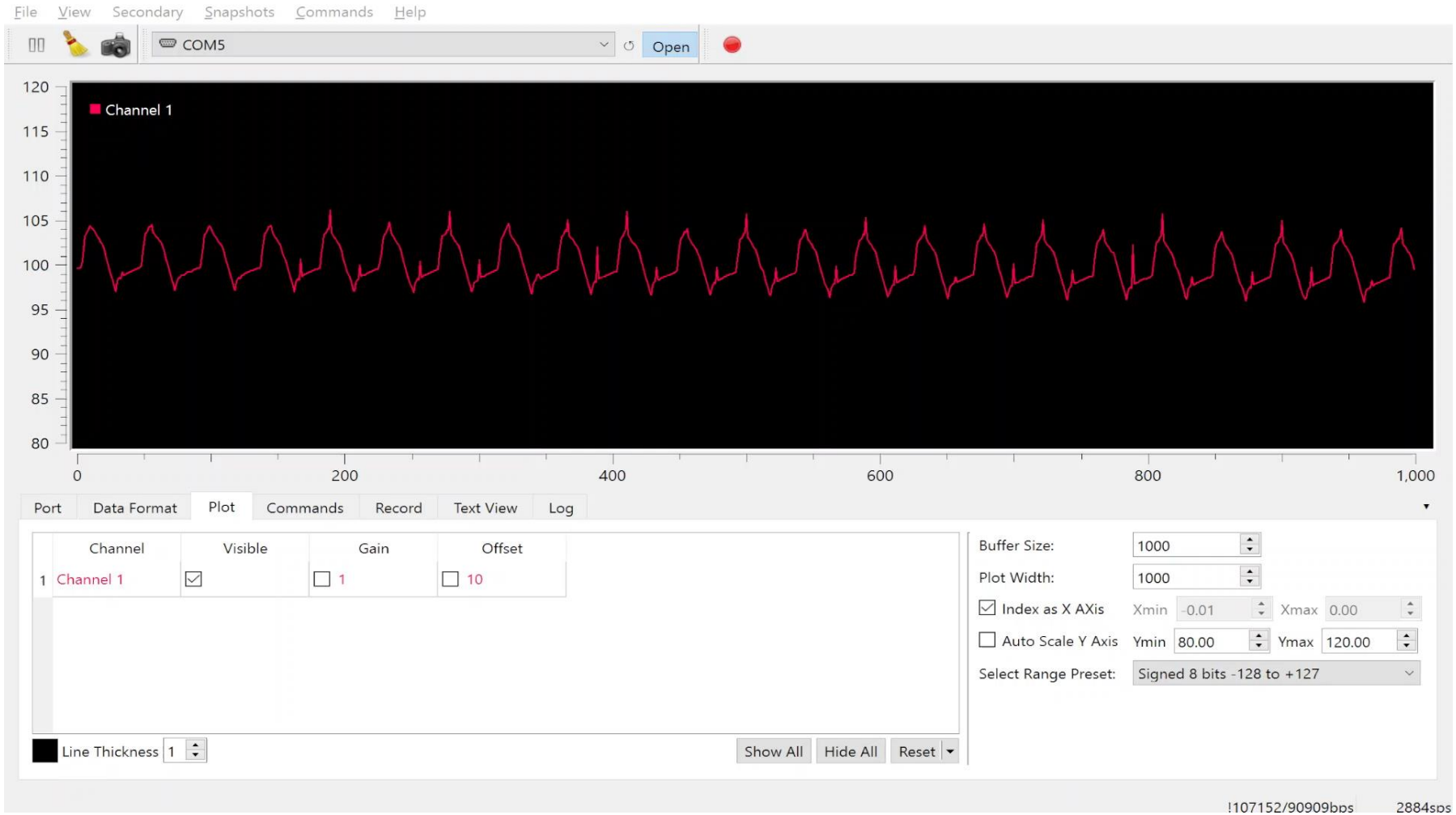
Launchpad

Power supply



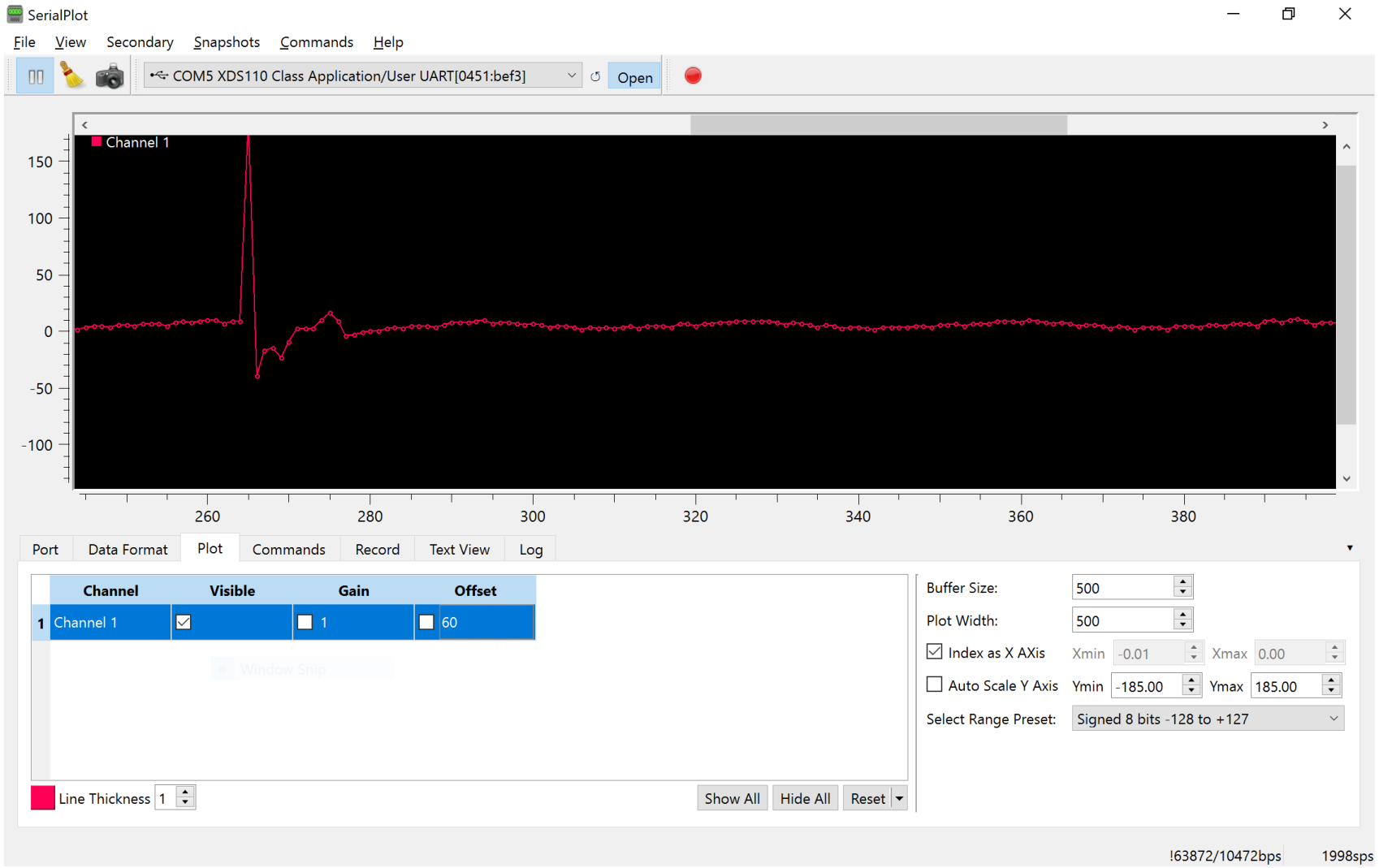
- Methodology: no signal -> 1Vpp 100 Hz sine wave -> 2Vpp 100 Hz sine wave -> 4Vpp 100 Hz sine wave
- With launchpad powering the board, noise is reduced, around half (~4mV) as that with the power supply (~8mV), but we cannot program the chip (low power mode error).

Results (Measurements of EMG test signal in PBS with launchpad powering)

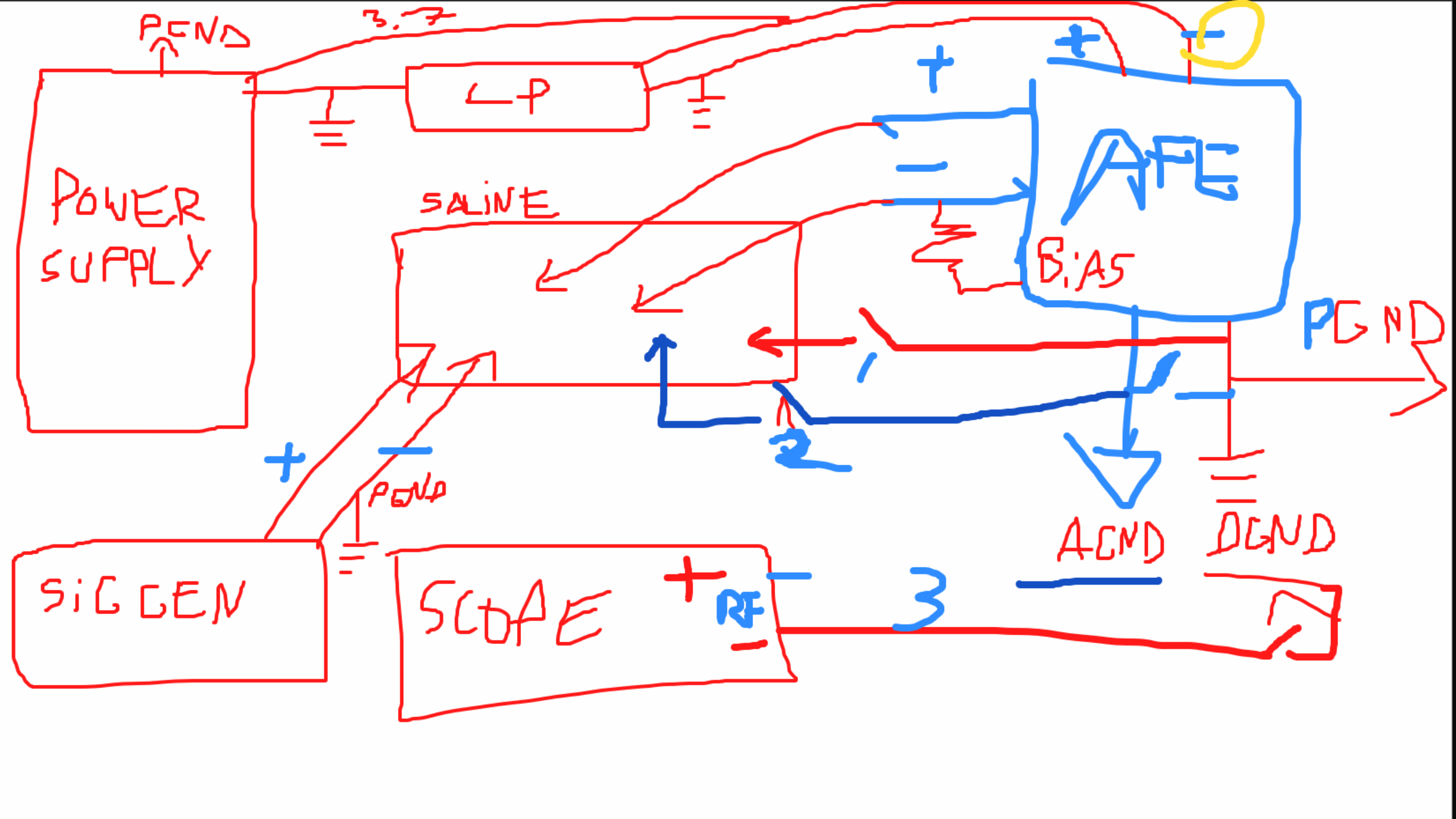


- The noise is still too large to make adequate measurements involving EMG signal

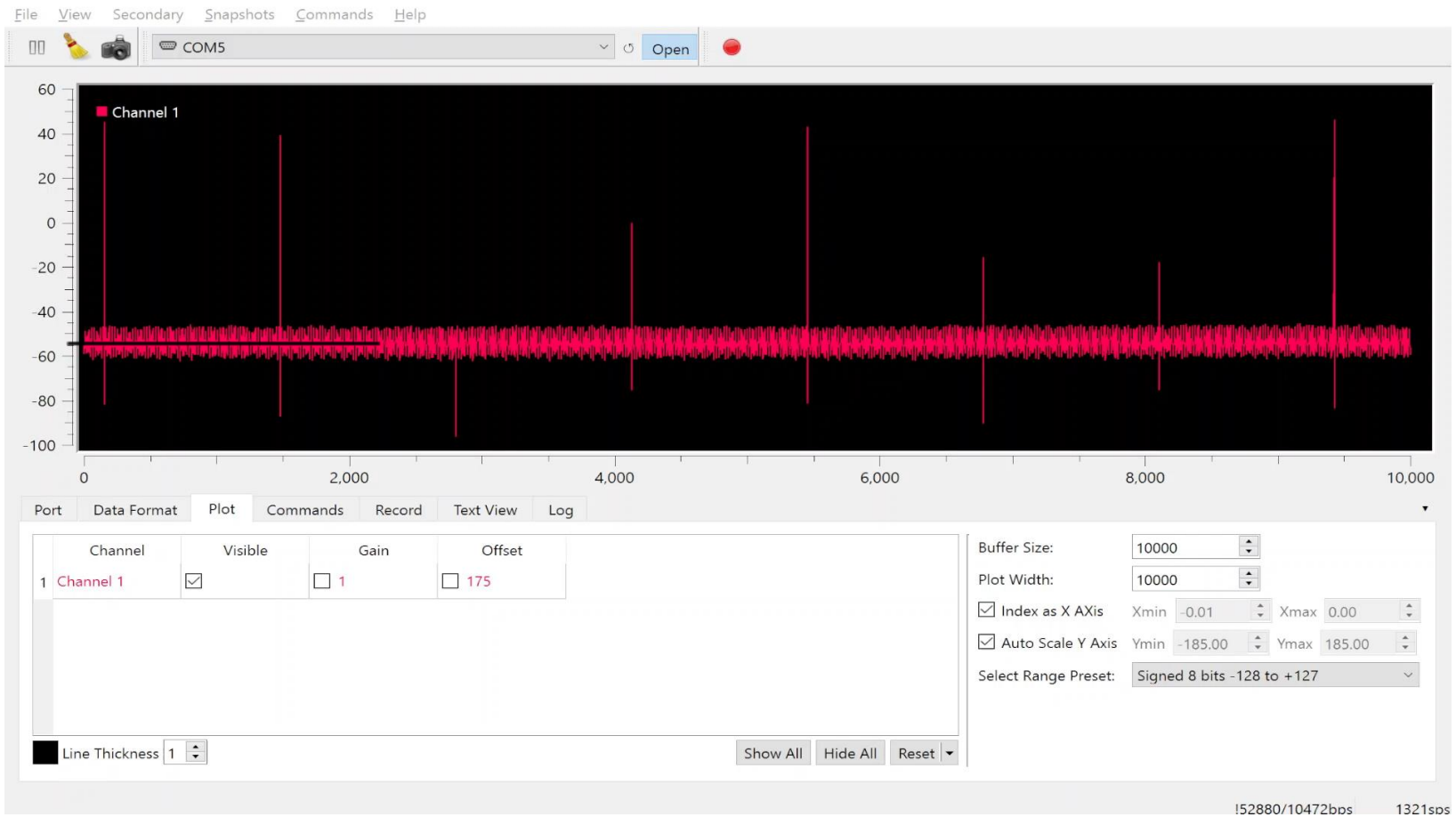
Results (No RF wire)



- Setup with the power supply
- After removing the RF output wire, placing the electrodes in a certain way, and zooming out in the plot, we were able to record this.
- However, the offset is not constant and actually overflowed to -185 mV later on, where we could only measure the initial spike


PGND
3.7
LP
POWER SUPPLY
SALINE
AFE
BIAS
PGND
1
2
PGND
SIGGEN
SCOPE
RF
3
ACND
DGND

Results (Connecting RF output)



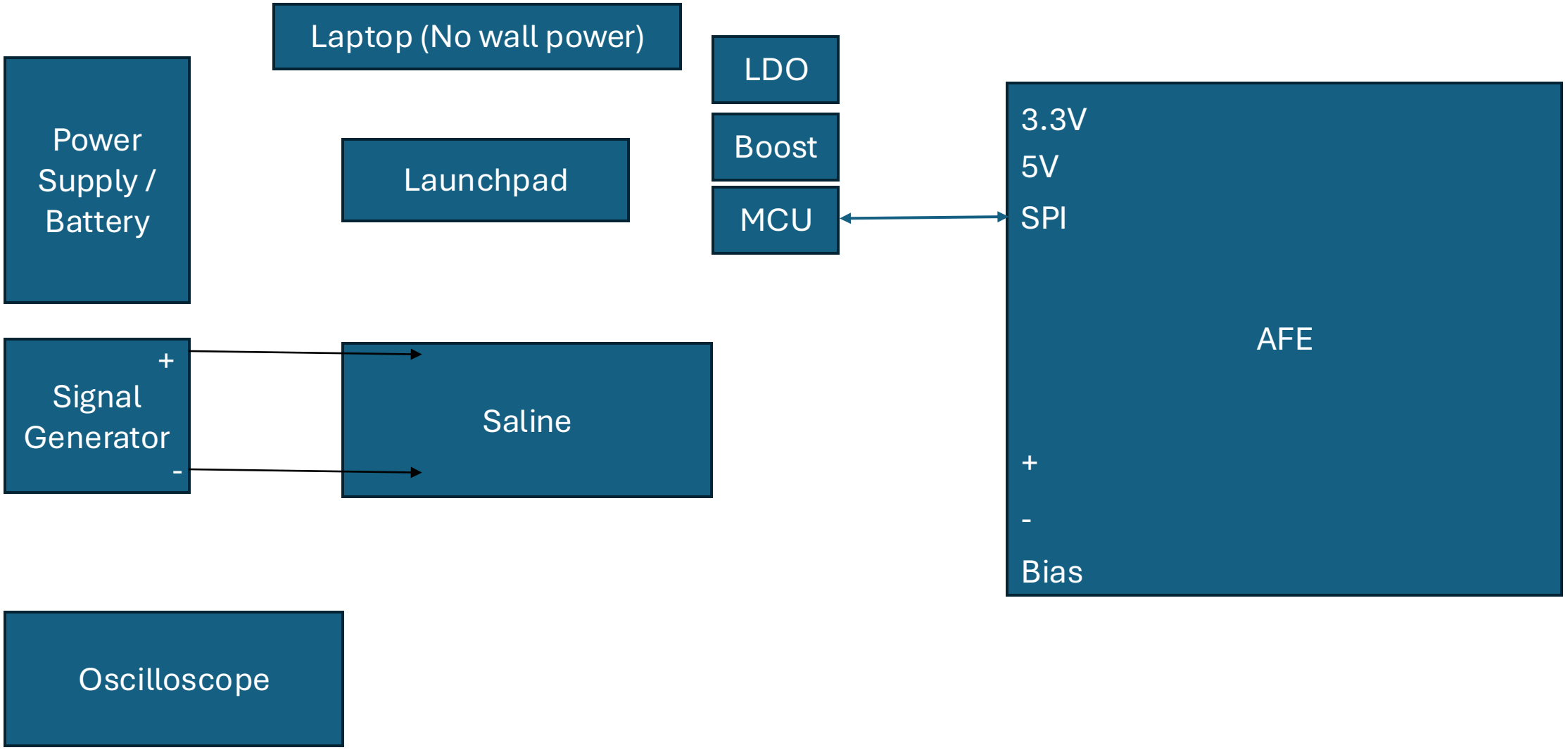
- Connecting RF cable (ground; the wire is connected in 6:00 in the video) (to oscilloscope) affects the AFE measurements

Observation / Conclusion (Measurements in PBS)

- Depending on the placement of the electrodes in the PBS, we get different measurements.
  - Some of the problems with the measurements might be due to using the function generator
- Connecting either analog or digital ground to PBS solution causes the same result as in slide 117

Next steps

- Test the battery
- Test the power transmitter and AFE circuits working together


Laptop (No wall power)
LDO
Boost
MCU
3.3V
5V
SPI
Power Supply / Battery
Launchpad
AFE
+
Signal Generator
-
Saline
+
-
Bias
Oscilloscope